

LM01 Fixed-Income Instrument Features

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Version 1.0

1. Introduction

This learning module introduces the features of fixed income instruments and the legal contracts that govern them.

2. Features of Fixed-Income Securities

Fixed-income instruments, such as loans and bonds, are debt instruments that represent a contractual agreement in which an issuer borrows money from investors in exchange for interest and future repayment of principal.

Basic Features of a Bond

The basic features of a fixed-income security include:

Issuer:

Issuer is the entity that has issued the bond, or the one who has borrowed money. It is responsible for servicing the debt (paying interest and principal payments). Bonds can be classified into the following categories based on the type of the issuer:

- *Supranational* organizations: Bonds issued by international organizations such as the World Bank and IMF.
- *Sovereign* governments: Debt of national governments such as government of the United States, Germany, or Italy.
- *Non-sovereign* governments: Bonds that are not issued by a national or central government. Instead, these are issued by states, provinces, municipalities, etc. For example, municipal bonds in the United States, etc.
- *Quasi-government* entities: Bonds issued by agencies that are financed either directly or indirectly by the government. They are also called sub-sovereign or agency bonds. For example, Ginnie Mae, Fannie Mae, and Freddie Mac in United States.
- *Companies*: Bonds issued by a corporate. A distinction is often made between financial issuers and non-financial issuers.

All bonds are exposed to credit risk. 'Credit risk' is the risk that interest and principal payments will not be made by the issuer as they come due. Credit rating agencies such as Moody's and S&P assign a rating to issuers based on this risk.

Bonds can be classified into two categories based on their creditworthiness:

- Investment grade.
- Non-investment grade, or high-yield or speculative bonds.

Maturity:

The maturity date is the date on which the issuer will redeem the bond by paying the outstanding principal amount. Once a bond has been issued, the time remaining until maturity is known as the tenor of the bond.

- If original maturity is one year or less, the bond is called **money market security**.

- If original maturity is more than a year, the bond is called **capital market security**.

Par value:

Also known as face value, maturity value, or redemption value.

- It is the principal amount that is repaid to bondholders at maturity.
- If market price > par value, the bond is trading at a premium.
- If market price < par value, the bond is trading at a discount.
- If market price = par value, the bond is trading at par.

Coupon rate and frequency:

Coupon rate is the percentage of par value that the issuer agrees to pay to the bondholder annually as interest. It can be a fixed rate or a floating rate. The coupon frequency may be annual, semi-annual, quarterly, or monthly. Based on the frequency of coupon payments, we can classify bonds into the following:

- *Plain vanilla bond*: It is the most basic type of bond with periodic fixed interest payments during the bond's life and the principal paid on maturity.
- **Floating-rate bond or floating-rate notes or floaters**: Unlike vanilla bonds, the interest rate of a floating-rate bond is not fixed. The coupon payments are based on a floating Market reference rate (MRR) at the start of the period. Some bonds specify the coupon rate as two components: a market reference rate, plus a spread. **The coupon rate and coupon interest paid in every period change as the reference rate changes.**
- **Zero-coupon bonds or pure discount bonds**: Bonds that have only one payment at maturity. These are bonds that do not make a coupon payment over a bond's life, and are sold at a discount (less than the par value) at issuance. At maturity, the investor receives the par value of the bond. Think of it as interest getting accumulated during the bond's life and being paid at maturity for a zero-coupon bond.

Seniority:

A single borrower may issue bonds with different seniority rankings. Seniority ranking determines who gets paid first, or who has the first claim on the cash flows of the issuer, in the event of default/bankruptcy/restructuring. Senior debt has a priority over other debt claims. Junior or subordinated debt has a lower priority and is paid only once the senior claim are satisfied.

Contingency provisions:

A contingency provision is a clause in a legal document that allows for some action if the event or circumstance does occur. It is also called an embedded option. The most common contingency provisions for bonds are: call, put, and conversion options.

Yield Measures

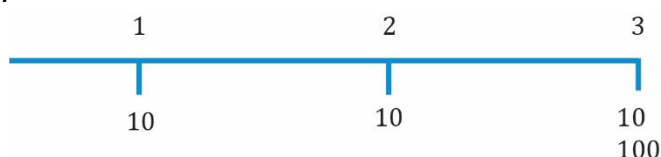
There are two widely used yield measures to describe a bond: current yield and yield to maturity.

Current yield or Running yield

Current yield is the annual coupon divided by the bond's price and is expressed as a percentage. For example, consider a three-year, annual coupon bond with a par value of \$100. The bond is issued at \$95. The coupon rate is 10%, so the coupon payments are \$10 every year. Current yield = $\frac{10}{95} = 10.5\%$

Yield to Maturity (YTM)

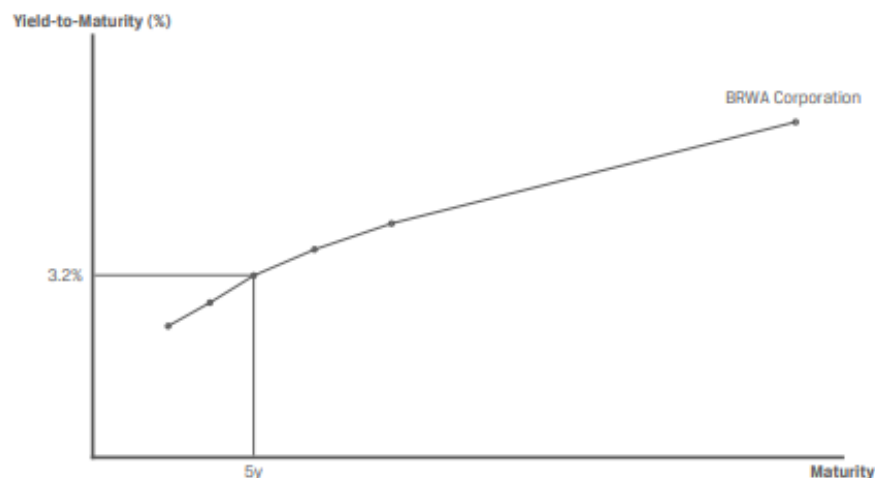
Yield to maturity is the internal rate of return on a bond's expected cash flows. In other words, it is the expected annual rate of return an investor will earn if the bond is held to maturity. It is also known as 'yield to redemption' or 'redemption yield'. The IRR can be calculated easily using the financial calculator. Let us consider the same bond as in the current yield example:



Input these values: $CF_0 = -95$; $CF_1 = 10$; $CF_2 = 10$; $CF_3 = 110$. Computing for IRR, you should get 12.08%. As you can see, YTM is higher than the coupon rate. For a discount bond such as this one, the YTM is higher than the coupon rate. A bond's price is inversely related to its yield to maturity.

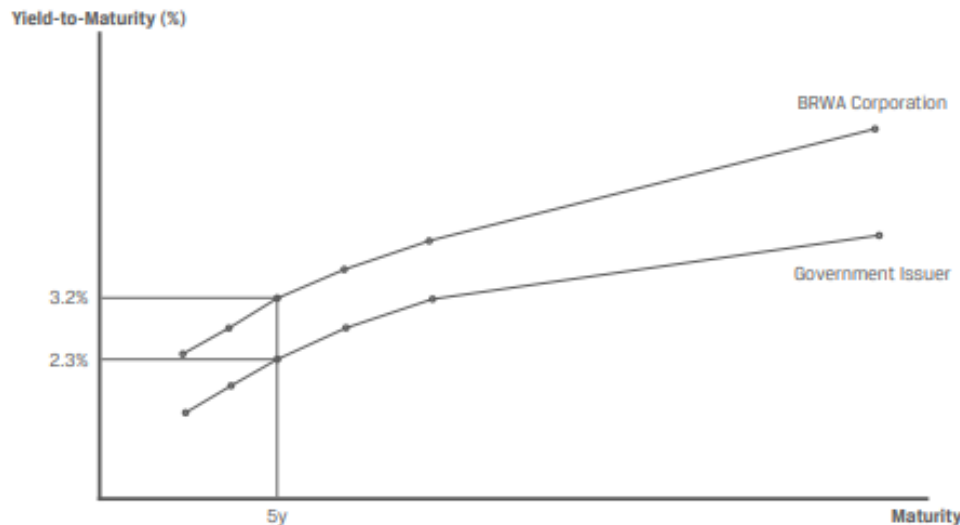
Yield Curves

The yield curve plots the YTM of bonds on the y-axis versus maturity on the x-axis. Exhibit 4 from the curriculum shows a sample yield curve. This curve is constructed using six bonds issued by a single corporation.



As seen in the graph, bonds with longer maturities have higher YTM, indicating that investors are demanding higher expected returns to compensate for higher risk associated with longer-maturity bonds.

To evaluate credit risk, we can compare an issuer's yield curve with the yield curve for comparable sovereign bonds (which have no credit risk). Exhibit 5 from the curriculum illustrates this.



The yield difference between the 5-year corporation bond and the 5-year US treasury bond is $3.2\% - 2.3\% = 0.9\%$. This reflects the compensation that investors demand for taking the additional credit risk associated with the corporation bond.

3. Bond Indentures and Covenants

Bond Indenture

A bond indenture is a written legal agreement between the bond issuer and the investor. The indenture has the following information:

- The name of the issuer.
- All the terms of a bond issue such as the type of bond.
- Its features such as the principal value, coupon rate, dates when interest payments will be made, and maturity date.
- Issuer's obligations.
- Bondholders' rights.
- If the bonds are secured or not.
- Covenants.
- Contingency provisions like call option.

Source of Repayment Proceeds

Investors need to know how the issuer intends to make interest payments and repay the principal. The bond indenture specifies the source of revenue or how the issuer intends to make interest and principal payments. We look at the repayment sources for each category now.

- For supranational bonds: From **repayment of previous loans** or paid-in capital from its members. For example, the World Bank may have given a loan to a particular country. When the country pays back the loan, this money is used to pay the bondholders. **Paid-in capital is the amount contributed by member nations.**
- For sovereign bonds: Tax revenues and printing new money. The government may also increase taxes to service debt.
- Three sources for non-sovereign government debt issues: the taxing authority of the issuer, cash flows of the project, special taxes or fees.
- For corporate bonds: Cash flow generated by the company's primary operations. The higher the credit risk, the higher the yield.
- For securitized bonds: Cash flows generated by the underlying assets such as mortgages, credit card receivables, auto loans, etc. **Securitized bonds are amortized, i.e., the principal is paid back gradually over the bond's life rather than in one payment at maturity.**

Asset or Collateral Backing

Collateral reduces credit risk.

- Secured vs. unsecured bonds: In a secured bond, assets are pledged as collateral to ensure debt repayment in the case of a default. Unsecured bonds have no collateral.
- Seniority ranking: What this means is that if a company faces bankruptcy and its assets are liquidated, the investors (or bondholders) are repaid based on seniority.

Covenants

Bond covenants are legally enforceable rules that borrowers and lenders agree on at the time of a new bond issue. Covenants can be affirmative (positive) or negative (restrictive). Affirmative covenants indicate what the issuer must do. These are generally administrative in nature. They do not impose additional costs on the issuer. For example, the issuer must:

- Make interest and principal payments on time.
- Comply with all laws and regulations.
- Maintain current lines of business.
- Insure and maintain assets.
- Pay taxes on time.
- **Other examples include a 'pari passu' clause which ensures that a debt obligation is treated the same as the borrower's other senior debt instruments, or a 'cross-default' clause which specifies that a borrower is considered in default if they default on another debt obligation.**

Negative covenants indicate what the issuer must not do. These are frequently costlier and materially constrain the issuer's potential business decisions. Examples include:

- **Restrictions on debt: Limits on maximum acceptable leverage ratios and minimum acceptable interest coverage ratios.**
- Negative pledges: No additional debt can be issued that is senior to existing debt.

- Restrictions on prior claims: The issuer cannot collateralize assets that were previously collateralized. The objective is to protect unsecured bondholders.
- Restrictions on distribution to shareholders: Restrictions on how much money can be spent on dividends and buy-backs.
- Restrictions on asset disposals: A limit on the total amount of assets that can be disposed during a bond's life. The objective is to ensure the company does not breakup.
- Restrictions on investments: This ensures no investment is made in a speculative business and that the capital is invested in a going-concern business.
- Restrictions on mergers and acquisitions.

Summary

LO: Describe the features of a fixed-income security.

The basic features of a fixed-income security include the specification of:

- Issuer: It is the entity that has issued the bond. The types of issuers include supranational organizations, sovereign governments, non-sovereign governments, quasi-government entities, and companies.
- Maturity: It is the date on which the last payment is made for a bond. Based on maturity, bonds can be classified into money market security, capital market security, and perpetual bonds.
- Par value: It is the amount an issuer agrees to pay the investor or bondholder on maturity date. A bond is premium if sold above par, discount if sold below par, and at par if sold at face value.
- Coupon rate and frequency: Coupon rate is the interest rate paid on a bond every year by the issuer until its maturity date. Bonds that have only one payment at maturity are called zero-coupon bonds.
- Seniority: Seniority ranking determines who gets paid first, or who has the first claim on the cash flows of the issuer, in the event of default/bankruptcy/restructuring.
- Contingency provisions: Some bonds may be issued with embedded call/put/conversion options.

LO: Describe the contents of a bond indenture and contrast affirmative and negative covenants.

Bond indenture is a legal agreement between the bond issuer and the investor. It includes all the terms of a bond issue such as the type of bond, its features such as principal value and coupon rate, and maturity date. It also includes the dollar amount of issue, issuer's obligations, bondholders' rights, when the coupon payments will be made, if the bonds are secured or not, covenants, and contingency provisions.

Bond covenants are legally enforceable rules that borrowers and lenders agree on at the time of a new bond issue. Covenants can be affirmative (positive) or negative (restrictive).

Affirmative covenants indicate what the issuer must do. These are generally administrative in nature. For example, issuer must make interest and principal payments on time, comply with laws and regulations, insure and maintain assets, etc.

Negative covenants indicate what the issuer must not do. These are frequently more costly and materially constrain the issuer's potential business decisions. Examples include restriction on debt, negative pledges, restrictions on prior claims, restrictions on distribution to shareholders, etc.

LM02 Fixed-Income Cash Flows and Types

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Version 1.0

1. Introduction

This learning module covers:

- Common fixed income instrument cash flow structures and their implications for issuers and investors
- Legal, regulatory, and tax considerations faced by fixed income issuers and investors

2. Fixed-Income Cash Flow Structures

Not all bonds are structured to make periodic interest payments and one lump-sum principal payment at the end. In this section we will look at the different ways in which principal and interest can be paid over the bond's life.

Principal Repayment Structures

Bullet bond: The principal is paid all at once at maturity. Such a type of bond is called a bullet bond.

<i>Bullet Bond: Payment Structure for a 5-year, \$1000 Bond with 6% Coupon Paid Annually</i>				
<i>Year</i>	<i>Cash flow in \$</i>	<i>Interest Payment (in \$)</i>	<i>Principal Repayment (in \$)</i>	<i>Outstanding principal (in \$)</i>
0	-1,000			1,000
1	60	60	0	1,000
2	60	60	0	1,000
3	60	60	0	1,000
4	60	60	0	1,000
5	1000 + 60 = 1060	60	1,000	0

Key points to be noted for a bullet bond (based on the table above):

- No part of the principal is paid before maturity. The \$1,000 amount towards principal is paid all at once at maturity.
- During the life of the bond, the principal remains outstanding.
- The last payment includes both the coupon payment of \$60 and principal payment of \$1,000.

Fully amortized: A fully amortized bond is one in which the principal is paid little by little in equal payments over the bond's life, so that it is repaid in full by the maturity date. The periodic payments made by the issuer consist of interest and a part of principal as shown for a sample bond in the table below.

<i>Fully Amortized Bond: Payment Structure for a 5-year, \$1,000 Bond with 6% Coupon Paid Annually, market interest rate = 6%</i>				
<i>Year</i>	<i>Investor cash flows in \$</i>	<i>Interest Payment</i>	<i>Principal Repayment</i>	<i>Outstanding principal</i>

	$a = b + c$	(in \$) b	(in \$) c	(in \$) $P_{t-1}-c$
0	-1,000			1,000
1	237.40	60	177.4	822.6
2	237.40	49.36	188.04	634.56
3	237.40	38.07	199.32	435.24
4	237.40	26.11	211.28	223.96
5	237.40	13.44	223.96	0

Partially amortized: A partially amortized bond is one in which only a part of the principal is repaid over the bond's life. The remaining big part of the principal is paid at maturity making it a balloon payment. This is a hybrid between the bullet and the fully-amortized bond. The table below shows a sample bond.

<i>Partially Amortized Bond: Payment Structure for a 5-year, \$1,000 Bond with 6% Coupon Paid Annually</i>				
Year	<i>Investor cash flows (Coupon) in \$</i>	<i>Interest Payment (in \$)</i>	<i>Principal Repayment (in \$)</i>	<i>Outstanding principal (in \$)</i>
0	-1,000			1,000
1	201.92	60	141.92	858.08
2	201.92	51.48	150.43	707.65
3	201.92	42.46	159.46	548.19
4	201.92	32.89	169.03	379.17
5	401.92	22.75	379.17	0

Sinking Fund Arrangements

This allows for full or partial amortization of a bond prior to its maturity. It specifies the portion of the bond's principal outstanding that must be repaid each year throughout the bond's life.

Three sinking fund arrangements:

- Standard: Issuer sends the repayment principal amount to the trustee. The trustee then either redeems bonds to this value or decides which bonds to retire through a lottery.
- Accelerated: Issuer retires more than the specified portion of the bond's notional principal. The amount redeemed steadily increases each year. If there is any remaining principal, it is redeemed at maturity.
- Call provision: Bonds with call provision give the issuer the right to call (repurchase) the bond before maturity. Callable bonds usually have higher yields as investors bear

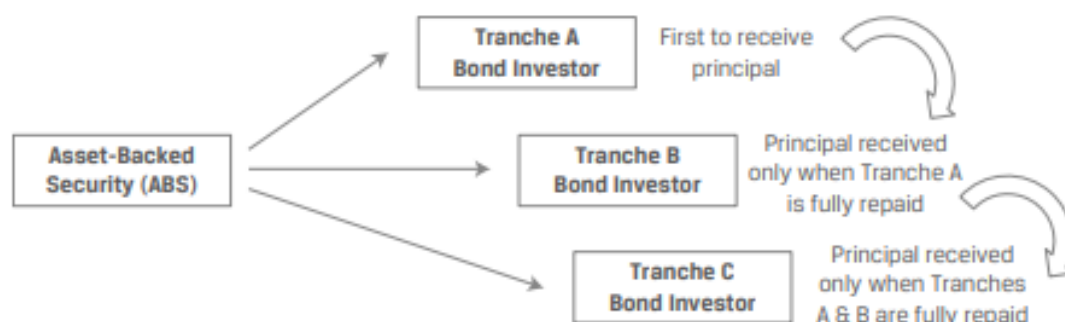
the risk that they may be called. It is beneficial to the issuer and disadvantageous to the bondholder. The bonds to be retired are selected randomly.

A sinking fund arrangement results in

- **Lower credit risk:** The objective of a sinking fund provision is to reduce credit risk for investors because the issuer does not have to pay a large payment at maturity. From an investor's perspective, there is less credit risk as the principal is being paid over the bond's term.
- **Higher reinvestment risk:** Receiving principal payments before maturity also means the investor has to bear reinvestment risk, i.e., if the money received cannot be invested at the same or higher expected return. In a declining interest rate environment, there is a risk of investing the proceeds at a lower rate.

Waterfall structures

This structure is commonly used in asset-backed securities (ABS) and mortgage-backed securities (MBS). Tranches with different priority of claims to the cash flows are created as shown in Exhibit 5 from the curriculum.



Interest payments are paid to all classed with no preference. However, the repayment of principal occurs sequentially – with the most senior tranche receiving principal payments first, followed by the second-highest tranche and so on.

Shortfalls in principal payment due to defaults are borne by the most junior tranches (since senior tranches are paid first). In the above diagram, Tranche A faces the lowest credit risk, while Tranche C faces the highest credit risk.

Coupon Payment Structures

Fixed periodic coupons

- This is the most basic form of coupon payment. A fixed interest is paid either semi-annually or annually.

Floating-rate notes (FRN) (a.k.a. Variable interest debt)

- A bond whose coupon is set based on some reference rate plus a spread.
- FRNs can have floors (minimum interest rate), caps (maximum interest rate), or

collars (both a minimum and maximum rate).

- An inverse FRN is a bond whose coupon has a negative relationship with the reference rate.

Other coupon structures

- Step-up coupons: Coupons increase by specified amounts on specified dates.
- Bonds with credit-linked coupons: Coupons change when the issuer's credit rating changes.
- Bonds with payment-in-kind coupons (also called a split coupon bond): Issuer can pay coupons with additional amounts of the bond issue instead of cash.
- Bonds with deferred coupons: No coupons paid in the initial years but higher coupons paid later.
- Zero coupon bond: No coupon payments are made and the principal is returned at maturity. These bonds are generally issued at a large discount to their face value. The difference between the issue price and the face value paid at maturity represents the cumulative interest that the investor will receive.
- Index-linked bonds: Coupon payments and/or principal repayments are linked to a price index, i.e., inflation-linked bonds. Examples of inflation-linked bonds include the following:
 - Zero-coupon-indexed bonds: The inflation adjustment is made via the principal repayment only.
 - Interest-indexed bonds: An index-linked coupon is paid during the bond's life but nominal principal amount at maturity is fixed. This is essentially a floating-rate note in which reference rate is the inflation rate instead of a market rate.
 - Capital-indexed bonds: Fixed coupon rate is paid but it is applied to a principal amount that increases in line with increases in the index during the bond's life.
 - Indexed-annuity bonds: The annuity payment, which includes both payment of interest and repayment of the principal, increases in line with inflation during the bond's life. These are fully amortized bonds, unlike interest-indexed and capital-indexed bonds that are non-amortizing coupon bonds.

Exhibit 8 from the curriculum summarizes the common cash flow structures for fixed-income instruments.

Standard, Fixed-Rate Bond ("bullet bond")	- Fixed coupon rate - Coupons are interest only - Par value or principal paid at maturity - Example: most long-term corporate and government bonds
Amortizing Principal	- Often uniform periodic payments, composed of both interest and principal - Principal paid over life of the instrument - Example: mortgage loans
Variable Interest	- Cash flows change based on specified market interest rate, index, inflation rate, etc. - Example: floating rate notes, most bank loans
Zero Coupon	- Fixed 0% coupon rate - Par value or principal paid at maturity - Typically sold at a discount - Example: short-term government bonds

3. Fixed-Income Contingency Provisions

A contingency provision is a clause in a legal document that allows for some action if the event or circumstance does occur. It is also called an embedded option.

Callable Bonds

A callable bond gives the issuer the right to redeem all or part of the bond before the specified maturity date. Investors face reinvestment risk with callable bonds, as it is not possible to reinvest the proceeds at the previous higher interest rates. To compensate this risk to an extent, issuers offer a higher yield and sell at a lower price than the respective non-callable bonds.

Why companies issue callable bonds?

- To protect the issuer when market interest rates drop.
- Interest rates drop when market interest rates fall or the credit quality of the issuer improves. The issuer has an opportunity to call the old bonds and replace them with new cheaper bonds by saving on otherwise higher interest expenses.
- Companies also issue callable bonds to signal the market about their credit quality.

The following details about a callable bond are included in the indenture:

- Call price: Price paid by the issuer to the bondholder when the bond is called.
- Call premium: The amount paid on top of the face value as compensation to bondholders as they will have to reinvest proceeds at a lower rate.
- Call schedule: The dates and prices at which the bond may be called.
- Call protection period: It is also known as the lockout period, deferment, or cushion period. During this period, a bond may not be called by the issuer. It is typically in the early days of a bond's life to encourage investors to invest in the issue.
- Call date: The earliest date at which a bond may be called.

The three types of callable bonds based on exercise styles are listed below:

- American call: The issuer has the right to call the bond any time after the first call date.
- European call: The issuer has the right to call the bond only once after the first call date.
- Bermuda-style call: The issuer has the right to call the bond on specific dates after the call protection period.

Example

Assume a hypothetical 20-year bond is issued on 1 December 2012 at a price of 97.315 (as a percentage of par). Each bond has a par value of \$100. The bond is callable in whole or in part every 1 December from 2017 at the option of the issuer. The callable prices are shown below.

Year	Call Price	Year	Call Price
2017	103.78	2023	101.47
2018	103.54	2024	101.21
2019	103.10	2025	100.68
2020	102.81	2026	100.32
2021	102.23	2026 and thereafter	100.00
2022	101.59		

1. What is the call protection period?
2. What is the call premium (per bond) in 2021?
3. What type of a callable bond is it *most likely*?

Solution:

1. The bonds were issued in 2012 and are first callable in 2017. The call protection period is $2017 - 2012 = 5$ years.
2. The call prices are stated as a percentage of par. The call price in 2021 is \$102.23 ($102.23\% \times \100). The call premium is the amount paid above par by the issuer. The call premium in 2021 is \$2.23 ($\$102.23 - \100).

3. It is a Bermuda call. The bond is callable every 1 December from 2017 – that is, on specified dates following the call protection period. Thus, the embedded option is a Bermuda call.

Puttable Bonds

A puttable bond gives the bondholder the right to sell the bond back to the issuer at a pre-determined price on specified dates. Puttable bonds offer a lower yield and sell at a higher price relative to otherwise non-puttable bonds. Puttable bonds are beneficial to the bondholder because:

- When interest rates rise, bond prices fall. If the selling price is pre-specified, bondholders may put (sell) back the bond to the issuer at that price, which is higher than the market price when interest rates rise.
- Cash can be reinvested at higher rates.

The following details about a puttable bond are included in the indenture:

- Redemption dates.
- Selling price; usually equal to the face value of the bond.
- How many times the issuer allows bondholders to sell the bond during the bond's life.
- One-time put: Gives bondholders a single sellback opportunity.
- Multiple put: More than one sellback opportunity available. Priced higher than one-time put bonds.

Like callable bonds, puttable bonds are also classified into three, based on their exercise styles:

- American put: Bondholder has the right to sell the bond back to the issuer any time after the first put date.
- European put: Bondholder has the right sell the bond back to the issuer only once on the put date.
- Bermuda-style put: Bondholder has the right sell the bond back to the issuer only on specified dates.

Convertible Bonds

A **convertible bond** is a hybrid security with both debt and equity features. It gives the bondholder the right to exchange the bond for a specified number of common shares in the issuing company.

Advantages of Convertible Bonds	
From investor's perspective	From issuer's perspective
Opportunity to convert into equity if share prices are increasing and participate in upside.	Reduced interest expense; lower yield than otherwise non-convertible bond because of the conversion provision given to bondholders.

Downside protection if shares prices are falling.	Elimination of debt if conversion option is exercised. So, they do not have to repay the debt.
	Convertible bonds are usually callable.

Some terms associated with the conversion provision are given below:

- **Conversion price:** Price per share at which the convertible bond can be converted into shares.
- **Conversion ratio:** Number of shares that each bond can be converted into.

$$\text{Conversion ratio} = \frac{\text{Par value}}{\text{Conversion price}}$$
- **Conversion value:** Current share price multiplied by the conversion ratio. It is also called the parity value.

$$\text{Conversion value} = \text{current share price} * \text{conversion ratio}$$
- **Conversion premium:** Difference between the convertible bond's price and its conversion value.

$$\text{Conversion premium} = \text{convertible bond's price} - \text{conversion value}$$

Warrant: A warrant is an attached option, not an embedded option. It gives the bondholder the right to buy the underlying common shares at a fixed price called the exercise price any time before the expiration date.

Contingent Convertible (CoCo) Bonds: These are bonds with contingent write-down provisions. The bonds can be converted into equity contingent to a specific condition. For example, a CoCo bond might be allowed to be converted into equity only after it reaches a certain price.

Example

Assume that a convertible bond issued in the United Kingdom has a par value of £1,000 and is currently priced at £1,200. The underlying share price is £56 and the conversion ratio is 25:1. What is the conversion condition for this bond?

Solution:

The conversion value of the bond is $£56 \times 25 = £1,400$. The price of the convertible bond is £1,200. Thus, the conversion value of the bond is more than the bond's price, and this condition is referred to as above parity.

4. Legal, Regulatory, and Tax Considerations

Legal and Regulatory Considerations

Fixed-income securities are subject to different legal and regulatory requirements depending on where they are issued and traded. National bond market is the bond market in a particular country.

- **Domestic bonds:** These are bonds that are issued and traded in a country, and denominated in the currency of that country. Bonds in domestic currency issued by a company incorporated in that country are called domestic bonds. For example, bonds denominated in Yen, issued by Toyota to be traded in Japan.
- **Foreign bonds:** These are bonds issued by a foreign company but traded in the domestic market. For example, bonds denominated in U.S. dollars issued by the Australian Rio Tinto Group.

Eurobonds are issued internationally, outside the jurisdiction of any single country and are denoted in currency other than that of the countries in which they trade. They are subject to less regulation than domestic bonds.

Bearer Bonds versus Registered Bonds

In the case of bearer bonds, the trustee does not maintain a record of who owns the bonds. That information is recorded in the clearing system. In the case of registered bonds, records of who owns the bond are maintained using a name or serial number. In the past, Eurobonds were typically bearer bonds. However, nowadays, **Eurobonds as well as domestic and foreign bonds are registered bonds.**

Global Bonds

Bonds that are issued simultaneously in multiple markets, such as the Eurobond market and in at least one domestic bond market. This ensures sufficient demand for the issue irrespective of the investors' location.

Sukuk

Sukuk are fixed-income instruments developed in accordance with Islamic law or *shari'a*. Payment of interest and financing of sectors such as alcohol or gambling is prohibited. Instead of interest, Sukuk pay investors a rental cash flow (or a profit rate) from the underlying assets.

Tax Considerations

There are two sources of return from a bond: income from coupon payments and capital gain. The way these two components are treated for tax purposes is different.

- The income portion of a bond investment is generally taxed at the ordinary income tax rate. For example, if you fall under the 30% income tax category, then the coupon income will be taxed at this rate.
- Assume you buy a bond for \$900 and later sell it for \$1,000. This \$100 is considered capital gain. Tax on capital gain may be different for long-term and short-term investments. Short-term is usually less than one year while more than one year is considered long-term. Often, the tax rate for long-term capital gains is lower than that for short-term capital gains.

- In some countries, a pro-rated portion of discount may be included in interest income. For example, assume you buy a 3-year zero-coupon bond, with a par value of \$1,000, at \$900. The gain of \$100 is over three years. Now, the taxing authority in some countries such as the U.S. may decide to tax this \$100 gain on a pro-rata basis over three years instead of taxing it all at once at the end of three years.

Summary

LO: Describe common cash flow structures of fixed-income instruments and contrast cash flow contingency provisions that benefit issuers and investors.

The cash flows to investors can be divided into principal repayment and coupon payments.

Principal can be repaid in three ways:

- Bullet bond: Principal is paid all at once at maturity.
- Fully amortized: In this method the principal is paid little by little, in equal payments over the bond's life, so that it is repaid in full by the maturity date.
- Partially amortized: Only a part of the principal is repaid over the bond's life. The remaining large part of the principal is paid at maturity, making it a balloon payment. This is a hybrid between the bullet and the fully-amortized bond.

The sinking fund arrangement allows for full or partial amortization of a bond prior to its maturity. Three sinking fund arrangements are standard, accelerated, and call provision.

The different types of coupon payments are listed below:

- Fixed periodic coupon: A fixed interest is paid either semi-annually or annually.
- Floating rate notes: The coupon payments are not fixed; instead, they are linked to a benchmark reference rate such as Libor, short-term Treasury bills, etc.
- Step-up bond: Can be a fixed-rate or floating-rate bond. The coupon rate increases over time.
- Credit-linked Coupon Bonds: The coupon rate changes when the bond's credit rating changes.
- Payment-in-kind Coupon Bonds: The issuer pays interest by issuing additional bonds.
- Deferred Coupon Bonds: Bonds that do not pay a coupon in the initial years, but compensate by paying a higher coupon in the later years.
- Index-Linked Bonds: Coupon and/or principal payments are linked to an index.
- TIPS: Treasury inflation-protected securities issued by the U.S. government; linked to the U.S.CPI.

A contingency provision is a clause in a legal document that allows for some action if the event or circumstance does occur.

Callable bond: It gives the issuer the right to redeem all or part of the bond before the specified maturity date. Investors face reinvestment risk with callable bonds, as it is not possible to reinvest the proceeds at the previous higher interest rates. When a bond is redeemed early, the issuer has to make the lump-sum payment equal to the present value of future coupon payments and principal. The three types of callable bonds are American call, European call, and Bermuda-style call.

Putable bond: It gives the bondholder the right to sell the bond back to the issuer at a pre-determined price on specified dates. Putable bonds offer a lower yield and sell at a higher

price relative to otherwise non-puttable bonds. Puttable bonds are beneficial to the bondholder. The three types of puttable bonds are American put, European put, and Bermuda-style put.

Convertible bond: It allows the bondholder the right to exchange the bond for a fixed number of common shares of the issuing company any time before maturity.

Advantages of Convertible Bonds	
From an investor's perspective	From an issuer's perspective
Opportunity to convert into equity if share prices are increasing and participate in upside.	Reduced interest expense; lower yield than otherwise non-convertible bond because of the conversion provision given to bondholders.
Downside protection if shares prices are falling.	Elimination of debt if conversion option is exercised. So, they do not have to repay the debt.
	Convertible bonds are usually callable.

Warrant: It is an attached option, not an embedded option. It gives the bondholder the right to buy the underlying common shares at a fixed price called the exercise price any time before the expiration date.

Contingent Convertible Bonds: These are bonds with contingent write-down provisions. The bonds can be converted into equity contingent to a specific condition.

LO: Describe how legal, regulatory, and tax considerations affect the issuance and trading of fixed-income securities.

Fixed-income securities are subject to different legal and regulatory requirements depending on where they are issued and traded. National bond market is the bond market in a particular country. It includes domestic bonds as well as foreign bonds. Eurobonds are international bonds that can be denominated in any currency.

Tax Considerations:

The two sources of return from a bond are treated differently for tax purposes:

- The income portion of a bond investment is generally taxed at the ordinary income tax rate.
- Tax on capital gain may be different for long-term and short-term investments.
- In some countries, a pro-rated portion of discount may be included in interest income.

LM03 Fixed-Income Issuance and Trading

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Version 1.0

1. Introduction

This learning module covers:

- Types of fixed income market segments
- Types of fixed-income indexes
- Primary and secondary fixed-income markets

2. Fixed-Income Segments, Issuers, And Investors

Fixed income markets are often categorized along three dimensions: issuer type, credit quality and time to maturity. Sometimes they may also be categorized along additional features such as currency, geography, and ESG characteristics.

Classification by Type of Issuer

Bond markets may be divided into four categories based on the type of issuers:

- 1) Households
- 2) Non-financial corporates
- 3) Government
- 4) Financial institutions

Classification by Credit Quality

Investors face credit risk, i.e., the risk of loss if the issuer fails to make timely payments of interest and principal as they come due. Rating agencies like Moody's, S&P, and Fitch assign credit ratings to bonds. Bonds with a rating of BBB or above are considered investment grade. Bonds below this rating are considered junk bonds.

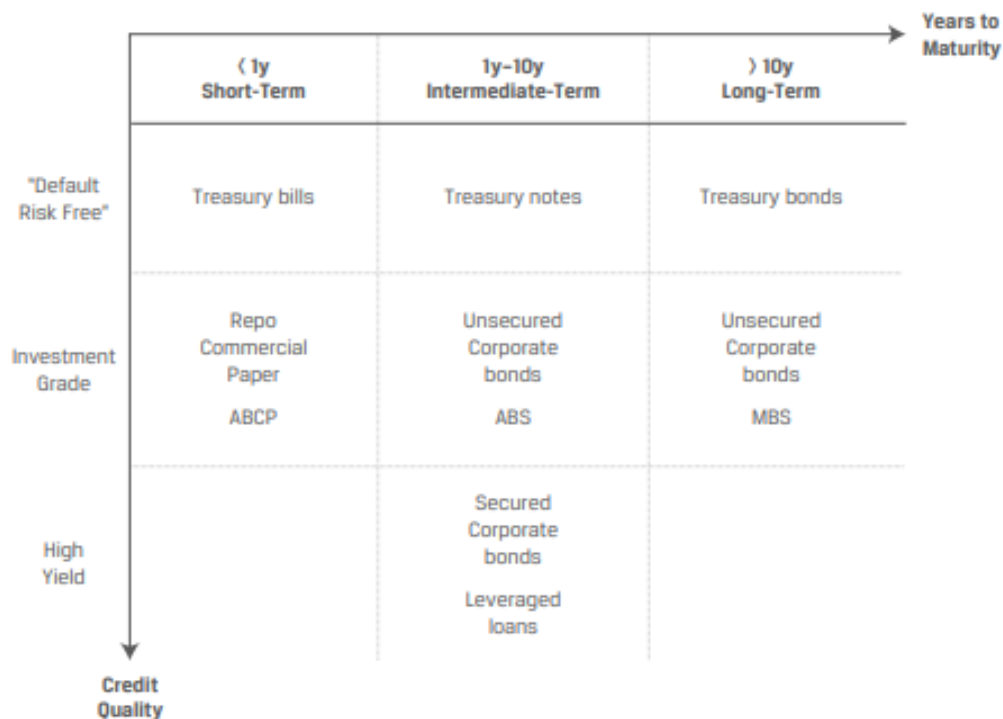
This differentiation is important as certain investors such as banks and life insurance companies may not be allowed to invest in junk bonds but only in investment grade bonds.

Classification by Maturity

Fixed-income securities can also be classified by the original maturity of the bonds when they are issued:

- Money market securities: They are issued with a maturity at issuance that ranges from overnight to one year. For example, T-bills issued by the US government or commercial paper with short maturities issued by the corporate sector.
- Capital market securities: The original maturity is usually longer than a year.

Exhibit 1 from the curriculum illustrates typical issuers and instrument types across credit and time-to-maturity segments.



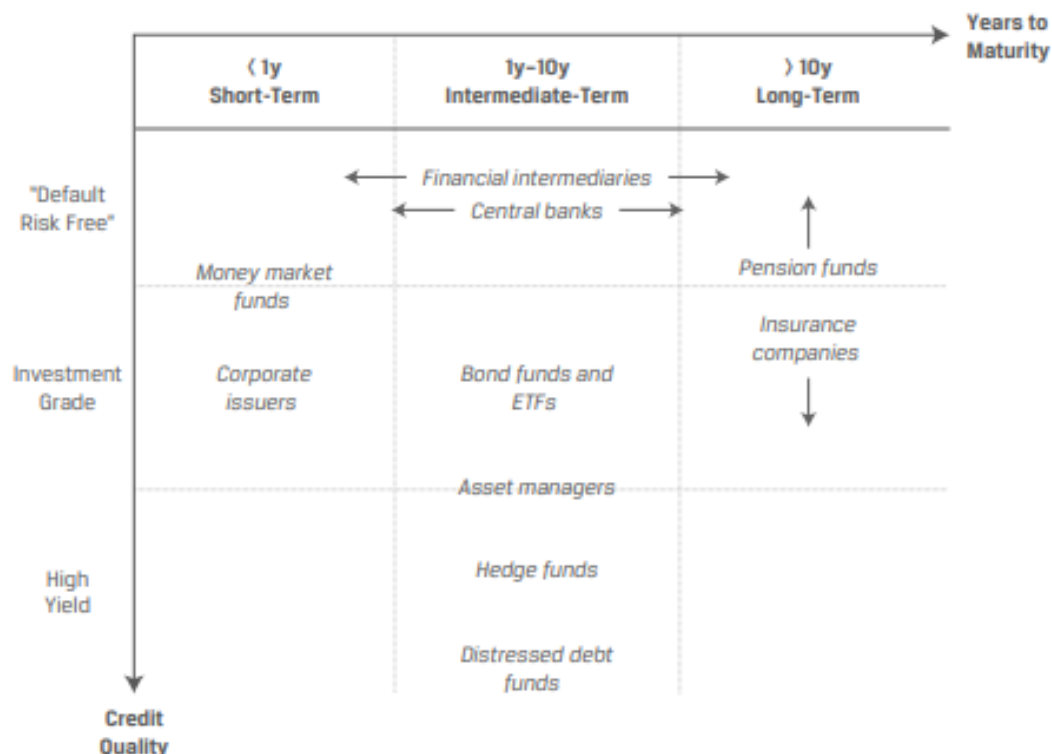
Note: ABCP is asset-backed commercial paper.

Fixed-income investors take positions across the credit and maturity spectrums in order to gain exposure to specific risks and match the cash flows of known future obligations.

Near-term obligations and liquid cash alternatives are often met with money market securities, while long-term bonds may be used to meet obligations further in the future or to achieve a higher expected return.

Pension funds and insurance companies with long investment time horizons prefer fixed-income instruments with fixed periodic coupon cash flows and a maturity profile that matches their long-term liabilities. Hedge funds and distressed debt funds may take additional credit risk at any point on the maturity spectrum to pursue higher expected return.

Exhibit 2 from the curriculum illustrates common investor positions across the credit and maturity spectrums.



Sovereign government issuers typically have the lowest credit risk (and the highest credit rating - AAA). Investors in investment-grade bonds anticipate more consistent cash flows. Defaults by investment grade issuers are extremely rare. High-yield investors expect a higher return in exchange for assuming a higher risk of default.

3. Fixed-Income Indexes

Like equity market indexes, fixed income indexes also track the returns of groups of securities that meet their inclusion criteria.

Fixed income indexes can be used to evaluate market performance, benchmark the performance of investments and investment managers, and serve as the foundation for indexed investment strategies.

Fixed income indexes differ from equity indexes in three ways:

1. In contrast to equities, where issuers typically issue just one or two instruments, issuers often have many fixed-income instruments outstanding. For example, at the end of 2021, Apple Inc had a single equity instrument but over 80 fixed income instruments outstanding. Therefore, fixed-income indexes tend to have many more constituent securities than equity indexes and are difficult to replicate.
2. Because bonds have a finite maturity and new bonds are issued on a regular basis, fixed income index constituents change far more frequently than equity index constituents. Therefore, fixed income indexes require frequent rebalancing.

3. Similar to the market capitalization weighting used in equity indexes, bond indexes are also typically weighted by the market value of debt outstanding. Since bonds are issued by many types of issuers, broad bond indexes will reflect changes to bond market composition over time—for example, increases in public versus private issuer debt, longer maturities, and deterioration in credit quality. Also, governments are usually the largest bond issuers in any market, therefore, many broad bond indexes have a high weight of government debt.

Because of the complexity of bond indexes, it is almost impossible to purchase all the constituent securities and a fund manager aiming to match the returns of an index will typically hold a representative sample of constituent securities.

Fixed-income indexes can be classified as broad-based, aggregate indexes with a vast number of constituents or narrower indexes that focus on criteria such as issuer type, credit quality, and time to maturity.

The most popular fixed-income indices include Barclays Capital Global Aggregate Bond Index, J.P. Morgan Emerging Market Bond Index, and FTSE Global Bond Index.

4. Primary and Secondary Fixed-Income Markets

Primary bond markets are markets in which bonds are sold for the first time by issuers to investors to raise capital. Bonds can be sold initially via a public offering or a private placement. Secondary bond markets are markets in which existing bonds are subsequently traded among investors. After the initial offering, bonds are bought and sold among investors in the secondary market.

Primary Fixed-Income Markets

A company/government/any entity issues bonds in two ways: public offering and private placement.

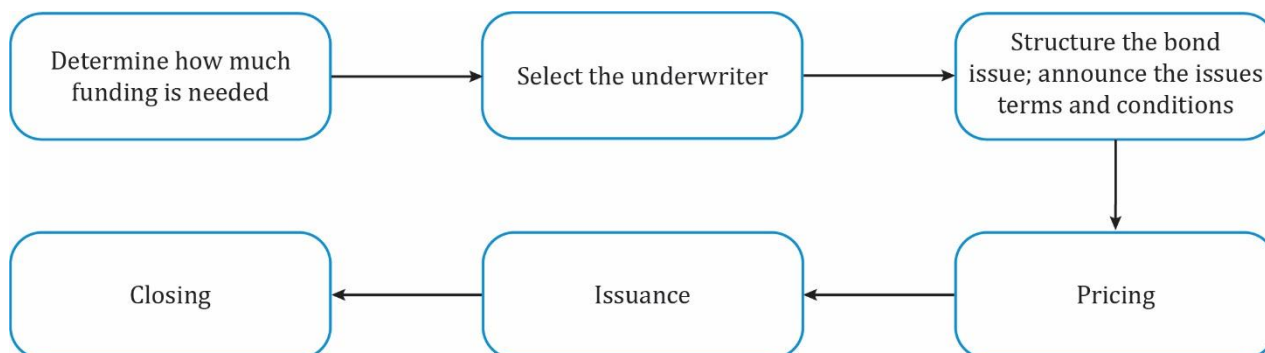
Public Offerings

As the name indicates, in a public offering, any member of the public may invest in a new bond issue. The issuer does not sell bonds directly to each investor. Instead, the issuer avails the services of an intermediary called the *underwriter* to facilitate the selling (placement) process. The underwriter is usually an investment bank because banks have a good understanding of how to market a new issue, can tap their networks to locate investors, and successfully place the issue.

The different bond issuing mechanisms are:

- Underwritten offering
- Best effort offering
- Shelf registration
- Auctions

Underwritten offerings: In an underwritten offering, an investment bank negotiates an offering price with the issuer; the offering price is the price at which the issue will be sold. It then buys the entire issue at the offering price and takes the risk of reselling it to investors or dealers. **Underwritten offering is also known as a firm commitment offering.** The underwriting process is graphically depicted below:



While small-size bond issue may be underwritten by a single investment bank, larger-size bond issues are often underwritten by a group (or syndicate) of investment banks. Such issues are called ‘syndicated offerings.’ A lead underwriter heads the syndicate and the group collectively establishes the pricing of the issue and takes the risk of reselling it to investors or dealers.

Best effort offering: Contrary to an underwritten offering, in a best effort offering issue, the investment bank acts as a broker and only sells as many securities as it can instead of committing to sell 100% of the issue. The unsold bonds are returned to the issuer. The investment bank gets a commission for bonds sold at the offering price, faces less risk and has less incentive to sell the issue than in an underwritten offering. Best effort offering is usually preferred for riskier issues and corporate bonds.

Shelf registration: Shelf registration is a type of public offering where the issuer is not required to sell the entire issue at once. The issuer files a single document with regulators that describe a range of future issuances. The advantage is that the issuer does not have to prepare a new document for every bond issue provided there is no change in the issuer’s business and financial terms stated in the prospectus. This allows the issuer to save on repeated administrative expenses and registration fees.

Auctions: Government bonds across the world are usually sold to investors via an auction. Governments finance public debt by borrowing money through the central bank. An auction is a public offering method that involves bidding, and is helpful in price discovery and allocating securities. The United States follows a single-price auction method for its sovereign securities such as T-bills, T-notes, TIPS, etc. In this method, all winning bids pay the same price for the security and receive the same coupon rate.

Private Placement

As the name implies, the securities are not sold to the public in this type of funding. Instead, they are sold only to a select group of investors such as institutional investors. Other characteristics are as follows:

- It is typically a non-underwritten, unregistered offering of bonds, i.e., a private issue need not comply with the registration requirements of a public offering such as preparing a prospectus.
- It is also exempt from securities laws that govern a public issue.
- It can be accomplished directly between the issuer and the investor(s) through an investment bank. Because privately placed bonds are unregistered and may be restricted securities that *can only be purchased by some types of investors*, there is usually no active secondary market to trade them.
- Institutional investors such as insurance companies and pension funds are typical investors of privately placed bonds.

Other points to note:

- Similar to primary equity issuance, which typically involves a transfer from private to public ownership, a first-time bond issuer is often replacing private debt, such as a bank loan, with bonds.
- Increasing the size of an existing bond with a price significantly different from par is referred to as the 'reopening' of an existing bond.
- Secured bond issuance by corporate high-yield, some special purpose entities, and other issuers is typically a more time-consuming and involved process than unsecured investment-grade bonds. Investors must become acquainted with unusual and complex covenants, as well as the use of operating cash flows and collateral as sources of bond repayment.

Secondary Fixed-Income Markets

Securities are traded among investors in the secondary market. Large institutional investors and banks are the primary participants. Retail investors are limited here, unlike in the equities market.

Unlike listed equities, for which secondary trading largely occurs on an electronic exchange, fixed-income markets consist mostly of quote-driven or over-the-counter markets.

It is important to understand the liquidity of a bond market:

- Liquidity is a measure of how quickly an investor can sell the bond and turn it into cash. Similarly, it should also measure how quickly one can buy a bond to cover a short position.
- Bid-ask or bid-offer spread reflects the liquidity of a market. The lower the spread, the better the liquidity.

- The most recently issued, on-the-run developed market sovereign bonds are typically the most liquid.
- Among corporate issuers, recently issued corporate bonds from frequent issuers with high credit quality have high liquidity. Whereas bonds of less frequent corporate issuers or more seasoned bonds of frequent issuers are rarely traded and have low liquidity.

Distressed debt refers to bonds issued by issuers who are on the verge of bankruptcy. Distressed debt typically trades in the secondary market at a price well below par.

Distressed debt is traded until either the corporate issuer's assets are liquidated or its outstanding bonds are restructured. An equity security, on the other hand, may be removed or delisted from secondary trading on an exchange if it fails to meet the exchange's listing requirements, which frequently include minimum share prices, net worth, and free float. By the time an issuer's debt has become distressed, its equity securities will likely have already been delisted.

Summary

LO: Describe fixed-income market segments and their issuer and investor participants.

Fixed income markets are often categorized along three dimensions: issuer type, credit quality and time to maturity. Sometimes they may also be categorized along additional features such as currency, geography, and ESG characteristics.

Fixed-income investors take positions across the credit and maturity spectrums in order to gain exposure to specific risks and match the cash flows of known future obligations.

Pension funds and insurance companies with long investment time horizons prefer fixed-income instruments with fixed periodic coupon cash flows and a maturity profile that matches their long-term liabilities. Hedge funds and distressed debt funds may take additional credit risk at any point on the maturity spectrum to pursue higher expected return.

LO: Describe types of fixed-income indexes.

Like equity market indexes, fixed income indexes also track the returns of groups of securities that meet their inclusion criteria.

Fixed income indexes can be used to evaluate market performance, benchmark the performance of investments and investment managers, and serve as the foundation for indexed investment strategies.

Fixed-income indexes can be classified as broad-based, aggregate indexes with a vast number of constituents or narrower indexes that focus on criteria such as issuer type, credit quality, and time to maturity.

LO: Compare primary and secondary fixed-income markets to equity markets.

Primary bond markets are markets in which bonds are sold for the first time by issuers to investors to raise capital. Bonds can be sold initially via a public offering or a private placement. Secondary bond markets are markets in which existing bonds are subsequently traded among investors. After the initial offering, bonds are bought and sold among investors in the secondary market.

LM04 Fixed-Income Markets for Corporate Issuers

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Version 1.0

1. Introduction

This learning module covers:

- Short term funding alternatives available to corporations and financial institutions
- Repurchase (repo) agreements – their uses, benefits and risks
- Long term corporate debt – similarities and differences between investment grade and high yield issuance

2. Short-Term Funding Alternatives

Both non-financial corporations and financial institutions use borrowed capital to support their short-term activities.

External Loan Financing

Non-financial corporations often rely on financial intermediaries for short-term financing.

Financial intermediaries can include both bank or non-bank lenders. The main type of financing options available through these sources are:

- Uncommitted lines of credit – They are the least reliable form of bank borrowing. The bank can refuse to honor the request to use the line. An uncommitted line is therefore not reliable.
- Committed lines of credit – Also called regular lines of credit, they are more reliable than uncommitted lines. The bank makes a formal commitment to honor the line of credit. The interest rate charged is usually the bank's prime rate or a money market rate plus a spread that depends on the borrower's creditworthiness. These lines are usually in effect for 364 days. They are unsecured and prepayable without penalty. Unlike uncommitted lines, regular lines require compensation. Banks typically charge a commitment fee e.g., 0.50% of the full amount or the unused amount of the line.
- Revolving credit agreements (Revolvers): They are the most reliable form of short-term bank borrowing. They involve formal agreements similar to those used for regular lines of credit. Revolvers differ from regular lines in two ways (1) they are in effect for multiple years and (2) they are often used for much larger amounts.
- Secured ("asset-based") loans: The options discussed above are unsecured loans. Secured loans are loans in which the lender requires the company to provide collateral in the form of assets. For example, a company can use its accounts receivables as a collateral to generate cash flows through the 'assignment of accounts receivable'.

A company can also sell its accounts receivables to a lender (called a factor), typically at a substantial discount. This is called a 'factoring arrangement'. In an assignment arrangement, the company retains the collection responsibilities, whereas in a

factoring arrangement, the company shifts the collection responsibilities to the lender (factor).

External, Security-Based Financing

Commercial Paper

It is a short-term, unsecured instrument typically issued by large, well-rated companies. It has maturities typically ranging from a few days to 270 days. Most maturing commercial paper is paid with proceeds from newly issued commercial paper (or "rolled over"). This practice increases the risk that an issuer will be unable to issue new paper at maturity, a risk known as rollover risk. To minimize rollover risk, issuers of commercial paper are often required to have a backup line of credit. The short duration, high creditworthiness of the issuing company, and the backup line of credit generally makes commercial paper a low-risk investment for investors.

Short-Term Funding Alternatives for Financial Institutions

This section looks at where banks get their funds from in the short-term.

Retail Deposits

One of the primary sources of funds for a bank is the money deposited by retail (like you and me) and commercial investors in their accounts. It is the lowest possible source of funding for a bank. The three types of retail accounts and characteristics of each of these accounts are discussed below:

Demand deposits or checking accounts

- Depositors have access to funds anytime.
- The funds may be used to pay for transactions.
- Little or no interest is paid.

Savings accounts

- Depositors have access to funds.
- Unlike checking account, savings accounts pay an interest. But they do not offer the same transactional convenience.

Money market accounts

- Funds are available at short or no notice.
- Offer money market rates of return.

Short-Term Wholesale Funds

Central Bank Funds

- When a bank receives deposits from customers, a certain percentage of this money must be kept as a reserve with the national central bank. The amount a bank keeps as reserves varies based on its financial position: some have a deficit and some have a surplus.

- The funds stashed in the central bank by all banks are collectively known as the central bank funds market.
- Assume a bank is running low on cash and a customer wants to withdraw money from this bank. The bank has two choices. It may withdraw cash from its reserve account in the central bank to pay the customer, if it has sufficient funds. Or it may borrow money from banks that have a surplus in their accounts at the central bank. The funds, known as central bank funds, may be borrowed for a period up to one year at rates known as central bank funds rates.
- If the borrowing is for one day, it is called overnight funds. If it is for more than one day, then it is called term funds.

Interbank Funds

- Banks lend to and borrow from each other in the interbank market.
- It is an unsecured system of lending.
- The term may vary from overnight to one year.
- The reference rate at which they borrow is called the interbank offered rate. Or, they may borrow at a fixed interest rate.
- Often large banks publish two rates: one at which they borrow and one at which they lend.

Certificates of Deposit

- A certificate of deposit is a savings instrument with a maturity date, a fixed interest rate, and can be issued in any denomination. The investor or bearer of the certificate receives an interest at the end of the deposit period. CDs can be issued to individuals, companies, trusts, funds, etc.
- There are two forms of CD: negotiable and non-negotiable CD.
- In a non-negotiable CD, the interest and deposit are paid at maturity. There is a penalty if the depositor withdraws funds before maturity.
- In a negotiable CD, depositors are allowed to sell the deposits before maturity.
- There are two types of negotiable CDs:
 - *Large-denomination CDs*: CDs of denomination of \$1 million or more; often traded among institutional investors.
 - *Small-denomination CDs*: of lower denominations and meant for retail investors.

Commercial paper

- Like non-financial corporations, banks may also issue unsecured commercial papers.
- Some banks and financial institutions also commonly issue a secured form of commercial paper known as “asset-backed commercial paper” (ABCP). This financing is not recorded on the balance sheet of the issuer and provides benefits to both the bank and investors. When the commercial paper is issued, the bank receives cash and

reduces its capital costs by providing undrawn backup liquidity rather than holding the short-term loans to maturity. Investors buy a liquid, short-term note that pays interest and principal from a loan portfolio to which they would not otherwise have direct access.

3. Repurchase Agreements

A repurchase agreement or repo is a sale and repurchase agreement. It is an agreement between two parties where the seller sells a security with a commitment to buy the same security back from the purchaser at an agreed-upon price at a future date. It is similar to borrowing funds against collateral. Assume there are two parties: banks A and B. A has a 90-day T-bill that it sells to B for \$99.50. It agrees to buy the same T-bill the next day for \$99.51. This was a means of borrowing \$99.50 overnight for A. The 1¢ can be considered as the interest for the borrowed amount.

Structure of Repurchase and Reverse Repurchase Agreements: We will look at some terms related to repo with the help of the A and B example we saw earlier.

- Reverse repo: It was a repo arrangement from bank A's (seller of security) perspective. But from bank B's perspective (purchaser of security), it is a reverse repo.
- Repurchase price: The price at which A will buy back the security from B the next day is called the repurchase price. The price at which the dealer repurchases the security is called the repurchase price.
- Repurchase date: The date on which the dealer (A) repurchases the security (from B) is called the repurchase date. In our example, it was the next day.
 - *Overnight repo*: If the repurchase happens the next day, i.e., if the agreement is for one day, it is called overnight repo.
 - *Term repo*: If the repurchase happens after more than a day, it is called term repo.
 - *Repo to maturity*: If the agreement is honored until maturity, then it is called repo to maturity.
- Repo rate: The interest rate negotiated between both the parties is called the repo rate. In our example, 1¢ was the interest paid.

The factors that affect the repo rate include:

- *The risk of the collateral*: Highly rated collateral such as sovereign bonds exhibit low risk and lower repo rates.
- *Term of the repurchase agreement*: The longer the term, the higher the rates.
- *Delivery requirement*: If the collateral is delivered to the lender, the rates are lower. If the underlying collateral is not delivered to the counterparty, then the risk is higher and so is the repo rate.
- *Supply and demand*: Collateral in high demand has lower repo rate. If the collateral is scarce/low supply, the repo rate is lower.

- Interest rates of alternative financing.

Repos include features to reduce the risk of a collateral shortfall over the life of the contract. These features are:

- **Initial margin:** Provision of collateral in excess of the cash exchanged. Initial margin is calculated as:

$$\text{Initial margin} = \frac{\text{Security Price}_0}{\text{Purchase Price}_0}$$

An initial margin of 100% indicates a fully collateralized loan. A value greater than 100% represents an even greater initial collateral protection.

- **Haircut:** Values of initial margin greater than 100% can also be expressed in the form of a haircut, which is a reduction of the underlying loan relative to the initial collateral value. Haircut is calculated as:

$$\text{Haircut} = \frac{\text{Security Price}_0 - \text{Purchase Price}_0}{\text{Security Price}_0}$$

- **Variation margin:** As the value of the collateral changes over the life of the repo contract, additional collateral may be required to maintain a security interest equal to the original initial margin terms. Variation margin is calculated as:

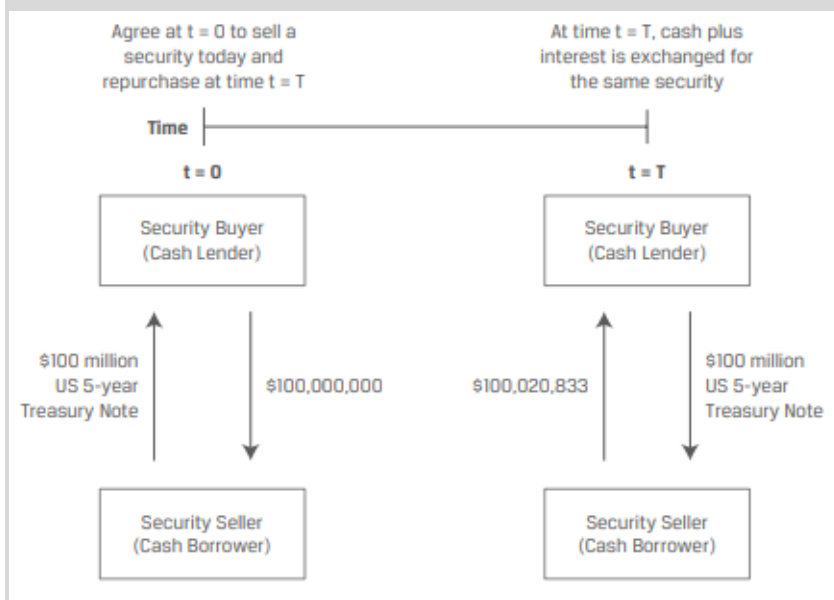
$$\text{Variation margin} = (\text{Initial margin} \times \text{Purchase Price}_t) - \text{Security Price}_t$$

The following examples illustrate these concepts:

Example:

(This is based on the Knowledge Check examples from the curriculum.)

Consider the following simple repurchase agreement.



Assume that today ($t = 0$) the current US five-year Treasury note trades at a price equal to the bond's face value of USD100,000,000. The security buyer takes delivery of the US Treasury note today and pays the security seller USD100,000,000. If we assume a repo term of 30 days (and 360 days in a year) and an annual interest rate (or repo rate) of 0.25%, then the buyer agrees to return the Treasury note 30 days from today ($t = T$) to the seller at a repurchase price of USD100,020,833, calculated as follows:

$$\text{USD100,000,000} \times [1 + (0.25\% \times 30/360)] = \text{USD100,020,833}.$$

1. Assume the repurchase agreement in Exhibit 4 is subject to a 102% initial margin.

Calculate the revised original purchase price (loan amount).

Solution: Using an initial margin of 102%, we can solve for a revised purchase price

$$\text{Initial margin} = \frac{\text{Security Price}_0}{\text{Purchase Price}_0}$$

$$102\% = \frac{100,000,000}{\text{Purchase Price}_0}$$

$$\text{Purchase Price}_0 = \text{USD98,039,216}.$$

Note that security seller must finance the remaining USD1,960,784 from other sources.

2. Calculate the associated repo haircut.

Solution:

$$\text{Haircut} = \frac{\text{Security Price}_0 - \text{Purchase Price}_0}{\text{Security Price}_0}$$

$$\text{Haircut} = \frac{100,000,000 - 98,039,216}{100,000,000} = 1.96\%$$

3. Calculate the repurchase price using the same 30-day repo term and 0.25% repo rate.

Solution: The repurchase price may be calculated as follows:

$$\text{Repurchase price} = \text{USD98,039,216} \times [1 + (0.25\% \times 30/360)].$$

$$\text{Repurchase price} = \text{USD98,059,641}$$

4. Calculate the variation margin five days after trade inception if the price of the five-year US Treasury note rises to 103% of the security's USD100 million face value.

Solution:

Recall the original loan amount (or purchase price at $t = 0$) is equal to USD98,039,216, or USD100,000,000/1.02. We may solve for the purchase price when $t = 5$ as follows:

$$\text{Purchase price} = \text{USD98,039,216} \times [1 + (0.25\% \times 5/360)].$$

$$\text{Purchase price} = \text{USD98,042,620}.$$

To calculate the variation margin, we use the following equation:

$$\text{Variation margin} = (\text{Initial margin} \times \text{Purchase price}_t) - \text{Security price}_t$$
$$\text{Variation margin} = (102\% \times \text{USD}98,042,620) - \text{USD}103,000,000.$$
$$\text{Variation margin} = -\text{USD}2,996,528.$$

The security price increase to USD103,000,000 resulted in an overcollateralization of the loan by USD2,996,528, allowing the cash borrower (security seller) to request the release of five-year US Treasury notes with a face value of approximately USD2.91 million ($= \text{USD}2,996,528 / 1.03$)

Repurchase Agreement Applications and Benefits

Financial market participants use the repo market for three specific purposes:

- Finance the ownership of a security
- Earn short-term income by lending funds on a secured basis
- Borrow a security in order to sell it short.

Risks Associated with Repurchase Agreements

The risks associated with repurchase agreements are:

- Default risk: The security lender may default and may not repurchase the collateral. Or the cash lender may default and cannot deliver the collateral. Under a default scenario, the parties may face illiquidity, adverse price changes, and legal or operational challenges.
- Collateral risk: In order to diversify credit exposure, the collateral should have little or no correlation with the credit risk of the repo counterparty.
- Margining risk: Adverse market conditions may cause large changes in collateral value, increasing margin calls among market participants and forcing liquidations.
- Legal risk: Refers to the ability to enforce legal rights under a repurchase agreement.
- Netting and settlement risk: Refers to the ability of repo contract participants to both offset or net the obligations of a non-defaulting party and take possession of collateral or cash in settlement of a trade.

Repo market participants may manage the above risks by engaging a third party. Repos executed directly between two parties are called bilateral repos; whereas, repos where both parties agree to use a third-party agent for the transaction are called triparty repos.

4. Long-Term Corporate Debt

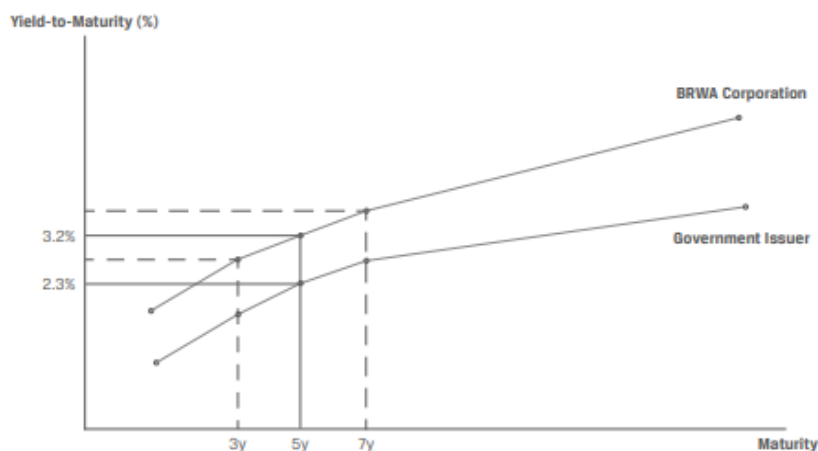
Corporate issuers use long-term debt as a more stable funding source for both short-term activities and long-term capital investments.

Investment-grade corporate issuers have a stronger ability to meet future fixed obligations,

whereas high yield issuers have a more vulnerable ability to meet debt interest and principal payments.

Similarities between Long-Term Investment-Grade and High-Yield Issuance

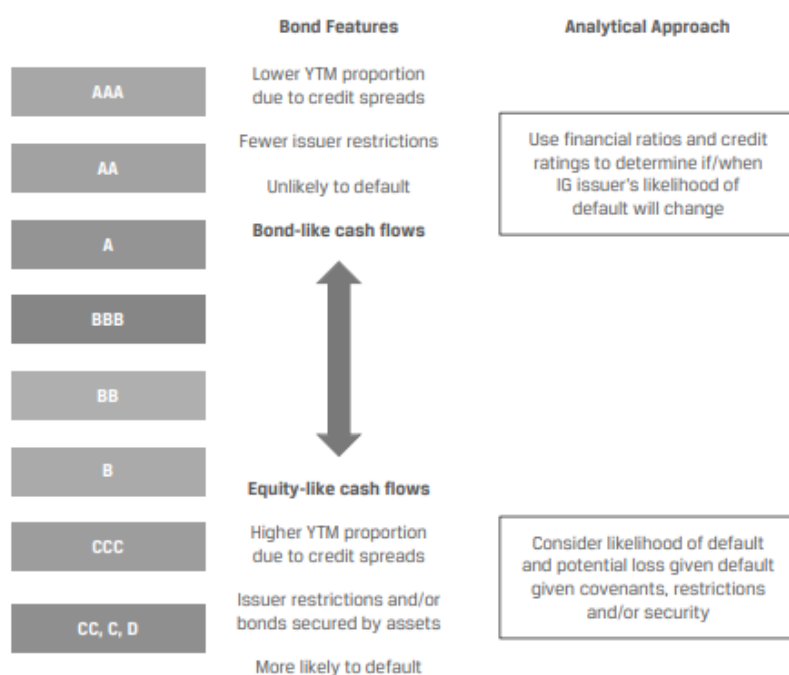
Under normal market conditions, longer maturities are associated with both higher interest rates and higher credit spreads for a given issuer. This is true for both IG and HY issuers. Exhibit 7 from the curriculum illustrates this concept.



Notice that as maturity increases the interest rates (YTM for government bonds) increases. Also, the credit spread i.e. the difference between the YTM of corporate bond and YTM of government bonds increases as maturity increases.

Differences between IG and HY Issuance

Exhibit 8 from the curriculum summarizes the differences between IG and HY bonds.



Because of its strong ability to meet promised interest and principal obligations from operating cash flows, an investment grade bond has a significant proportion of its YTM attributed to the government benchmark yield. Also, investors accept fewer or no restrictive covenants on the issuer.

High-yield issuers are characterized by a higher expected likelihood of financial distress, with a greater proportion of a bond's YTM attributable to an issuer-specific spread over the benchmark yield. Given the high chance of default, these instruments are more equity like in nature. Therefore, investors generally impose more restrictions and covenants for HY issuers.

These investor restrictions and constraints result in far less HY issuer flexibility and market availability than for IG issuers. Issuers in the high-yield market often seek to retain financial flexibility by borrowing under leveraged loans with prepayment features or issuing bonds with contingency features. Borrowers who believe their creditworthiness will improve frequently issue callable debt because the value of this contingency feature increases as a firm's borrowing costs fall. High yield investors also benefit from higher bond prices as issuer specific credit spreads fall; however, in case of callable bonds these gains are capped at the call price.

Summary

LO: Compare short-term funding alternatives available to corporations and financial institutions.

Non-financial corporations use the following short-term funding alternatives: uncommitted lines of credit, committed lines of credit, revolving credit agreements, secured loans, commercial papers.

Financial institutions such as banks use the following short-term funding alternatives: demand deposits or checking accounts, savings account, money market accounts, central bank funds, interbank funds, certificates of deposits, unsecured commercial paper and asset backed commercial paper.

LO: Describe repurchase agreements (repos), their uses, and their benefits and risks.

A repurchase agreement (repo) is a form of short-term secured lending that involves the sale and simultaneous agreement to buy back a security at a pre-agreed future price.

Financial market participants use the repo market for three specific purposes:

- Finance the ownership of a security
- Earn short-term income by lending funds on a secured basis
- Borrow a security in order to sell it short.

The risks associated with repos are:

- Default risk
- Collateral risk
- Margining risk
- Legal risk
- Netting and settlement risk

LO: Contrast the long-term funding of investment-grade versus high-yield corporate issuers.

Because of its strong ability to meet promised interest and principal obligations from operating cash flows, an investment grade bond has a significant proportion of its YTM attributed to the government benchmark yield. Also, investors accept fewer or no restrictive covenants on the issuer.

High-yield issuers are characterized by a higher expected likelihood of financial distress, with a greater proportion of a bond's YTM attributable to an issuer-specific spread over the benchmark yield. Given the high chance of default, these instruments are more equity like in nature. Therefore, investors generally impose more restrictions and covenants for HY issuers.

LM05 Fixed-Income Markets for Government Issuers

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2. Sovereign Debt.....	2
3. Sovereign Debt Issuance and Trading.....	5
4. Non-Sovereign, Quasi-Government, and Supranational Agency Debt.....	5
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Version 1.0

1. Introduction

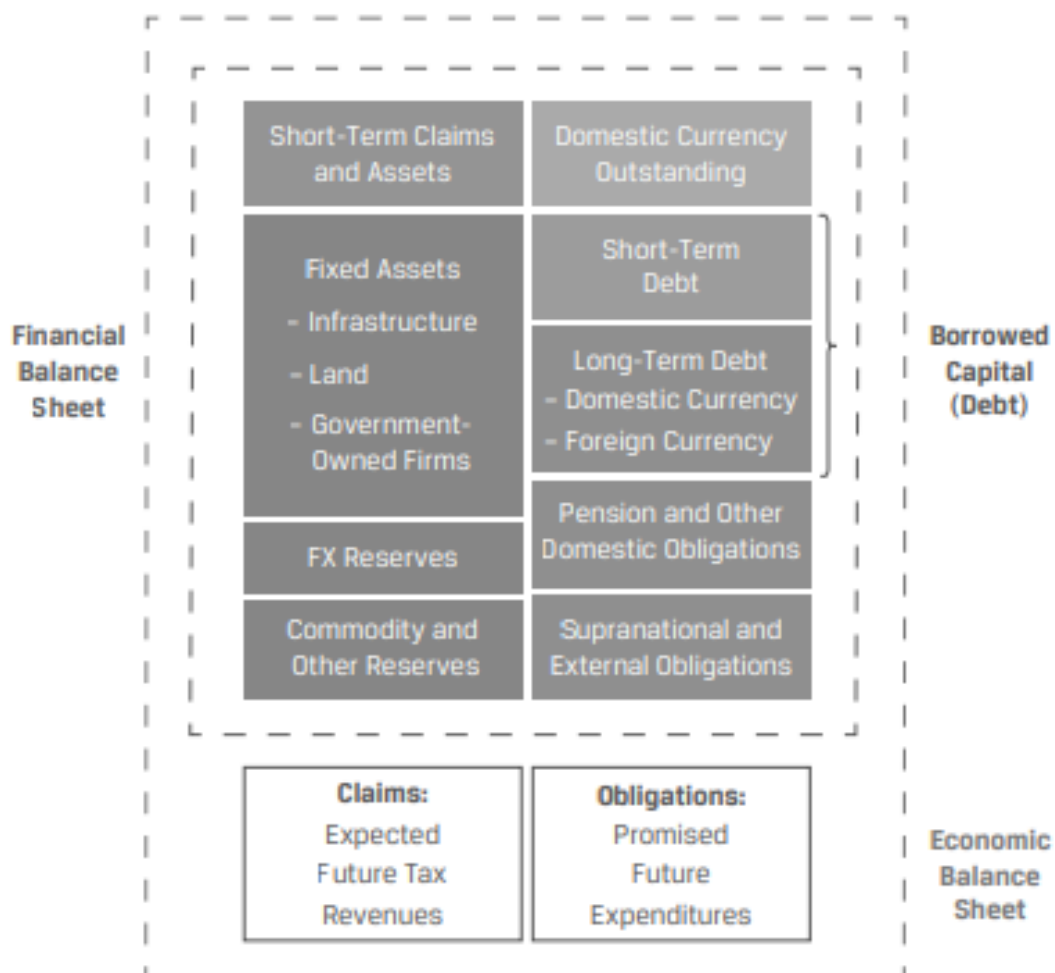
This learning module covers:

- Fixed income securities issued by sovereign and non-sovereign governments, quasi-government entities, and supranational agencies
- Differences between the issuance of government versus corporate fixed-income instruments

2. Sovereign Debt

Sovereign debt is issued by national governments primarily for fiscal reasons. Taxes are the primary source of revenue for a government. If tax revenue is insufficient, then a government raises money by issuing sovereign debt. Sovereign debt is not backed by collateral. The primary source of repayment is tax collections. Additional sources include tariffs, usage fees, and cash flows from government owned enterprises.

Exhibit 1 from the curriculum presents a government's economic balance sheet.



While corporate financial statements are accrual based, financial accounting for governments is often cash based. Items such as depreciation of fixed public goods (e.g., national highways), or accrual of unfunded liabilities (e.g., government pension obligations) are typically excluded from the balance sheet. However, expected future claims and obligations are included because they are of greater relevance for public issuers as compared to corporates.

Developed market versus emerging market sovereign issuers:

An important distinction among sovereign debt is the difference between developed market (DM) and emerging market (EM) sovereign issuers.

- DMs are characterized by a strong, stable, well-diversified domestic economy. They typically have a stable and transparent fiscal policy. For these reasons, the debt issued by DM sovereign issuers is often referred to as default-risk free.
- EMs are usually characterized by higher growth but less stable and less well-diversified economies that may be subject to greater fluctuations over the economic cycle. Rating agencies distinguish between a DM sovereign bond issued in local currency and one in foreign currency. Local currency bonds generally have a higher credit rating than foreign currency bonds, because if needed the national government can print local currency to repay the bond, however it cannot print the foreign currency.

Investors based in developed markets who purchase external debt of emerging and frontier market sovereign issuers face indirect exposure to currency fluctuations, because their returns depend on an issuer's ability to generate foreign currency revenue to meet foreign currency interest and principal payments through international capital, goods, and services flows.

Government debt management policies:

Government debt management policies determine the composition of sovereign debt, i.e. short-term versus long-term, as well as other features. Sovereign debt issues include the following:

- Short-term securities (often known as Treasury bills) with maturities ranging from 1 to 12 months. They are usually issued as zero-coupon instruments and are sold at a discount to par.
- Medium- and long-term securities (often known as Treasury notes and Treasury bonds). They are often issued as fixed-rate domestic currency instruments. But sovereigns may also issue floating rate, inflation linked, and foreign currency instruments.

- Sometimes sovereign governments do not directly issue securities but instead provide a guarantee that effectively makes the securities a type of sovereign debt. For example, Mortgage-backed securities in US.

Ricardian equivalence:

According to the Ricardian equivalence, a government's choice of debt maturity is irrelevant in determining the present value of future tax cash flows. Taxpayers expect government debt to be offset by higher future taxes, implying that a sovereign government should be indifferent between collecting taxes today or raising debt of any maturity based on the following assumptions:

- Taxpayers smooth consumption over time, saving expected future taxes today for future payment.
- Taxpayers form rational expectations that today's tax cuts will result in future tax increases.
- Capital markets are perfect with no transaction costs, and taxpayers can freely borrow and lend.
- Taxpayers are altruistic on an intergenerational basis; that is, they pass on tax savings to descendants.

Based on this theory governments should fund themselves with the shortest maturity to minimize borrowing costs (long term bonds have higher interest rates due to term premiums). However, this strategy will introduce significant rollover risk. Also, the assumptions rarely hold perfectly. Therefore, in practice, governments seek to minimize interest rate and rollover risks by distributing debt across maturities, while issuing debt in regular, predictable intervals.

Benefits of longer-term sovereign government securities:

As mentioned above, long-term sovereign debt has higher borrowing costs, however, it provides the following benefits:

- *Establishment of a risk-free benchmark for all debt of specific maturities:* Sovereign bond yields are often used as a benchmark to calculate a corporate issuer's credit risk premium. Therefore, government debt policies frequently require the regular issuance of benchmark securities with varying maturities in order to improve capital market efficiency and transparency for private issuers.
- *Use in managing and hedging market interest rate risk:* These instruments and related derivatives are often used by financial intermediaries and asset managers to manage interest rate risk separately from credit risk.

- *Preferred use as collateral in repo and derivative transactions:* These instruments are the most common form of collateral for both repo and derivative transactions due to their high degree of liquidity and safety.
- *Government bond use in monetary policy and foreign exchange reserves:* Central banks buy and sell these instruments (open market operations) to implement their monetary policy mandate. Foreign market participants often hold their foreign currency reserves in the form of liquid foreign government bonds.

3. Sovereign Debt Issuance and Trading

While corporate debt issuances are managed by investment banks, sovereign debt is typically issued through a public auction using standard procedures led by the national treasury or finance ministry.

Once a government debt auction is announced, prospective investors submit a **competitive** or **non-competitive bids**. A competitive bidder specifies an acceptable price and number of securities to be purchased. If the price determined at auction is above the bid, a competitive bidder will not be offered any securities. In contrast, a non-competitive bidder agrees to accept the price determined at auction and always receives securities.

A competitive bid process can be a single-price or multiple-price auction. In both cases, the issuer ranks bids by prices, choosing bids from highest to lowest until the desired issuance amount is reached. In a single-price auction, all winning bidders pay the same price and receive the same coupon rate for the bonds, regardless of their bid. A multiple-price auction, on the other hand, generates different prices among bidders for the same bond issue. The single-price auction process may result in a lower cost of funds and broader distribution among investor. Multiple price auction may result in a narrower distribution of large bids because investors must accept bonds at their bid prices.

Sovereign governments appoint a group of financial intermediaries to serve as **primary dealers**, and they must participate in all auctions with competitive prices.

Once issued, sovereign debt is typically traded on OTC markets by financial intermediary brokers/dealers in a manner similar to that of private sector debt securities.

Sovereign securities are classified into two categories based on when they were issued: **on-the-run** and **off-the-run**. On-the-run are recently issued sovereign securities that trade frequently. They are also called benchmark bonds because the yields of other bonds are determined relative to these bonds. Off-the-run refers to securities that were issued some time ago. They are less liquid compared to on-the-run securities.

4. Non-Sovereign, Quasi-Government, and Supranational Agency Debt

The level and type of non-sovereign government funding vary widely among countries, depending on whether specific goods and services are provided and financed at the national, regional, or local level.

Government Agencies

Government agencies are quasi-government entities that issue debt to fund specific public goods or services. For example, the Airport Authority of Hong Kong is a government agency that operates and develops the Hong Kong International Airport. This agency issues a combination of short-and-long term debt to meet the airport working capital and capital investment needs. The primary source of repayment is cash flows from airport operations, while its sovereign government backing is a secondary source of repayment.

Local and Regional Government Authorities

This category includes local governments such as states, provinces, and cities (i.e. all government authorities except the national government).

They may issue debt for general purposes known as *general obligation bonds (GO bonds)* which is repaid from local tax cash flows.

They may also issue debt for specific project financing (such as highways, bridges, or tunnels) known as *revenue bonds*. The source of repayment is usually linked to the project's revenue stream (tolls, fees etc.) They are usually longer-dated with a maturity matching the expected life of the project being financed.

Supranational Organizations

Supranational organizations are created and supported by sovereign governments in pursuit of a common objective, such as fostering economic cooperation and development, promoting trade, or providing financing to emerging economies in pursuit of sustainable growth. Examples include the World Bank, IMF, Asian Development Bank etc.

Member states typically share decision-making authority and provide these organizations with implicit and explicit financial support, resulting in the highest credit quality among these issuers and a strong ability to access capital markets across maturities.

Summary

LO: Describe funding choices by sovereign and non-sovereign governments, quasi-government entities, and supranational agencies.

Sovereign debt issues include short-term securities (with maturities ranging from 1 to 12 months), medium- and long-term notes and bonds, and bonds that are guaranteed but not directly issued by the sovereign government.

Non-sovereign governments may issue general obligation or revenue bonds.

Government agencies are quasi-government entities that issue debt to fund specific public goods or services. For example, the Airport Authority of Honk Kong.

Supranational organizations are created and supported by sovereign governments in pursuit of a common objective.

LO: Contrast the issuance and trading of government and corporate fixed-income instruments.

Sovereign debt is typically issued through a public auction. Once a government debt auction is announced, prospective investors submit a competitive or non-competitive bids.

Sovereign governments appoint a group of financial intermediaries to serve as primary dealers, and they must participate in all auctions with competitive prices.

Once issued, sovereign debt is typically traded on OTC markets by financial intermediary brokers/dealers in a manner similar to that of private sector debt securities.

LM06 Fixed-Income Bond Valuation - Prices and Yields

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Version 1.0

1. Introduction

This learning module covers:

- How to price a bond on or between coupon dates
- Relationship between a bond's price and bond features such as coupon rate, maturity, and yield-to-maturity
- Matrix pricing

2. Bond Pricing and The Time Value of Money

Bond Pricing with a Market Discount Rate

A bond's price is the present value of all future cash flows at the market discount rate. The discount rate is the rate of return required by investors given the risk of investment in the bond. It is also known as the **required yield**, or **required rate of return**.

$$\text{PV of bond} = \frac{\text{PMT}}{(1+r)^1} + \frac{\text{PMT}}{(1+r)^2} + \cdots + \frac{\text{PMT}+\text{FV}}{(1+r)^N}$$

where:

PMT = coupon payment per period

FV = par value of the bond paid at maturity

r = market discount rate

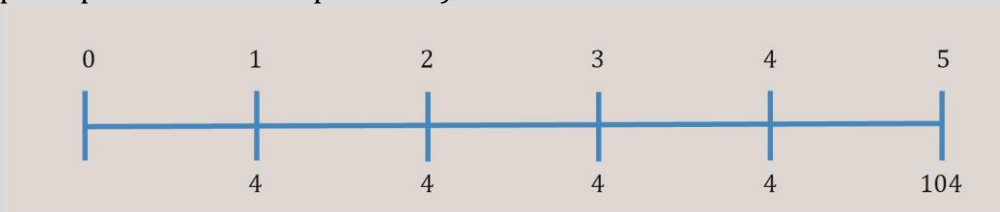
N = number of periods until maturity

Example

The coupon rate on a bond is 4% and the payment is made once a year. The time to maturity is five years and the market discount rate is 6%. What is the bond price per 100 of par value?

Solution:

Start by drawing a timeline for the cash flows. The par value of the bond or principal is \$100. A coupon payment of \$4 is made every year. At maturity (at the end of five years), a payment of \$104 (principal of \$100 + coupon of \$4) is made.



We are required to calculate the present value of bond at time $t = 0$. For that, we discount all the future cash flows at the market discount rate of 6%.

End of year	Type of cash flow	Amount	Present value
1	Coupon	4	3.77
2	Coupon	4	3.55

3	Coupon	4	3.36
4	Coupon	4	3.17
5	Coupon + principal	104	77.71
			91.56

Using the formula, $PV = \frac{4}{1.06} + \frac{4}{1.06^2} + \frac{4}{1.06^3} + \frac{4}{1.06^4} + \frac{104}{1.06^5} = 91.575$

Note: The table and formula above were just for your understanding.

On the exam, solve this quickly using a financial calculator:

N = 5; I/Y = 6; FV = 100; PMT = 4; PV = ? PV = -91.575

Notice the present value of the bond (91.575) is lower than the par value of the bond (100).

Premium, Par, and Discount Bonds

Example

Assume we have another bond with a coupon rate of 8%, paid annually. If the market discount rate is again 6%, what is the price of the bond?

Solution:

Note that the coupon rate here is higher than the market rate. Calculate PV using these input values: N = 5; I/Y = 6; PMT = 8; FV = 100; CPT PV

PV = -108.425. In this case, the present value of the bond is higher than its par value. Such a bond is called a premium bond. Since the coupon rate (8%) is higher than the required return of 6%, investors pay more than the par value for the bond.

What if the coupon rate is 6%?

N = 5; I/Y = 6; PMT = 6; FV = 100; CPT PV

PV = -100. The present value is equal to the par value of the bond.

What if the coupon rate is 2%?

N = 5; I/Y = 6; PMT = 2; FV = 100; CPT PV

PV = -83.151. This type of bond, which sells at a price below its par value, is called a discount bond. Investors require a return of 6% but it pays a coupon of only 2%, so investors pay less.

Par, premium, or discount	Relationship between coupon rate & market discount rate
Par bond	Coupon rate = market discount rate
Discount bond	Coupon rate < market discount rate
Premium bond	Coupon rate > market discount rate

Yield-to-Maturity

The yield to maturity (YTM) is the internal rate of return on the cash flows – the uniform interest rate that will make the sum of the present values of future cash flows equal to the

price of the bond. It is the implied market discount rate. In simpler terms, it is a bond's internal rate of return - the rate of return on a bond including interest payments and capital gain if the bond is held until maturity. Yield to maturity is based on three important assumptions:

- The investor holds the bond to maturity.
- The issuer does not default on payments and pays coupon and principal as they come due.
- The investor is able to reinvest all proceeds (coupons) at the YTM. (This is an unrealistic assumption as the interest rates may increase or decrease after the bond is purchased. So, the coupon may be reinvested at a higher or lower rate.)

Example

A \$100 face value bond with a coupon rate of 10% has a maturity of 4 years. The price of the bond is \$80. What is its yield to maturity?

Solution:

Based on YTM's definition, we come up with the equation below:

$$80 = \frac{10}{(1+r)^1} + \frac{10}{(1+r)^2} + \frac{10}{(1+r)^3} + \frac{110}{(1+r)^4}$$

This equation can be solved using trial and error but it is much easier to use a financial calculator: N = 4; PMT = 10; FV = 100; PV = -80; CPT I/Y = 17.34%. The bond is trading at a discount. So, the coupon rate (10%) must be lower than market rate (17.34%).

Example

Calculate the yields to maturity for the following bonds:

Bond	Coupon Payment per Period	Number of Periods to Maturity	Price
A	2.50	3	102.80
B	2.00	5	97.76
C	0.00	48	23.425

Solution:

$$\text{Bond A: } 102.80 = \frac{2.5}{(1+r)} + \frac{2.5}{(1+r)^2} + \frac{102.5}{(1+r)^3}, r = 1.54\%$$

$$\text{Bond B: } 97.76 = \frac{2}{(1+r)} + \frac{2}{(1+r)^2} + \frac{2}{(1+r)^3} + \frac{2}{(1+r)^4} + \frac{102}{(1+r)^5}, r = 2.48\%$$

$$\text{Bond C: } 23.425 = \frac{100}{(1+r)^{48}}, r = 3.07\%$$

Flat Price, Accrued Interest, and the Full Price

When a bond is between coupon dates, its price has two parts: flat price and accrued interest. The sum of the two parts is called the full price or dirty price.

Full price or dirty price = Flat price + Accrued interest

$$PV^{\text{full}} = PV^{\text{flat}} + AI$$

Flat price = Full price – Accrued interest

$$\text{Accrued interest} = \frac{t}{T} \times PMT$$

where:

t = number of days since the last coupon payment

T = number of days between coupon payments

PMT = interest payment per period

As indicated in the above equation, the accrued interest is based on the number of days since the last coupon payment and the number of days between coupon payments. Here are two conventions for calculating the number of days:

- Actual convention: Uses the actual number of days, including weekends, holidays and leap days.
- 30/360 convention: Assumes 30 days in a month and 360 days in a year.

Illustration of Flat Price and Dirty Price through an Example

Assume you own a two-year bond issued on 10 January 2017 with a par value of \$100. The bond pays a coupon of 8% half-yearly. The first coupon is due on 10 July 2017, the next on 10 January 2018, the third on 10 July 2018, and the final one is paid along with principal at maturity date on 10 January 2019. The market discount rate is equal to the coupon rate and so the flat price is 100. An investor wants to purchase the bond from you on 31 May 2017. What is the full price he must pay?

You have held the bond for 140 days (since the bond was issued) and earned an interest for this period even though it has not been paid. The next coupon payment is \$4. Assuming each period is 180 days the accrued interest is $4 \times (140/180) = 3.11$. The full price is $100.00 + 3.11 = 103.11$. This is the amount you will receive on selling the bond.

Note that it is the flat (or clean) price which is quoted. The accrued interest increases every day until the coupon payment. So, if the accrued interest was included there would be a variation in bond quotes on a daily basis and a sudden drop after the coupon payment is made, which is misleading.

The full price of a bond can also be calculated using the equation shown below.

$$PV_{\text{full}} = PV \times (1 + r)^{\frac{t}{T}}$$

where:

t = number of days since the last coupon payment

T = number of days between coupon payments

PV = present value of the bond (not the flat price)

Example

A 5% French corporate bond is priced for settlement on 14 July 2015. The bond makes semi-annual coupon payments on 1 March and 1 September of each year and matures on 1 September 2020. Assuming a 30/360 day count convention for accrued interest, calculate the full price, the accrued interest, and the flat price per EUR 100 of par value for the following three yields to maturity: (I) 4.75%, (II) 5.00% and (III) 5.10%.

Solution:

Given the 30/360 day-count convention, there are 134 days between the last coupon on 1 March 2015 and the settlement date on 14 July 2015 (120 days for full months of March, April, May, and June, plus 14 days in July). Therefore, the fraction of the coupon period that has gone by is assumed to be 134/180. At the beginning of the period, there are 11 periods to maturity.

I) Stated annual YTM of 4.75%

The price at the beginning of the period is 101.198 per 100 of par value.

$$PV = \frac{2.5}{1.02375} + \frac{2.5}{1.02375^2} + \dots + \frac{102.5}{1.02375^{11}} = 101.198$$

$$\text{The full price on 14 July is } PV(\text{full}) = 101.198 \times \left(1.02375^{\frac{134}{180}}\right) = 102.979$$

$$\text{The accrued interest is } AI = 2.5 \times \left(\frac{134}{180}\right) = 1.861$$

$$\text{The flat price is } PV(\text{flat}) = 102.979 - 1.861 = 101.118$$

II) Stated annual YTM of 5.00%

The price at the beginning of the period is 100.00 per 100 of par value.

$$PV = \frac{2.5}{1.025} + \frac{2.5}{1.025^2} + \dots + \frac{102.5}{1.025^{11}} = 100.00$$

$$\text{The full price on 14 July is } PV(\text{full}) = 100.00 \times \left(1.025^{\frac{134}{180}}\right) = 101.855$$

$$\text{The accrued interest is } AI = 2.5 \times \left(\frac{134}{180}\right) = 1.861$$

$$\text{The flat price is } PV(\text{flat}) = 101.855 - 1.861 = 99.9942$$

III) Stated annual YTM of 5.10%

The price at the beginning of the period is 99.5256 per 100 of par value.

$$PV = \frac{2.5}{1.0255} + \frac{2.5}{1.0255^2} + \dots + \frac{102.5}{1.0255^{11}} = 99.5256$$

$$\text{The full price on 14 July is } PV(\text{full}) = 99.5256 \times \left(1.0255^{\frac{134}{180}}\right) = 101.4088$$

$$\text{The accrued interest is } AI = 2.5 \times \left(\frac{134}{180}\right) = 1.861$$

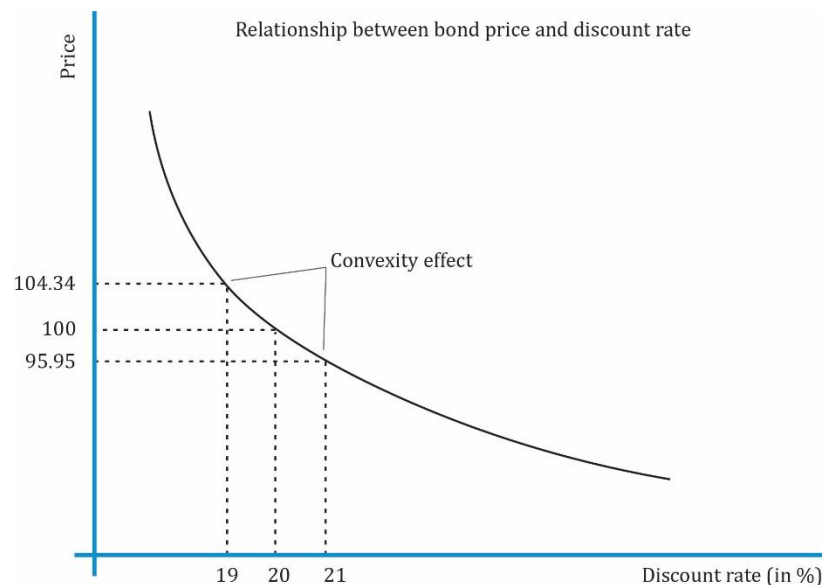
The flat price is $PV(\text{flat}) = 101.4088 - 1.861 = 99.5478$

The flat price of the bond is a little below par value, even though the coupon rate and the yield to maturity are equal, because the accrued interest does not take into account the time value of money. The accrued interest is the interest earned by the owner of the bond for the time between the last coupon payment and the settlement date. However, that interest income is not received until the next coupon date. The calculation of accrued interest in practice neglects the time value of money. Therefore, compared to theory, the reported accrued interest is a little “too high” and the flat price is a little “too low.”

3. Relationships Between Bond Prices and Bond Features

In this section, we will study the relationship between the bond price and market discount rate. The underlying principle is that a bond's price changes whenever there is a change in market discount rate.

Inverse Effect and Convexity Effect



The bond price is inversely related to the market discount rate. As you can see from the exhibit above, when the discount rate goes down, the price of the bond increases. This is called the inverse effect. Remember that the price-yield relationship of a bond is not a straight line. Also note that price changes are not linear for the same amount of change in discount rate. At an interest rate of 20%, the price of the bond is 100. If the interest rate decreases from 20% to 19%, the bond price increases by 4.34%. However, if the interest rate increases from 20% to 21%, the bond price decreases by only 4.05%. This difference in price change is called the convexity effect.

Coupon Effect

Consider three bonds (A, B, C) which have the same time to maturity but different coupon rates.

Bond	Coupon Rate	Price At 20%	Price At 19%	% Change	Price At 21%	% Change
A	10.00%	58.075	60.950	4.95%	55.405	-4.60%
B	20.00%	100.000	104.339	4.34%	95.946	-4.05%
C	30.00%	141.925	147.728	4.09%	136.487	-3.83%

Note that the bond with lowest coupon rate has the highest interest rate sensitivity. In other words, a 1% change in interest rates causes a greater % change in the price of bond A (10% coupon) relative to bonds B (20% coupon) and C (30% coupon). This is called the coupon effect.

Maturity Effect

For the same coupon rate, a longer-term bond has a greater percentage price change than a shorter-term bond when their market discount rates change by the same amount. This is called the maturity effect.

Summary of relationship between bond prices and discount rates	
Inverse effect	Bond price is inversely related to the discount rate.
Convexity effect	The percentage price change is more when discount rate goes down than when it goes up.
Coupon effect	For the same time to maturity and same change in discount rate the price of a low-coupon bond changes more than a high-coupon bond.
Maturity effect	For the same coupon rate and same change in discount rate, the price of a long-term bond changes more than a short-term bond.
Note: Low-coupon and Long-term bonds are more sensitive to changes in discount rates.	

Constant-Yield Price Trajectory

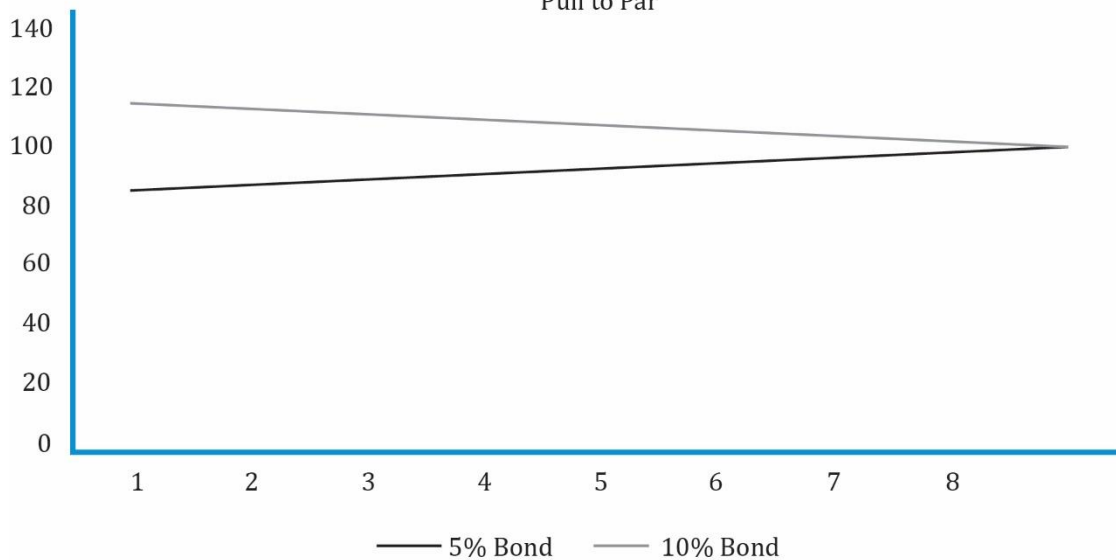
Even if the yield stays constant, bond prices change and come closer to par as time passes and as they near maturity date. The 'constant-yield price trajectory' illustrates the change in a bond's price over time if yields remain constant. This trajectory shows the "pull to par" effect. For a premium bond, the price decreases over time to par and for a discount bond, the price increases over time to par.

The exhibit below shows the constant-yield price trajectories for 5% and 10% annual coupon payment, 8-year bonds. Both bonds have a market discount rate of 7.5%. The 5% bond's initial price is 85.3567 and the 10% bond's initial price is 114.6433.

Year	5% Bond	10% Bond
1	85.3567	114.6433
2	86.7585	113.2415

3	88.2654	111.7346
4	89.8853	110.1147
5	91.6267	108.3733
6	93.4987	106.5013
7	95.5111	104.4889
8	97.6744	102.3256
	100.0000	100.0000

Pull to Par



Instructor's Note

If the yield stays constant:

- A premium bond's price decreases to par value as its time to maturity approaches zero.
- A discount bond's price increases to par value as its time to maturity approaches zero.
- A par bond's value remains unchanged as it approaches maturity.

4. Matrix Pricing

Matrix Pricing Process

One can easily get information at what price a stock is trading. But it is not easy to determine the price of most bonds as they do not trade as often as equity. Matrix pricing is an estimation process to find the market price of a not-so-frequently traded bond based on the prices of comparable bonds with similar times to maturity, type of issuer, coupon rates, and credit quality. This is also applicable for bonds that have not been issued yet.

For example, consider a three-year, 4% semi-annual bond X that you are trying to value. Comparable bonds whose prices are known are shown in the matrix below. Based on the four bonds, we can determine the price of bond X.

Matrix Pricing				
	2% coupon	3% coupon	4% coupon	5% coupon

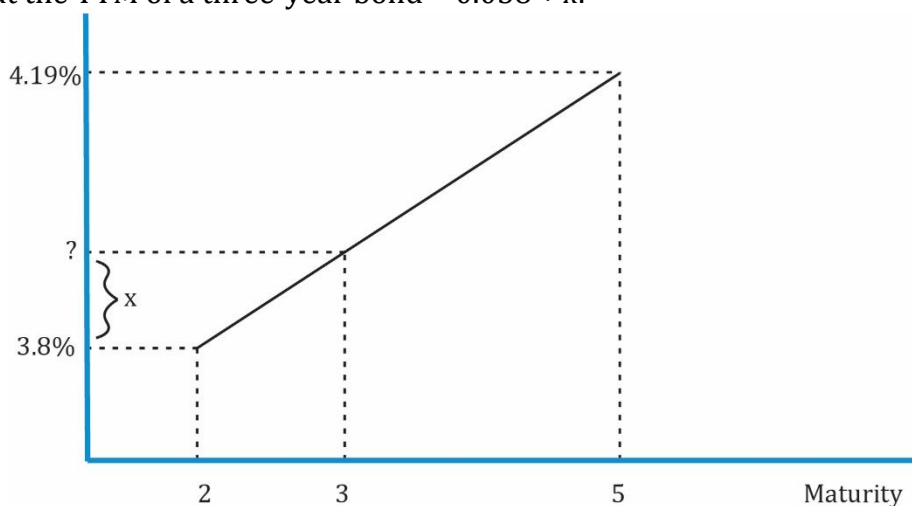
Two years		98.5		102.25
		3.786%		3.821%
Three years			Bond X	
Four years				
Five years	90.25		99.125	
	4.181%		4.196%	

There are five steps to determine the value of a bond using the matrix pricing method:

1. Determine the YTM of comparable bonds, which is given to us already. If it were not given, you can calculate the YTM using the price and coupon. For example, YTM of a two-year, 3% coupon is: $N = 4$; $PMT = 1.5$; $FV = 100$; $PV = -98.5$; $CPT I/Y$. $I/Y = 3.786\%$.
2. Determine the average yield for two-year bonds:

$$\frac{0.03786 + 0.03821}{2} = 0.038035$$
3. Determine the average yield for five-year bonds:

$$\frac{0.04181 + 0.4196}{2} = 0.041885$$
4. Calculate the three-market discount rate using linear interpolation. We have the data (yield) for two-year and five-year bonds, but not a three-year bond. From the graph you can see that the YTM of a three-year bond = $0.038 + x$.



x can be calculated as $\frac{3-2}{5-2} \times (0.041885 - 0.038035) = 0.001283$. Discount rate for the three-year bond = $0.038035 + 0.001283 = 0.039318 = 3.39\%$.

5. Using 0.039318 as the discount rate, compute the price for the three-year bond.
 $N = 6$; $I/Y = 1.966$; $PMT = 2$; $FV = 100$; $CPT PV$. $PV = -100.1906$.

A few points to be noted on matrix pricing:

- Matrix pricing is used when underwriting new bonds to estimate the required yield spread over the benchmark rate.
- Benchmark rate is a market reference rate, or a government bond with the same maturity as the bond being priced.
- Assume the YTM for a new bond is calculated using the matrix pricing method as 2.2% and a comparable government bond has a yield of 2.0%. The difference between the yield on a new bond over its benchmark rate is called the **required yield spread** or spread over the benchmark. In this case, it is 0.2%.
- Yield spreads are always specified in basis points where 1 basis point is one-hundredth of a percentage point. In this case, it is 20 basis points.

Example

An analyst decides to value an illiquid 3-year, 3.75% annual coupon payment corporate bond. Given the following information and using matrix pricing, what is the estimated price of the illiquid bond? Additional information: 2-year, 4.75% annual coupon payment bond priced at 105.60 and 4-year, 3.5% annual coupon payment bond priced at 103.28.

Solution:

The required yield on the 2-year, 4.75% bond priced at 105.60 is 1.871%.

The required yield on the 4-year, 3.50% bond priced at 103.28 is 2.626%.

The estimated market discount rate for a 3-year bond having the same credit quality is the average of two required yields: $\frac{1.871\% + 2.626\%}{2} = 2.249\%$.

The estimated price of the illiquid 3-year, 3.75% annual coupon payment corporate bond is 104.3078 per 100 of par value.

$$\frac{3.75}{1.02249} + \frac{3.75}{1.02249^2} + \frac{103.75}{1.02249^3} = 104.3078$$

Summary

LO: Calculate a bond's price given a yield-to-maturity on or between coupon dates.

A bond's price is the present value of all future cash flows at the market discount rate, which is the rate of return required by investors given the risk of investment in the bond.

$$\text{PV of bond} = \frac{\text{PMT}}{(1+r)^1} + \frac{\text{PMT}}{(1+r)^2} + \cdots + \frac{\text{PMT}+\text{FV}}{(1+r)^N}$$

When a bond is between coupon dates, its price has two parts: flat price and accrued interest. The sum of the two parts is called the full price or dirty price.

Full price or dirty price = flat price + accrued interest

$$\text{Accrued interest} = \frac{t}{T} * \text{PMT}$$

Full price of a fixed-rate bond between coupon payments:

$$\text{PV}_{\text{full}} = \text{PV} * (1 + r)^{\frac{t}{T}}$$

LO: Identify the relationships among a bond's price, coupon rate, maturity, and yield-to-maturity.

Inverse effect	Bond price is inversely related to discount rate.
Convexity effect	For the same coupon/maturity bond: the percentage price change is more when discount rate goes down than when it goes up.
Coupon effect	For the same time to maturity and same change in discount rate: prices of low-coupon bonds change more than a high-coupon bond.
Maturity effect	For the same coupon rate and same change in discount rate: prices of long-term bonds change more than short-term bonds.
Note: Low-coupon and Long-term bonds are more sensitive to discount rates.	

LO: Describe matrix pricing.

Matrix pricing is an estimation process to find the market price of a not-so-frequently traded bond based on the prices of comparable bonds with similar times to maturity, type of issuer, coupon rates, and credit quality.

The steps involved in matrix pricing are:

- Determine the YTM of comparable bonds (which is usually given in the question).
- Determine the average yields of different comparable bonds.
- Calculate the market discount rate using linear interpolation.
- Using the discount rate, compute the price of the bond.

LM07 Yield and Yield Spread Measures for Fixed-Rate Bonds

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3. Other Yield Measures, Conventions, and Accounting for Embedded Options	3
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Version 1.0

1. Introduction

This learning module covers:

- How to calculate the annual yield on a bond for varying compounding periods in a year
- Yield and yield spread measures for fixed-rate bonds

2. Periodicity and Annualized Yields

Periodicity is the number of compounding periods in a year, or number of coupon payments made in a year.

The **stated annual rate** for a bond will depend on the periodicity we are assuming. The stated annual rate is also called the annual percentage rate or APR.

Instructor's Note

A quarterly coupon paying bond has a periodicity of four, while a semi-annual bond has a periodicity of two, and a monthly-pay bond with a given annual yield would have a periodicity of twelve.

"Compounding more frequently within the year results in a lower (more negative) yield-to-maturity."

Consider a 5-year, zero-coupon bond priced at 80 per 100 par value. What is the stated annual rate for periodicity = 4, periodicity = 2, and periodicity = 1?

When periodicity = 4: compounding happens four times a year. $N = 20$; (5 years $\times$ 4 = 20). $PMT = 0$ as it is a zero-coupon bond. $PV = -80$; $FV = 100$; $CPT I/Y = 1.12$. This is the rate for each quarter. The stated annual rate is $1.12 \times 4 = 4.487\%$.

When periodicity = 2: $N = 10$; $PV = -80$; $PMT = 0$; $FV = 100$; $CPT I/Y = 2.2565$. The stated annual rate is $2.25 \times 2 = 4.51\%$.

When periodicity = 1: $N = 5$; $PV = -80$; $PMT = 0$; $FV = 100$; $CPT I/Y = 4.56\%$. With a periodicity of 1, the stated annual rate is the same as the effective annual rate.

The formula for conversion based on periodicity is

$$\left(1 + \frac{APR_m}{m}\right)^m = \left(1 + \frac{APR_n}{n}\right)^n$$

Example

A 4-year, 3.75% semi-annual coupon payment government bond is priced at 97.5. Calculate the annual yield to maturity stated on a semi-annual bond basis and convert the annual yield to:

1. An annual rate comparable to bonds that make quarterly coupon payments.
2. An annual rate comparable to bonds that make annual coupon payments.

Solution to 1:

The stated annual yield to maturity on a semiannual bond basis can be calculated using a financial calculator: N = 8; PMT = 1.875; FV = 100; PV = -97.5; CPT I/Y. I/Y = 2.2195%.

Hence, the stated annual yield to maturity = 2.2195% x 2 = 4.439%.

$$\left(1 + \frac{0.04439}{2}\right)^2 = \left(1 + \frac{APR_4}{4}\right)^4$$

$$APR_4 = 4.415\%$$

The annual rate of 4.439% for compounding semiannually compares with 4.415% for compounding quarterly.

Solution to 2:

$$\left(1 + \frac{0.04439}{2}\right)^2 = (1 + APR_1)$$

$$APR_1 = 4.488\%$$

The annual percentage rate of 4.439% for compounding semiannually compares with an effective annual rate of 4.488%.

The **effective annual rate (EAR)** is the yield on an investment in one year taking into account the effects of compounding. This rate has a periodicity of one as there is only one compounding period per year. EAR is used to compare the rate of return on investments with different frequency of compounding (periodicities).

Semiannual bond equivalent yield: Yield per semi-annual period times two. If the yield per semi-annual period is 2%, then the semi-annual bond equivalent yield is 4%.

3. Other Yield Measures, Conventions, and Accounting for Embedded Options

Other Yield Measures and Conventions

- **Street convention:** It is the yield to maturity using a 30/360-day convention assuming payments are made on scheduled dates, even if the payment date fell on a weekend or a holiday.
- **True yield:** Yield to maturity calculated using an actual calendar of weekends and holidays. For instance, assume the coupon date falls on 15 March 2015, which is a Sunday. Street convention assumes the payment is made on that date, whereas true yield assumes the payment is made on 16 March if it is a business day. The coupon payment is discounted back from 16 March instead of 15 March.
- **Government equivalent yield:** Yield to maturity calculated using the actual day/count convention used for U.S. Treasuries.
- **Current yield (Simple yield):** Sum of the coupon payments received over the year divided by the flat price. It is also called the income or interest yield. Example: A 5-year, 8%

semiannual coupon payment bond is priced at \$960. Its current yield is $80/960 = 0.0833 = 8.33\%$. Current yield is not an accurate measure of the rate of return as it ignores the frequency of coupon payments, reinvestment income, and capital gain/loss on a bond.

$$\text{Current yield} = \frac{\text{Annual cash coupon payment}}{\text{Bond price}}$$

Bonds with Embedded Options

For bonds with embedded options the following yield measures are used.

- **Yield-to-call:** Calculates the rate of return on a callable bond if it is bought at market price and held until the call date. The difference between YTM and the yield-to-call is that YTM assumes the bond is held to maturity. Calculation of yield-to-call is the same as YTM where N = number of periods to call date and FV = call price.
- **Yield-to-first call (YTFC):** It is the internal rate of return if the bond was bought at market price and held until the first call date.
- **Yield-to-second call:** Similarly, the yield on a callable bond if it was bought at market price and held to the second call date is called yield-to-second call.
- **Yield-to-worst:** Yield is calculated for every scenario. The lowest yield is called the yield-to-worst.
- **Option adjusted yield:** The option-adjusted yield is the required market discount rate whereby the price is adjusted for the value of the embedded option. For example, investors pay a lower price for the callable bond than if it were option-free. If the bond were non-callable, its price would be higher. The option-adjusted price is used to calculate the option-adjusted yield.

Example

An analyst observes the following statistics for two bonds:

	Bond A	Bond B
Annual Coupon Rate	6.00%	10.00%
Coupon Payment Frequency	Semi-annually	Quarterly
Years to Maturity	4 years	4 years
Price (per 100 par value)	95	110
Current Yield	?	?
Yield to Maturity	?	?

1. Calculate both yield measures for the two bonds.
2. How much additional compensation, in terms of yield to maturity, does a buyer of Bond A receive for bearing additional risk compared with Bond B

Solution to 1:

The current yield for Bond A is $6/95 = 6.316\%$ and the yield to maturity for Bond A is

7.469%.

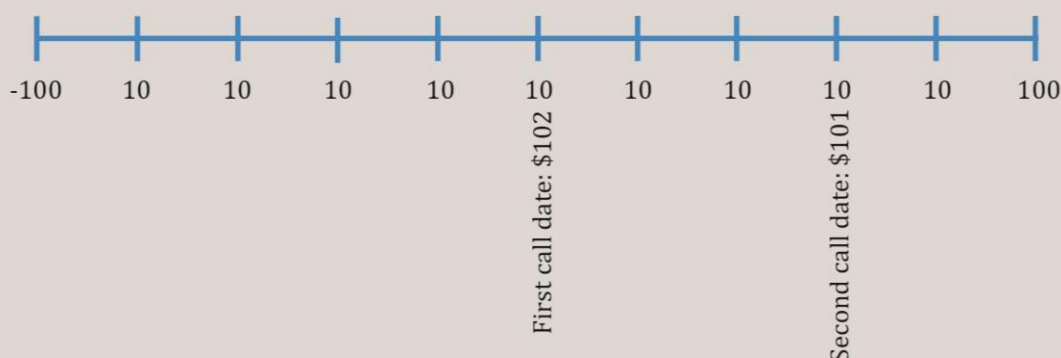
The current yield for Bond B is $10/110 = 9.091\%$ and the yield to maturity for Bond B is 7.106%.

Solution to 2:

Compare the yields for the same periodicity to answer this question. 7.106% for a periodicity of four converts to 7.169% for a periodicity of two. The additional compensation for the greater risk in Bond A is 30 bps ($0.07469 - 0.07169$).

Example

Consider a bond which is selling for \$100 and has the following cash flows:



Calculate the yield-to-first call and yield-to-second call.

Solution:

The yield-to-first call can be calculated as follows: $PV = -100$; $N = 5$; $PMT = 10$; $FV = 102$; CPT I/Y; I/Y = 10.325%. YTFC = 10.325%.

The yield-to-second call in our example will be 10.09%: $PV = -100$; $N = 8$; $PMT = 10$; $FV = 101$; CPT I/Y; I/Y = 10.09%.

Example

A bond with 4 years remaining until maturity is currently trading for 101.75 per 100 of par value. The bond offers a 5% coupon rate with interest paid semiannually. The bond is first callable in 2 years and is callable after that date on coupon dates according to the following schedule:

End of Year	Call Price
2	102.50
3	101.50
4	100.00

1. What is the bond's annual yield-to-first-call?

2. What is the bond's yield-to-worst?

Solution to 1:

The yield-to-first-call can be calculated with the following key strokes:

PV = -101.75, FV = 102.5, N = 4, PMT = 2.5, CPT I/Y = 2.6342.

To arrive at the annualized yield-to-first-call, the semiannual rate must be multiplied by two. ($2.6342 \times 2 = 5.2684$)

Solution to 2:

The yield-to-worst is 4.52%. The bond's yield to worst is the lowest of the sequence of yields-to-call and the yield to maturity. Yield-to-first-call = 5.27%, Yield-to-second-call = 4.84%, and Yield to maturity = 4.52%.

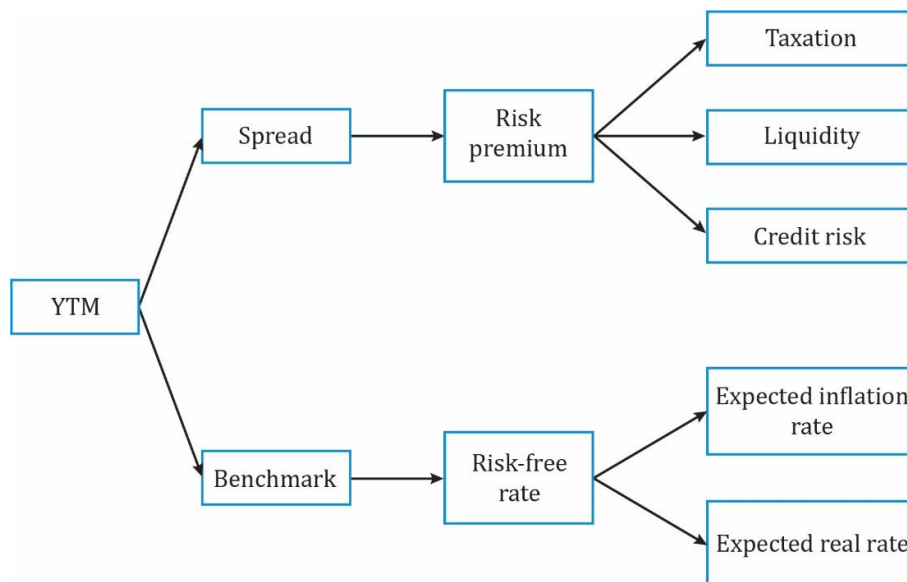
4. Yield Spread Measures for Fixed-Rate Bonds and Matrix Pricing

Yield Spreads over Benchmark Rates

The yield spread is the difference in yield between a fixed-income security and a benchmark. Say the YTM of a 3-year corporate bond is 7.00%. The benchmark rate is 3-year Libor, which is 5.00%. The yield spread of the corporate bond relative to the benchmark is 2.00%.

Generally, the benchmark reflects macroeconomic factors. A spread reflects microeconomic factors and aspects specific to the issuer, such as credit quality of the issuer and bond, tax status, etc.

The YTM of a bond can be broken down into the following components:



Spreads enable an analyst to more accurately assess a bond's historical and comparative value by controlling for (excluding) macroeconomic factors such as a broad increase or decrease in benchmark rates. An analyst might compare the current spread to its historical average, highs and lows, and other bonds to determine whether a bond is under- or overpriced relative to others.

Two related concepts are the G-spread and the I-spread.

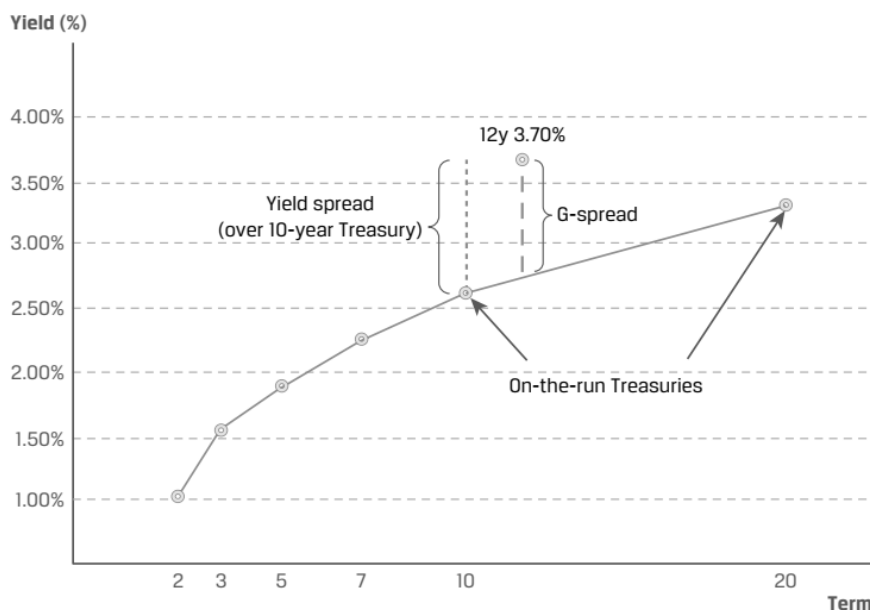
- The 'G-spread' is the yield spread in basis points over an interpolated government bond. The spread is higher for bearing higher credit, liquidity, and other risks relative to the government bond.
- The 'I-spread' is the yield spread of a specific bond over the standard swap rate in that currency of the same tenor.

Calculating G-Spread using Matrix Pricing

The G-spread measures the spread over interpolated government bonds. It uses constant maturity Treasury yields to maturity as the benchmark.

Say we want to calculate the G-spread of a 12-year bond. In the market on the run government bonds are available with 7 years, 10 years and 20 years maturities. Since we do not have a government bond with exactly 12 years maturity, we will use linear interpolation on the 10-year and 20-year government bonds. In this case:

The following exhibit illustrates the G-spread.



Example

A 5% annual coupon corporate bond with 3 years remaining to maturity is trading at 100.175. The 3-year, 3% annual payment government benchmark bond is trading at 100.50. The 1-year and 2-year government spot rates are 2.05% and 3.425% respectively. Calculate the G-spread, the spread between the yields to maturity on the corporate bond and the government bond having the same maturity.

Solution:

Solve for the time value of money approach on the corporate bond using the following keystrokes:

PV = -100.175, PMT = 5, FV = 100, N = 3, CPT I/Y = 4.936

The yield to maturity for the corporate bond is 4.936%.

The yield to maturity of the government bond can be calculated with the following keystrokes:

PV = -100.50, PMT = 3, FV = 100, N = 3, CPT I/Y = 2.824

The G-spread is $4.936\% - 2.824\% = 2.11\%$ or 211 bps.

Example:

A portfolio manager considers the following annual coupon bonds:

Issuer	Term	Coupon	Yield	ModDur
Bank	8y	2.75%	2.68%	7.10
Government	7y	1.5%	1.39%	6.61
Government	10y	1.625%	1.66%	9.16

Calculate the G-spread for the bank bond.

Solution:

The G-spread is the difference between the bank bond YTM and a linear interpolation of the YTM's of the 7-year government bond and the 10-year government bond.

The interpolated 8-year government rate is:

$1.39\% + (8-7)/(10-7) \times (1.66 - 1.39) = 1.48\%$

G-spread = $2.68\% - 1.48\% = 1.20\%$

Yield Spreads over the Benchmark Yield Curve

The Z-spread (zero-volatility spread) is based on the entire benchmark spot curve. It is the constant spread that is added to each spot rate such that the present value of the cash flows matches the price of the bond. The Z-spread is also called the static spread as it is constant for all periods.

The option-adjusted spread (OAS) is the Z-spread adjusted for the value of an embedded option. Consider a callable bond with a Z-spread of 90 basis points, of which 10 basis points are due to the embedded call option. In this case, the OAS is $90 - 10 = 80$ basis points. If a puttable bond has a Z-spread of 180 basis points and the value of the put option is 10 basis points, the OAS is $180 + 10 = 190$ basis points. Based on these simple scenarios, it should be clear that for a callable bond, the OAS is lower than the Z-spread, and for a puttable bond, the OAS is higher than the Z-spread.

Summary

LO: Calculate annual yield on a bond for varying compounding periods in a year.

- The stated annual rate for a bond will depend on the periodicity we are assuming. The stated annual rate is also called the annual percentage rate or APR.
- The effective annual rate (EAR) is the yield on an investment in one year taking into account the effects of compounding. EAR is used to compare the rate of return on investments with different frequency of compounding (periodicities).
- The semiannual bond equivalent yield is the yield per semi-annual period times two.

LO: Compare, calculate, and interpret yield and yield spread measures for fixed-rate bonds.

- Street convention is the yield to maturity using a 30/360-day convention assuming payments are made on scheduled dates.
- True yield is the yield to maturity calculated using an actual calendar with weekends and holidays.
- Current yield is the sum of the coupon payments received over the year divided by the flat price.
- Yield-to-call calculates the rate of return on a callable bond if it is bought at market price and held until the call date.

The yield spread is the difference in yield between a fixed-income security and a benchmark.

- G-spread is the yield spread in basis points over an interpolated government bond. The spread is higher for bearing higher credit, liquidity, and other risks relative to the government bond.
- I-spread is the yield spread of a specific bond over the standard swap rate in that currency of the same tenor.
- Z-spread (zero-volatility spread) is based on the entire benchmark spot curve. It is the constant spread that is added to each spot rate such that the present value of the cash flows matches the price of the bond.
- Option-adjusted spread (OAS) is the Z-spread adjusted for the value of an embedded option.

LM08 Yield and Yield Spread Measures for Floating-Rate Instruments

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Version 1.0

1. Introduction

This learning module covers:

- Yield spread measures for floating rate instruments – instruments with variable rather than fixed coupons
- Yield measures for money market instruments – instruments with original maturities of one year or less

2. Yield and Yield Spread Measures for Floating-Rate Notes

Floating-rate notes (FRN) are instruments where coupon/interest payments change from period to period based on a reference interest rate. Some important points to note about floating-rate notes:

- The objective is to protect the investor from volatile interest rates.
- The reference rate, usually a money market instrument such as a T-bill or a market reference rate (MRR), is used to calculate the interest payments. This rate is determined at the beginning of each period, but the interest is actually paid at the end of the period. This payment structure is called ‘in arrears.’
- Often, the coupon rate of an FRN is not just the reference rate, but a certain number of basis points, called the spread, is added to the reference rate.
- The specified yield spread over the reference rate is called the quoted margin.
- The spread remains constant throughout the life of the bond. The amount of spread depends on the credit quality of the issuer.

Example of a Floating Rate Note

Moody’s assigned a long-term credit rating of A2 to Nationwide, U.K.’s largest building society. Nationwide issued a perpetual floating-rate bond with a coupon rate of 6-month MRR+ 240 basis points. The 2.4% quoted margin is a reflection of its credit quality. On the other hand, AAA-rated Apple sold a three-year bond at 0.05% over 3-month MRR in 2013 as its credit risk was very low.

Coupon rate of a FRN = reference rate + quoted margin

The **required margin** is the spread demanded by the market. We saw that the quoted margin, or the spread over the reference rate, is fixed at the time of issuance. But what happens if the floater’s credit risk changes and investors demand an additional spread for bearing this risk? The required margin is the additional spread over the reference rate such that the FRN is priced at par on a rate reset date. If the required margin increases (decreases) because of a credit downgrade (upgrade), the FRN price will decrease (increase).

For example, assume a floater has a coupon rate of 3-month MRR plus 50 basis points. Six months after issuance, the issuer’s credit rating is downgraded and the market demands a required spread of 75 basis points. The coupon paid by the floater is lower than what the

market demands. As a result, the floater would be priced at a discount to par as the cash flow is now discounted at a higher rate. The amount of the discount will be the present value of differential cash flows, i.e., the difference between the required and quoted margins. Conversely, if the credit rating of the issuer improves, the required margin would be below the quoted margin, and the market will demand a lower spread.

How required margin affects a floater's price at reset date	
Relationship between quoted and required margin	Floater's price at reset date
Required margin = quoted margin	Par
Required margin > quoted margin	Discount (below par)
Required margin < quoted margin	Premium (above par)

The required margin is also called the **discount margin**. FRNs can be valued using the model shown below.

$$PV = \frac{(MRR + QM) * \frac{FV}{m}}{\left(1 + \frac{MRR+DM}{m}\right)^1} + \frac{(MRR + QM) * \frac{FV}{m}}{\left(1 + \frac{MRR+DM}{m}\right)^2} + \dots + \frac{(MRR + QM) * \frac{FV}{m} + FV}{\left(1 + \frac{MRR+DM}{m}\right)^N}$$

where:

PV = present value of the FRN

MRR = the market reference rate, stated as an annual percentage rate

QM = quoted margin, stated as an annual percentage rate

FV = future value paid at maturity, or the par value of the bond

m = periodicity of the floating-rate note, or the number of payment periods per year

DM = discount margin; the required margin stated as an annual percentage rate

N = number of evenly spaced periods to maturity

Equation 1 for reference:

$$PV \text{ of bond} = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \dots + \frac{PMT+FV}{(1+r)^N}$$

How to interpret the floating-rate note equation:

- Think of it as an extension of Equation 1, we will draw similarities between the two equations.
- $(MRR + QM) * \frac{FV}{m}$ is the first interest payment similar to PMT of Equation 1, which is the coupon payment per period.
- $(MRR + QM)$ is the annual rate.
- Since it is divided by periodicity we get the interest payment for that period.
- In Equation 1, cash flows are discounted at $1 + r$. For FRN, the cash flow for the first period is discounted at $1 + \frac{MRR+DM}{m}$, for the second period at $\left(1 + \frac{MRR+DM}{m}\right)^2$, and so on.

This is considered a simple model because of the following assumptions:

- The value is calculated only on reset dates. There is no accrued interest, so the flat price is the full price.
- The model uses a 30/360 day-count convention, which means periodicity is always an integer.
- The same reference rate is used in the numerator and denominator for all payment periods.

Example

A 3-year Italian floating-rate note pays 3-month MRR plus 0.75%. Assuming that the floater is priced at 99, calculate the discount margin for the floater if the 3-month MRR is constant at 1% (assume 30/360 day-count convention).

Solution:

The interest payment for each period is $(1.00\% + 0.75\%) / 4 = 0.4375\%$.

The keystrokes to calculate the market discount rate are: PV = -99, PMT = 0.4375, FV = 100, N = 12, CPT I/Y = 0.5237×4 , I/Y = 2.09%

The discount margin for the floater is $2.09\% - 1\% = 1.09\%$ or 109 bps.

3. Yield Measures for Money Market Instruments

Money market instruments are short-term debt securities. They have maturities of one year or less, ranging from overnight repos to one-year certificates of deposit.

Differences in Bond Market and Money Market Yields	
Bond Market Yields	Money Market Yields
YTM is annualized and compounded.	Rate of return is annualized but not compounded; stated on a simple interest basis.
YTM calculated using the standard time value of money approach using a financial calculator.	Non-standard calculation using discount rates and add-on rates.
YTM stated for a common periodicity for all times to maturity.	Instruments with different times to maturity have different periodicities for the annual rate.

The calculation of interest of a money market instrument is different from calculating accrued interest on a bond. Money market instruments can be classified into two categories based on how the rates are quoted:

- **Discount rates:** T-bills, commercial paper (CP), and banker's acceptances are discount instruments. It means they are issued at a discounted price, and pay par value at maturity. They do not make any payments before maturity. The difference between the

purchase price and par value at redemption is the interest earned. *Note: Do not confuse this discount rate with the rate used in TVM calculations.*

Price of a money market instrument quoted on a discount basis

$$PV = FV \times \left(1 - \frac{\text{days to maturity}}{\text{year}} \times DR\right)$$

where:

PV = present value of the money market instrument

FV = face value of the money market instrument

days to maturity = actual number of days between settlement and maturity

year = number of days in the year. Most markets use a 360-day year.

DR = discount rate, stated as an annual percentage rate (APR)

Money market discount rate

$$\text{Money market discount rate DR} = \left(\frac{\text{Year}}{\text{days to maturity}}\right) * \frac{FV - PV}{FV}$$

FV-PV = interest earned on the instrument (this is the discount)

$\frac{\text{year}}{\text{days}}$ = periodicity of the instrument

- **Add-on rates:** Bank term deposits, repos, certificates of deposit, and indices such as Libor/Euribor are quoted on an add-on basis. For a money market instrument quoted using an add-on rate, interest is added to the principal to calculate the redemption amount at maturity. In simple terms, if PV is the initial principal amount, days is the days to maturity, and year is the number of days in a year, then the amount to be paid at maturity is: $FV = PV + PV \times AOR \times \frac{\text{days to maturity}}{\text{year}}$ where AOR is the add-on rate stated on an annualized basis.

Present value or price of a money market instrument quoted on an add-on basis

$$PV = \frac{FV}{1 + \frac{\text{Days to maturity}}{\text{Year}} \times AOR}$$

where:

PV = present value of the money market instrument

FV = amount paid at maturity including interest

days = number of days between settlement and maturity

year = number of days in a year

AOR = add-on rate stated as an annual percentage rate

Add-on rate

$$AOR = \left(\frac{\text{Year}}{\text{Days}}\right) * \frac{FV - PV}{PV}$$

Instructor's Note

The primary difference between a discount rate (DR) and an add-on rate (AOR) is that the interest is included on the face value of the instrument for DR whereas it is added to the principal in case of AOR.

Example

Suppose that a banker's acceptance will be paid in 91 days. It has a face value of \$1,000,000. It is quoted at a discount rate of 5%. What is the price of the banker's acceptance?

Solution:

$$PV = FV * \left(1 - \frac{\text{Days to maturity}}{\text{Year}} * DR\right)$$

$$FV = 1,000,000$$

$$\text{Days} = 91$$

$$\text{Year} = 360 \text{ days}$$

$$DR = 5\%$$

$$PV = 1,000,000 * \left(1 - \frac{91}{360} * 0.05\right) = \$987,361$$

Suppose that a Canadian pension fund buys a 180-day banker's acceptance (BA) with a quoted add-on rate of 4.38% for a 365-day year. If the initial principal amount is CAD 10 million, what is the redemption amount due at maturity?

Solution:

$$AOR = \left(\frac{\text{Year}}{\text{Days}}\right) * \frac{FV - PV}{PV}$$

$$0.0438 = \frac{365}{180} * \frac{FV - 10,000,000}{10,000,000}$$

$$FV = \$10,216,000$$

Comparing Discount Basis with Add-On Yield

There are two approaches to compare the return of two money market instruments if one is quoted on a discount basis and the other on an add-on basis.

First approach: If you don't want to memorize one more formula, follow this approach:

1. Determine the present value of the instrument quoted on a discount basis.
2. Use the present value to determine the AOR.
3. Compare the two AORs to see which instrument offers a better return.

Second approach: Use the following relationship between AOR and DR:

Relationship between AOR and DR

$$\text{AOR} = \frac{\text{DR}}{1 - \frac{\text{Days to maturity}}{\text{Year}} * \text{DR}}$$

Example

A T-bill with a maturity of 90 days is quoted at a discount rate of 5.25%. Its par value is \$100. Calculate the add-on rate.

Solution:

Using the first approach:

FV = 100; Days = 90; Year = 360 days; DR = 5.25%

$$\text{PV} = 100 * \left(1 - \frac{90}{360} * 0.0525\right) = \$98.687$$

$$\text{AOR} = \frac{360}{90} * \frac{100 - 98.687}{98.687} = 5.32\%.$$

Using the second approach:

$$\text{AOR} = \frac{0.0525}{1 - \frac{90}{360} * 0.0525} = 5.3198\%$$

Example

Money market Instrument	Quotation Basis	Number of Days in the Year	Quoted Rate
A	Discount Rate	360	3.23%
B	Discount Rate	365	3.46%
C	Add-on Rate	360	3.25%
D	Add-on Rate	365	3.35%

Given the four 90-day money market instruments, calculate the bond equivalent yield for each of them. Which instrument offers the highest rate of return if the credit risk is the same?

Solution:

A. The price of instrument A is $100 - \left(\frac{90}{360} \times 3.23\right) = 99.1925$ per 100 of par value. The bond equivalent yield is $\left(\frac{365}{90}\right) \times \frac{100 - 99.1925}{99.1925} = 3.30\%$.

B. The price of instrument B is $100 - \left(\frac{90}{365} \times 3.46\right) = 99.1468$ per 100 of par value. The bond equivalent yield is $\left(\frac{365}{90}\right) \times \frac{100 - 99.1468}{99.1468} = 3.49\%$.

C. The redemption amount per 100 of principal is $100 + \left(\frac{90}{360} \times 3.25\right) = 100.8125$. The bond equivalent yield is $\frac{365}{90} \times (100.8125 - 100) = 3.295\%$.

D. The quoted rate for instrument D of 3.35% is the bond equivalent yield. Instrument B offers the highest rate of return on a bond equivalent yield basis.

Periodicity of the Annual Rate

Another difference between yield measures in the money market and the bond market is the periodicity of the annual rate. Because bond yields to maturity are computed using interest rate compounding, there is a well-defined periodicity. For instance, bond yields to maturity for semi-annual compounding are annualized for a periodicity of two. Money market rates are computed using simple interest without compounding. In the money market, the periodicity is the number of days in the year divided by the number of days to maturity. Therefore, money market rates for different times to maturity have different periodicities.

$$\text{Periodicity of a money market instrument} = \frac{\text{Number of days in the year}}{\text{Number of days to maturity}}$$

Example

A 90-day T-bill has a BEY of 11%. Calculate its semiannual bond yield.

Solution:

The 11% BEY of the T-bill is based on a periodicity of 365/90. The periodicity of a semiannual bond is 2. Give this information, we can create the equation shown below and solve for r . We will get $r = 0.1115 = 11.15\%$.

$$\left(1 + \frac{0.11}{\frac{365}{90}}\right)^{\left(\frac{365}{90}\right)} = \left(1 + \frac{r}{2}\right)^2$$

Summary

LO: Calculate and interpret yield spread measures for floating-rate instruments.

Floating-rate notes (FRN) are instruments where coupon/interest payments change from period to period based on a reference interest rate. The coupon rate of a FRN is equal to the sum of reference rate and quoted margin.

$$PV = \frac{(MRR + QM) * \frac{FV}{m}}{\left(1 + \frac{MRR+DM}{m}\right)^1} + \frac{(MRR + QM) * \frac{FV}{m}}{\left(1 + \frac{MRR+DM}{m}\right)^2} + \dots + \frac{(MRR + QM) * \frac{FV}{m} + FV}{\left(1 + \frac{MRR+DM}{m}\right)^N}$$

The quoted margin is a specified spread over or under the reference rate. The required margin, also known as the discount margin, is the spread required by investors.

LO: Calculate and interpret yield measures for money market instruments.

Money market instruments are short-term debt securities. They have maturities of one year or less, ranging from overnight repos to one-year certificates of deposit. Money market instruments can be classified into two categories based on how the rates are quoted; discount rate and add on rate.

$$PV = FV * \left(1 - \frac{\text{days to maturity}}{\text{year}} * DR\right)$$

$$PV = \frac{FV}{1 + \frac{\text{days to maturity}}{\text{year}} * AOR}$$

$$AOR = \frac{DR}{1 - \frac{\text{days to maturity}}{\text{year}} * DR}$$

LM09 The Term Structure of Interest Rates Spot, Par, and Forward Curves

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Version 1.0

1. Introduction

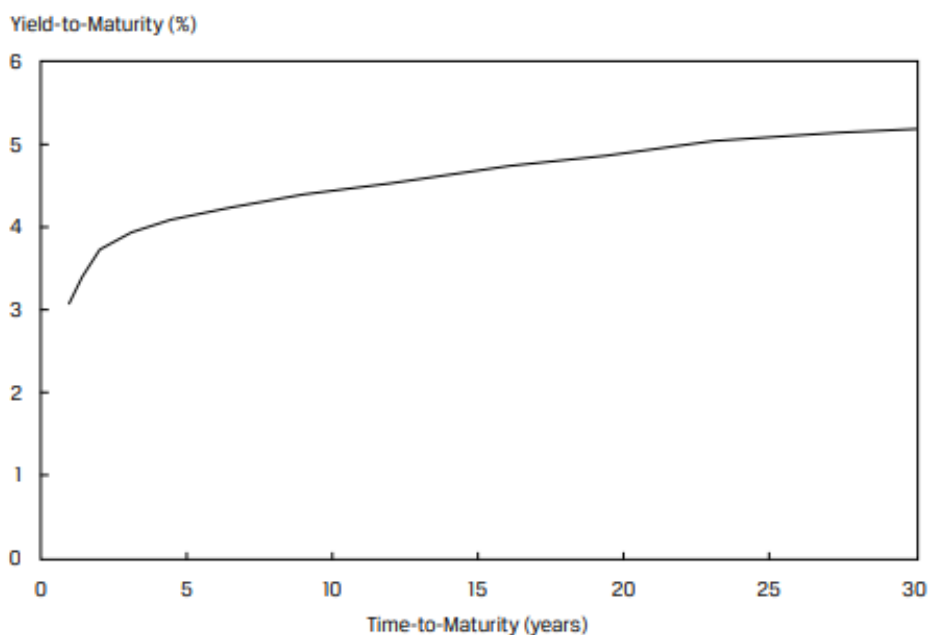
In the previous learning modules, we priced fixed income securities by discounting future cash flows using a single interest rate (such as the yield-to-maturity or MRR plus a discount margin). In this learning module, we will cover how to price fixed income securities using a sequence of interest rates. We will introduce the concept of term structure of interest rates, and the three common ways of representing the term structure – spot rate curve, par curve, and forward curve.

2. Maturity Structure of Interest Rates and Spot Rates

Maturity Structure of Interest Rates

The term structure of interest rates is the relationship between interest rates and bonds with different times to maturity. The basic premise is that interest rates change when inflation rates are expected to change over a period of time. For example, a one-year zero-coupon bond may have an interest rate of 7.50%, while a two-year zero-coupon bond may have an interest rate of 9.75%, assuming all the other factors are the same (currency, credit rating, periodicity, etc.). The three common ways of representing the term structure – spot rate curve, par curve, and forward curve.

Spot rate curve: In our example above, the interest rates of zero-coupon bonds, 7.50% and 9.75% are called spot rates. Spot rates are yields to maturity (or return earned) on zero-coupon bonds maturing at the date of each cash flow, if the bond is held to maturity. So, the spot rate curve is also called the zero or strip curve. The spot rate curve plots different maturities on the x-axis and corresponding spot rates on the y-axis. Exhibit 1 from the curriculum illustrates a spot curve.

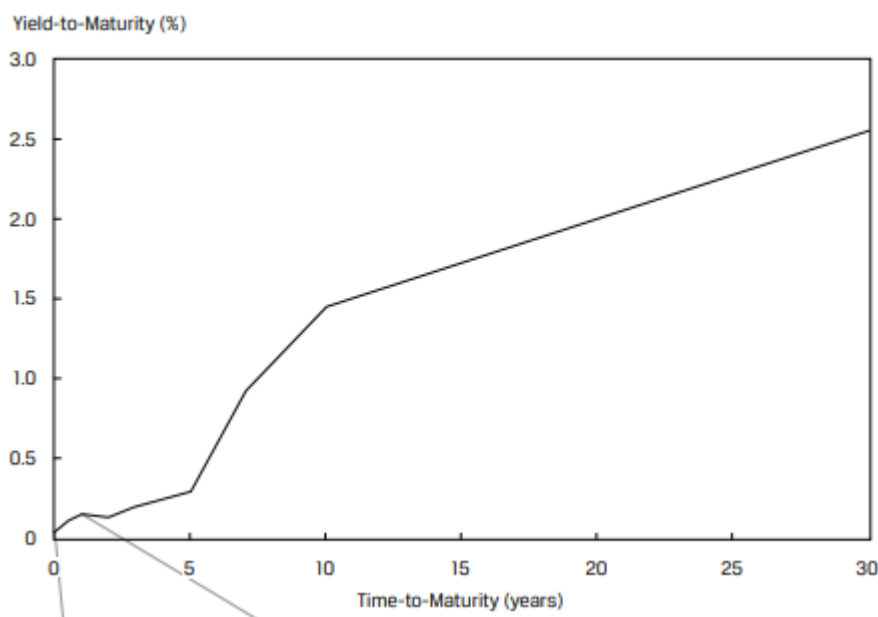


This spot curve is upward sloping, which means that longer-term government bonds have higher yields than the shorter-term bonds. This is the pattern typically observed under normal market conditions. However, unique circumstances can also result in a spot curve that is flat or downward sloping.

While a zero-coupon government bond spot curve is ideal for analysis, there are several practical issues to consider.

- Most actively traded government bonds make coupon payments (i.e. they are not zero-coupon).
- Older bonds tend to be less liquid than newly issued debt.

In practice, only the most recently issued actively traded government bonds are used to build a yield curve. Exhibit 2 illustrates a yield curve for a government that issues 2, 3, 5, 7, 10, and 30 year bonds. Straight line interpolation is used to complete the curve.



Bond Pricing Using Spot Rates

Spot rates are yields to maturity on zero-coupon bonds maturing at the date of each cash flow. Since zero-coupon bonds have no intermediate cash flows, the actual yield on a zero-coupon bond is used as the discount rate for a cash flow occurring at the same maturity date. Bond price (or value) determined using spot rates is sometimes referred to as the bond's 'no-arbitrage' value. If a bond's price differs from its no-arbitrage value, an arbitrage opportunity exists in the absence of transaction costs.

$$PV = \frac{PMT}{(1+Z_1)^1} + \frac{PMT}{(1+Z_2)^2} + \dots + \frac{PMT+FV}{(1+Z_N)^N}$$

where:

PMT = coupon payment

FV = par value of the bond

Z_1 = spot rate or yield of zero-coupon bond for period 1

Z_2 = spot rate or yield of zero-coupon bond for period 2

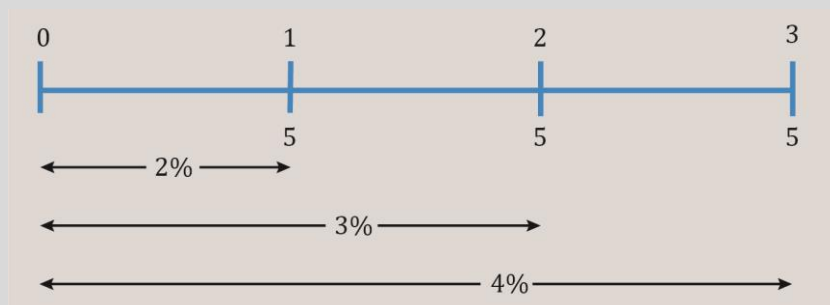
Z_N = spot rate or yield of zero-coupon bond for period N

Example

The one-year spot rate is 2%, the two-year spot rate is 3%, and the three-year spot rate is 4%. What is the price of a three-year bond that makes a 5% annual coupon payment?

Solution:

Think of the bond as a portfolio of three zero-coupon bonds with one, two and three-year maturities with yields of 2%, 3%, and 4% respectively. Draw a timeline for the cash flows.



$$\text{Bond's no-arbitrage value} = \frac{5}{1.02} + \frac{5}{1.03^2} + \frac{105}{1.04^3} = 102.96$$

Is there a single rate (yield) for the bond that equals the present value of cash flows to its purchase price? The YTM can be calculated as:

$$102.96 = \frac{5}{(1+r)^1} + \frac{5}{(1+r)^2} + \frac{105}{(1+r)^3}$$

Computing for r , we get 3.93%, which is the bond's yield to maturity.

3. Par and Forward Rates

Par Rates from Spot Rates

Spot rates are often used to determine par rates. A par rate is a yield-to-maturity that makes the present value of a bond's cash flows equal to par (100% of face value). Par rates derived for hypothetical government bonds with different maturity dates are commonly used for term structure analysis because they incorporate tax, trading, and other potential distortions associated with actual bonds priced at a discount or premium.

Given a sequence of spot rates $Z_1, Z_2, \dots, Z_N$, we can use the following equation to calculate a par rate by solving for PMT.

$$100 = \frac{\text{PMT}}{(1+Z_1)^1} + \frac{\text{PMT}}{(1+Z_2)^2} + \dots + \frac{\text{PMT}+100}{(1+Z_N)^N}$$

Notice that this equation is similar to the 'bond pricing using spot rates' equation, expect $PV = FV = 100$. Also, for a bond to trade at par, its coupon rate and yield-to-maturity must be equal. So by solving for PMT, we also solve for the YTM for the bond to trade at par. This single rate is divided by 100, to obtain the par rate per period.

The following example from the curriculum illustrates the calculation of par rates given spot rates.

Example:

Suppose the current one-, two-, three-, and four- year spot rates on government bonds are 5.263%, 5.616%, 6.359%, and 7.008%, respectively. These are effective annual rates. Calculate the par rates.

Solution:

The one-year par rate is 5.263%:

$$100 = \frac{PMT + 100}{(1.05263)^1}; PMT = 5.263$$

The two-year par rate is 5.606%.

$$100 = \frac{PMT}{1.05263^1} + \frac{PMT + 100}{(1.05616)^2}; PMT = 5.606$$

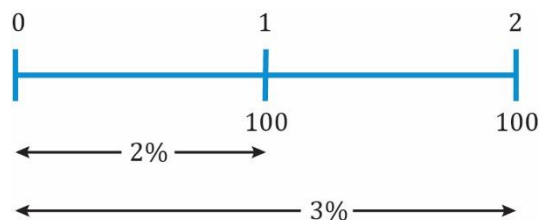
The three-year and four-year par rates are 6.306% and 6.899%, respectively.

$$100 = \frac{PMT}{1.05263^1} + \frac{PMT}{(1.05616)^2} + \frac{PMT + 100}{(1.06359)^3}; PMT = 6.306$$

$$100 = \frac{PMT}{1.05263^1} + \frac{PMT}{(1.05616)^2} + \frac{PMT}{(1.06359)^3} + \frac{PMT + 100}{(1.07008)^4}; PMT = 6.899$$

Forward Rates from Spot Rates

A forward rate is an interest rate in the future. The one-year interest rate after one year is an example of a forward rate. The 'implied forward rate' can be calculated using spot rates. As always, this is best understood with the help of an example. Consider a scenario where the one-year spot rate is 2%, and the two-year spot rate is 3%. This is illustrated below:



The forward rate f_2 is the rate over the second year. This can be calculated if we assume the following: \$1 invested for two years at the 2-year spot rate of 3% should give same result as \$1 invested for 1 year at the one-year spot rate of 2% and then again at the forward rate, f_2 . Mathematically, this can be expressed as: $(1 + 0.03)^2 = (1 + 0.02)(1 + f_2)$

Solving for f_2 , we get: $f_2 = 4.0098\%$.

The notation for a forward rate is expressed like this: 1y1y, 2y5y etc.; the first number refers to the length of the forward period in years from today and the second number refers to the tenor of the underlying bond. Thus, 1y1y is the rate for a 1-year loan one year from now; 2y5y is the rate for a 5-year loan, two years from now.

Spot Rates from Forward Rates and Bond Pricing with Forward Rates

Just like forward rates can be calculated from spot rates, spot rates can also be calculated from forward rates because they are interconnected.

Suppose the current sequence of one-year rates today and the one-year rate one, two, and three years from now are as follows:

Time Period	Forward Rate
0y1y	1.88%
1y1y	2.77%
2y1y	3.54%
3y1y	4.12%

Using the forward rates 0y1y and 1y1y, we can calculate the two-year spot rate as:

$$(1.0188)(1.0277) = (1 + z_2)^2; Z_2 = 2.32\%$$

The three-year spot rate can be calculated as:

$$(1.0188 \times 1.0277 \times 1.0354) = (1 + Z_3)^3; Z_3 = 2.73\%$$

The four-year spot rate can be calculated as:

$$(1.0188 \times 1.0277 \times 1.0354 \times 1.0412) = (1 + Z_4)^4; Z_4 = 0.030741$$

Bond Pricing with Forward Rates

Suppose that an analyst needs to value a four-year 3.75% annual coupon payment bond that has the same risks as the bonds used to obtain the forward curve. Using the implied spot rates, the value of the bond is 102.637 per 100 of par value.

$$\frac{3.75}{(1.0188)^1} + \frac{3.75}{(1.023240)^2} + \frac{3.75}{(1.027278)^3} + \frac{103.75}{(1.030741)^4} = 102.637$$

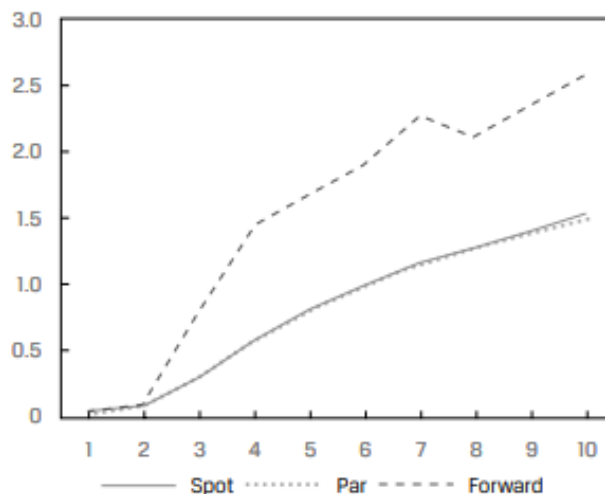
The bond also can be valued using the forward rates as follows:

$$\frac{3.75}{(1.0188)} + \frac{3.75}{(1.0188 \times 1.0277)} + \frac{3.75}{(1.0188 \times 1.0277 \times 1.0354)} + \frac{103.75}{(1.0188 \times 1.0277 \times 1.0354 \times 1.0412)} = 102.637$$

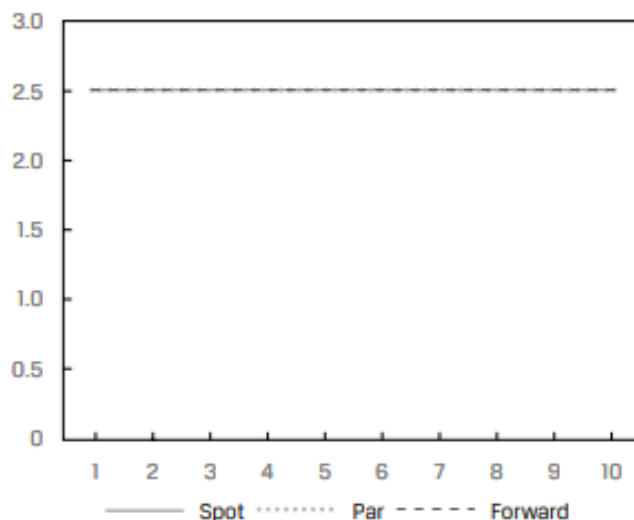
4. Spot, Par, And Forward Yield Curves and Interpreting Their Relationship

Since par and forward rates can be derived from spot rates, the shape of the spot curve is closely related to the shape of the par and forward curves.

When the spot curve is upward sloping, the par rates will be near, but below the spot rates and the forward rates will be greater than the spot rates. This is illustrated in Exhibit 6 from the curriculum.



When the spot curve is flat, the par rates and forward rates will be equal to the spot rates. This is illustrated in Exhibit 7 from the curriculum.



When the spot curve is downward sloping, the par rates will be close to but greater than the spot rates and the forward rates will be lower than the spot rates. This is illustrated in Exhibit 8 from the curriculum.

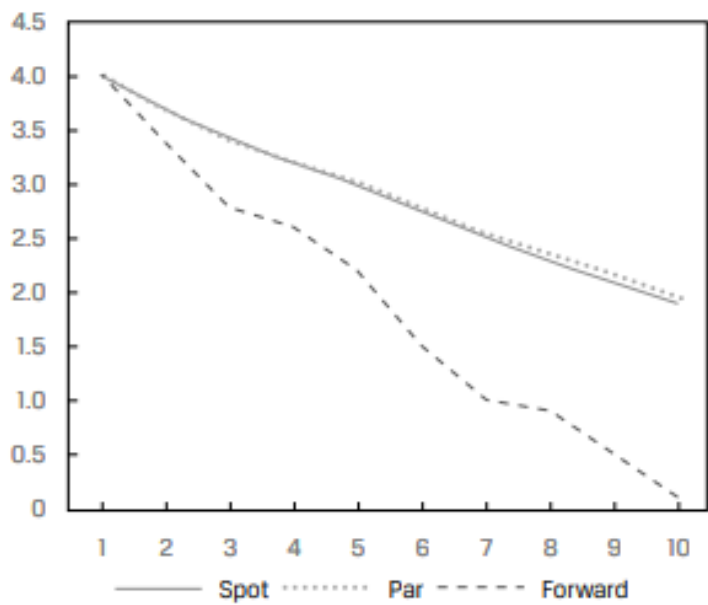


Exhibit 9 from the curriculum summarizes these relationships.

Spot Curve Shape	Par Curve	Forward Curve
Upward Sloping	Below spot curve	Above spot curve
Flat	Equal to spot curve	Equal to spot curve
Downward Sloping (Inverted)	Above spot curve	Below spot curve

Summary

LO: Define spot rates and the spot curve, and calculate the price of a bond using spot rates.

Spot rates are yields to maturity on zero-coupon bonds maturing at the date of each cash flow. Since zero-coupon bonds have no intermediate cash flows, the actual yield on a zero-coupon bond is used as the discount rate for a cash flow occurring at the same maturity date.

$$PV = \frac{PMT}{(1 + Z_1)^1} + \frac{PMT}{(1 + Z_2)^2} + \dots + \frac{PMT + FV}{(1 + Z_N)^N}$$

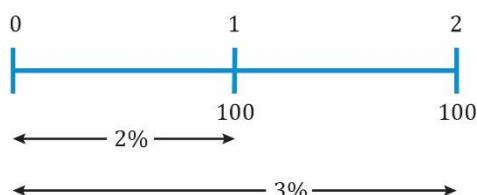
where: Z_N are the spot rates.

LO: Define par and forward rates, and calculate par rates, forward rates from spot rates, spot rates from forward rates, and the price of a bond using forward rates.

A par rate is the market discount rate for a specific maturity that would result in a bond priced at par. A par rate is derived from the spot rates up to and including the maturity date.

A forward rate is an interest rate in the future. The one-year interest rate after one year is an example of a forward rate. The implied forward rate can be calculated using spot rates.

Consider a scenario where the one-year spot rate is 2%, and the two-year spot rate is 3%. This is illustrated below:



The forward rate f_2 is the rate over the second year. This can be calculated if we assume the following: \$1 invested for two years at the 2-year spot rate of 3% should give the same result as \$1 invested for 1 year at the one-year spot rate of 2% and then again at the forward rate, f_2 .

Mathematically, this can be expressed as:

$$(1 + 0.03)^2 = (1 + 0.02) (1 + f_2)$$

Solving for f_2 , we get: $f_2 = 4.0098\%$.

LO: Compare the spot curve, par curve, and forward curve.

Spot Curve Shape	Par Curve	Forward Curve
Upward Sloping	Below spot curve	Above spot curve
Flat	Equal to spot curve	Equal to spot curve
Downward Sloping (Inverted)	Above spot curve	Below spot curve

LM10 Interest Rate Risk and Return IFT Notes

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Version 1.0

1. Introduction

This learning module covers:

- Sources of return from investing in a fixed rate bond
- How investors are exposed to different interest rate risk if the same bond is held for different time periods.
- Macaulay duration

2. Sources of Return from Investing in a Fixed-Rate Bond

The total return is the future value of reinvested coupon interest payments and the sale price (or redemption of principal if the bond is held to maturity). The horizon yield (or holding period rate of return) is the internal rate of return between the total return and the purchase price of the bond.

Total return on a bond = reinvested coupon interest payments + sale/redemption of principal at maturity

A bond investor has three sources of return:

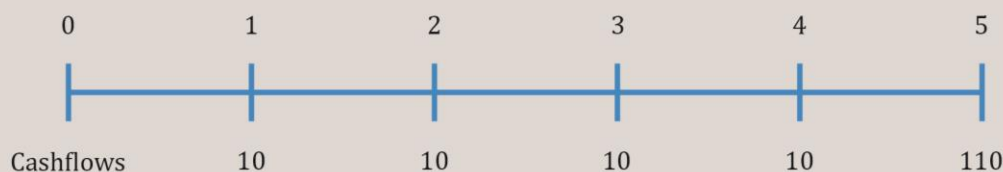
- Receiving the full coupon and principal payments on the scheduled dates.
- Reinvesting the interest payments. This is also known as interest-on-interest.
- Potential capital gain or loss on sale of the bond, if the bond is sold before maturity date.

Now, we will look at a series of examples that demonstrate the effect on two investors' realized rate of returns when one of these variable changes: time horizon, interest rate at which the coupons are reinvested, and market discount rates at the time of purchase and at the time of sale.

In Examples 1 and 2, we look at the realized rate of return for two investors with different time horizons for the same bond.

Example 1: Calculating the total return on a bond that is held until maturity

Investor 1: A "buy-and-hold" investor purchases a 5-year, 10% annual coupon payment bond at 92.79 per 100 of par value and holds it until maturity. Calculate the total return on the bond.



Solution:

$$92.79 = \frac{10}{(1+r)^1} + \frac{10}{(1+r)^2} + \frac{10}{(1+r)^3} + \frac{10}{(1+r)^4} + \frac{110}{(1+r)^5}$$

Use the following keystrokes to calculate the bond's yield to maturity:

N = 5; PV = -92.79; PMT = 10; FV = 100; CPT I/Y;

Hence, $r = 12\%$.

This is the yield to maturity at the time of purchase. However, this holds good only if all of the following three conditions are true:

- The bond is held to maturity.
- The coupon and final principal payments are made on time (no default or delay).
- The coupon payments are reinvested at the same rate of interest.

To calculate the total return on the bond, we first need to calculate the interest earned when coupon payments are reinvested.

Coupon reinvestment:

- The investor receives 5 coupon payments of 10 (per 100 of par value) for a total of 50, plus the redemption of principal (100) at maturity. The investor has the opportunity to reinvest the cash flows. If the coupon payments are reinvested at 12% (i.e., yield to maturity), the future value of the coupons on the bond's maturity date is 63.53 per 100 of par value.

$$[10 \times (1.12)^4] + [10 \times (1.12)^3] + [10 \times (1.12)^2] + [10 \times (1.12)^1] + 10 = 63.53$$

- The first coupon payment of 10 is reinvested at 12% for 4 years until maturity, the second is reinvested for 3 years, and so on. The future value of the annuity is obtained easily using a financial calculator: N = 5; PV = 0; PMT = 10; I/Y = 12; CPT FV. FV = 63.53
- The amount in excess of the coupons, 13.53 (= 63.53 - 50), is the 'interest-on-interest' gain from compounding.
- The investor's total return is 163.53, the sum of the reinvested coupons (63.53) and the redemption of principal at maturity (100). The realized rate of return is 12%.

$$92.79 = \frac{163.53}{(1+r)^5}, r = 12\%$$

Example 2: Calculating the total return on a bond that is sold before maturity

Now let us consider investor 2 who buys the same 5-year, 10% annual coupon payment bond but sells the bond after three years. Assuming that the coupon payments are reinvested at 12% for three years, calculate the total return on the bond.

Solution:

The future value of the reinvested coupons is 33.74 per 100 of par value.

$$[10 \times (1.12)^2] + [10 \times (1.12)^1] + 10 = 33.74$$

The interest-on-interest gain from compounding is 3.74 (= 33.74 - 30). After three years, when the bond is sold, it has two years remaining until maturity. If the yield to maturity remains 12%, then the sale price of the bond is 96.62.

$$\frac{10}{(1.12)^1} + \frac{110}{(1.12)^2} = 96.62$$

The total return is 130.36 (= 33.74 + 96.62) and the realized rate of return is 12%.

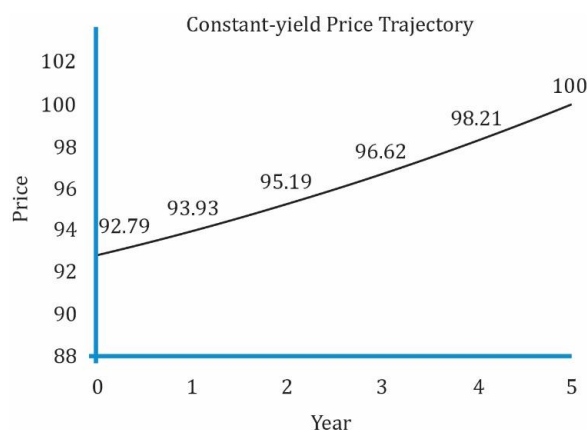
$$92.79 = \frac{130.36}{(1+r)^3}, r = 12\%$$

This 'r' is called the horizon yield, the internal rate of return between the total return and the purchase price of the bond. Horizon yield is equal to the original yield to maturity if:

- Coupon payments are reinvested at the same yield to maturity calculated at the time of purchase of the bond. In our case, it is 12% as calculated in Example 1.
- There are no capital gains or losses when the bond is sold. It is sold at a price on the constant-yield price trajectory. We arrive at the price 96.62 by taking 12% as the constant yield for the remaining two years. If the yield is more than 12%, then losses occur. Similarly, if the yield is less than 12%, then capital gains occur. This concept is elaborated below:

Constant-yield price trajectory for a 5-year, 10% annual payment bond

The price of the bond at different time periods for a yield of 12% is plotted in the graph below:



The calculations for determining the price are shown below:

Year	Price	Calculation
0	92.79	N = 5 ; I/Y = 12 ; PMT = 10 ; FV = 100 ; CPT PV
1	93.93	N = 4 ; I/Y = 12 ; PMT = 10 ; FV = 100 ; CPT PV
2	95.19	N = 3 ; I/Y = 12 ; PMT = 10 ; FV = 100 ; CPT PV
3	96.62	N = 2 ; I/Y = 12 ; PMT = 10 ; FV = 100 ; CPT PV
4	98.21	N = 1 ; I/Y = 12 ; PMT = 10 ; FV = 100 ; CPT PV
5	100	

Observation:

There will be a capital gain if the bond is sold at a price above the trajectory at any point in time during the bond's life. This will happen if the yield is below 12%. Remember a bond's price and interest rates are inversely related. Similarly, there will be a capital loss if the bond is sold at a price below the trajectory at any point in time during the bond's life. This will happen if the yield is above 12%. **Any point on the trajectory represents the carrying value of the bond at that time.**

Next, we analyze what happens to the investors' realized rate of return if interest rates go up after the bond is purchased.

Example 3: Calculating an investor's realized return when interest rates go up and bond is held until maturity

The buy-and-hold investor purchases the same 5-year, 10% annual payment bond at 92.79. After the bond is purchased and before the first coupon is received, interest rates go up to 15%.

Calculate the investor's realized rate of return.

Solution:

Use the following keystrokes to calculate the future value of the reinvested coupons at 15% for 5 years:

$$[10 \times (1.15)^4] + [10 \times (1.15)^3] + [10 \times (1.15)^2] + [10 \times (1.15)^1] + 10 = 67.42$$

The total return is 167.42 (= 67.42 + 100).

$$92.79 = \frac{167.42}{(1 + r)^5} ; r = 12.53\%$$

The investor's realized rate of return is 12.53%.

Observation:

Compared to the investor in Example 1 (12%), the realized return of this investor is higher because the coupons are reinvested at a higher rate. There is no capital gain or loss because the bond is held to maturity and the principal of 100 is redeemed.

Example 4: Calculating an investor's realized return when interest rates go up and bond is sold before maturity

The second investor buys the same 5-year, 10% annual payment bond at 92.79 and sells it in three years. After the bond is purchased, interest rates go up to 15%. Calculate the investor's realized gain.

Solution:

The future value of the reinvested coupons at 15% after three years is:

$$[10 \times (1.15)^2] + [10 \times (1.15)^1] + 10 = 34.73$$

$$N = 3; PV = 0; PMT = 10; I/Y = 15; CPT FV; FV = -34.73$$

The sale price of the bond after three years is 91.87.

$$\frac{10}{(1.15)^1} + \frac{110}{(1.15)^2} = 91.87$$

The total return is 126.60 (= 34.73 + 91.87), resulting in a realized three-year horizon yield of 10.91%.

$$92.79 = \frac{126.60}{(1 + r)^3}, r = 10.91\%$$

N = 3; PV = 92.79; FV = 126.60; CPT I/Y

Hence, r = 10.91%.

Observation:

Compared to the investor in Example 2 with a similar time horizon (12%), the realized return to this investor is lower at 10.91% because there is a capital loss. Even though the coupons are reinvested at a higher rate, the capital loss is greater than the gain from reinvesting coupons.

Increase in the value of reinvested coupons = 34.73 – 33.74 = 0.99

Capital loss = 91.87 - 96.62 = -4.75. Capital gain or loss is always calculated relative to the carrying value at that point in time.

Next, we look at the effect of realized returns when interest rates go down after the bond is issued.

Example 5: Calculating an investor's realized return when interest rates go down

The buy-and-hold investor purchases the same 5-year, 10% annual payment bond at 92.79 and holds the security until it matures. After the bond is purchased, and before the first coupon is received, interest rates go down to 8%. Calculate the investor's realized return.

Solution:

The future value of reinvesting the coupon payments at 8% for 5 years is 58.67 per 100 of par value.

$$[10 * (1.08)^4] + [10 * (1.08)^3] + [10 * (1.08)^2] + [10 * (1.08)^1] + 10 = 58.67$$

The total return is 158.67 (= 58.67 + 100), the sum of the future value of reinvested coupons and the redemption of par value.

$$92.79 = \frac{158.67}{(1 + r)^5}, r = 11.33\%$$

The investor's realized rate of return is 11.33%.

Observation:

The realized return is lower than that in Example 1 (12%) because the coupons are reinvested at a lower rate of return. Since the bond is held to maturity, there is no capital gain or loss.

Decrease in the value of reinvested coupons = $58.67 - 63.53 = -4.86$

Example 6: Calculating an investor's realized return when interest rates go down

The second investor buys the same 5-year, 10% annual payment bond at 92.79 and sells it after three years. After the bond is purchased, interest rates go down to 8%. Calculate the investor's realized return.

Solution:

The future value of the reinvested coupons at 8% after three years is:

$$[10 * (1.08)^2] + [10 * (1.08)^1] + 10 = 32.46$$

This reduction in the future value of coupon reinvestments is offset by the higher sale price of the bond, which is 103.57 per 100 of par value.

$$\frac{10}{(1.08)^1} + \frac{110}{(1.08)^2} = 103.57$$

The total return is 136.03 (= 32.46 + 103.57), resulting in a realized three-year horizon yield of 13.60%.

$$92.79 = \frac{136.03}{(1 + r)^3}, \quad r = 13.60\%$$

Observation:

The realized return is greater than that of the investors in Examples 2 and 4 with a similar time horizon. It is primarily due to the capital gains.

$$\text{Capital gain} = 103.57 - 96.62 = 6.95$$

$$\text{Decrease in the value of reinvested coupons} = 32.46 - 33.74 = -1.28$$

As you can see, the capital gain is far greater than the decrease in the value of reinvested coupons.

Interest rate risk affects the realized rate of return for any bond investor in two ways: coupon reinvestment risk and market price risk. But what is interesting is that these are offsetting types of risk. Two investors with different time horizons will have different exposures to interest rate risk. From the examples above, let us sum up what happens when interest rates go up or down:

When interest rates go up or down:

- Reinvestment income is directly proportional to interest rate movements. The value of reinvested coupons increases when the interest rate goes up.
- Bond price is inversely proportional to interest rate movements. Bond price decreases when the interest rate goes up.

When does coupon reinvestment risk matter?

- Coupon reinvestment risk matters when an investor has a long-term horizon. If the investor buys a bond and sells it before the first coupon payment, then this risk is irrelevant. But a buy-and-hold investor as in Examples 1, 3, and 5 has only coupon reinvestment risk.

When does market price risk matter?

- Market price risk matters when an investor has a short-term horizon relative to the time to maturity. If the investor buys a bond and sells it before the first coupon payment, then he is exposed to market price risk only. But a buy-and-hold investor such as those in Examples 1, 3, and 5 has only coupon reinvestment risk, and no market price risk.

Example 7: Calculating the purchase price of the bond for various YTM

An investor buys a five-year, 10% annual coupon payment bond priced to yield 8%. The investor plans to sell the bond in three years once the third coupon payment is received. Calculate the purchase price for the bond and the horizon yield assuming that the coupon reinvestment rate after the bond purchase and the yield to maturity at the time of sale are (1) 7%, (2) 8%, and (3) 9%.

Solution:

The purchase price is:

$$\frac{10}{(1.08)^1} + \frac{10}{(1.08)^2} + \frac{10}{(1.08)^3} + \frac{10}{(1.08)^4} + \frac{110}{(1.08)^5} = 107.99$$

N = 5; PMT = 10; FV = 100; I/Y = 8; CPT PV, PV = -107.99.

Solution to 1:

With YTM at sale = 7%:

N = 3; PV = 0; PMT = 10; I/Y = 7; CPT FV, FV = -32.15.

The future value of the reinvested coupons after three years is 32.15.

$$\frac{10}{(1.07)^1} + \frac{110}{(1.07)^2} = 105.42.$$

N = 2; I/Y = 7; PMT = 10; FV = 100; CPT PV, PV = 105.42.

The sale price of the bond after three years is 105.42.

The total return after three years is 137.57 (= 32.15 + 105.42).

The realized return calculated using the keystrokes: N = 3; PMT = 0; PV = 107.99; FV = 137.57; CPT I/Y,
r = 8.4%.

$$107.99 = \frac{137.57}{(1+r)^3}, r = 8.40\%.$$

If interest rates go down from 8% to 7%, then the realized rate of return over the three-year investment horizon is 8.40%, higher than the original yield to maturity of 8%.

Solution to 2:

With YTM at sale = 8%, the future value of the reinvested coupons after three years is:

$N = 3$; $PV = 0$; $PMT = 10$; $I/Y = 8$; CPT FV, $FV = 32.46$.

The sale price of the bond is calculated as:

$N = 2$, $I/Y = 8$, $PMT = 10$, $FV = 100$; CPT PV

The sale price of the bond after three years is 103.57.

The total return after three years is 136.03 ($= 32.46 + 103.57$).

The realized return calculated using the keystrokes: $N = 3$; $PMT = 0$; $PV = -107.99$; $FV = 136.03$; CPT I/Y .

$r = 8\%$.

$$107.99 = \frac{136.03}{(1 + r)^3}, r = 8.00\%$$

If interest rates remain 8% for reinvested coupons and for the required yield on the bond, the realized rate of return over the three-year investment horizon is equal to the yield to maturity of 8%.

Solution to 3:

With YTM at sale = 9%: the future value of the reinvested coupons after three years is:

$N = 3$; $PV = 0$; $PMT = 10$; $I/Y = 9$; CPT FV, $FV = 32.78$.

The sale price of the bond is calculated as:

$N = 2$; $I/Y = 9$; $PMT = 10$; $FV = 100$; CPT PV

The sale price of the bond after three years is 101.76.

The total return after three years is 134.54 ($= 32.78 + 101.76$).

The realized return calculated using the keystrokes: $N = 3$; $PV = -107.99$; $FV = 134.54$; CPT I/Y

$$107.99 = \frac{134.54}{(1+r)^3}, r = 7.60\%;$$

If interest rates go up from 8% to 9%, the realized rate of return over the three-year investment horizon is 7.60%; lower than the yield to maturity of 8%.

3. Investment Horizon and Interest Rate Risk

The impact of a sudden change in yield on the price of a bond is of particular concern to short-term investors (price risk). Long-term investors will also be concerned about the impact of a change in yield on the reinvestment income (reinvestment risk). An investor who plans to hold the bond to maturity will only be concerned about reinvestment risk.

Consider another 10-year, 8% annual coupon bond priced at 85.5 and YTM of 10.4%. If the investment horizon is 10 years, the only concern is reinvestment risk. Consider a scenario where interest rates go down. In this case, reinvestment income goes down. When the price of the bond goes up, it does not matter to the investor because at maturity he will simply receive par value. When interest rates go up, the reinvestment income goes up.

If the investment horizon is 4 years, then the major concern is price risk. In this case, the price effect dominates relative to the gain/loss from reinvestment of coupons. If the investment horizon is 7 years, the reinvestment risk and price risk offset each other. For this particular bond the Macaulay duration is 7 years.

Macaulay duration (explained in the next section) is the investment horizon for which coupon reinvestment risk and market price risk offset each other. The assumption is a one-time parallel shift in the yield curve.

Note: Investment horizon is different from the bond's maturity. In this case, the maturity is 10 years while the horizon is 7 years.

The **duration gap** of a bond is defined as the Macaulay duration – investment horizon.

Duration gap = Macaulay duration – Investment horizon

- If Macaulay duration < investment horizon, the duration gap is negative: coupon reinvestment risk dominates.
- If investment horizon = Macaulay duration, the duration gap is zero: coupon reinvestment risk offsets market price risk.
- If Macaulay duration > investment horizon, the duration gap is positive: market price risk dominates.

Example 8: Calculating duration gap and assessing interest rate risk

An investor plans to retire in 8 years. As part of the retirement portfolio, the investor buys a newly issued, 10-year, 6% annual coupon payment bond. The bond is purchased at par value, so its yield to maturity is 6.00% stated as an effective annual rate. The bond's Macaulay duration is 7.8016.

1. Calculate the duration gap at the time of purchase.
2. Does this bond at purchase entail the risk of higher or lower interest rates? Interest rate risk here means an immediate, one-time, parallel yield curve shift.

Solution to 1:

Given an investment horizon of 8 years, the duration gap for this bond at purchase is negative: $7.8016 - 8 = -0.1984$

Solution to 2:

A negative duration gap entails the risk of lower interest rates. To be precise, the risk is an immediate, one-time, parallel, downward yield curve shift because the coupon reinvestment risk dominates market price risk. The loss from reinvesting coupons at a rate lower than 6% is larger than the gain from selling the bond at a price above the constant-yield price trajectory.

4. Macaulay Duration

Macaulay duration is a weighted average of the time to receipt of the bond's promised payments, where the weights are the shares of the full price that correspond to each of the bond's promised future payments.

Let us consider a 10-year, 8% annual payment bond. To determine the Macaulay duration, we calculate the present value of each cash flow, multiply by weight and add, as shown in the below exhibit.

Macaulay Duration of a 10 – Year, 8% Annual Payment Bond				
Period	Cash flow	Present Value	Weight	Period x Weight
1	8	7.246377	0.08475	0.0847
2	8	6.563747	0.07677	0.1535
3	8	5.945423	0.06953	0.2086
4	8	5.385347	0.06298	0.2519
5	8	4.878032	0.05705	0.2853
6	8	4.418507	0.05168	0.3101
7	8	4.002271	0.04681	0.3277
8	8	3.625245	0.04240	0.3392
9	8	3.283737	0.03840	0.3456
10	108	40.154389	0.46963	4.6963
		85.503075	1.00000	7.0029

We can also use the following formula to calculate Macaulay Duration:

$$\text{MacDur} = \left(\frac{1 + r}{r} - \frac{1 + r + [N * (c - r)]}{c * [(1 + r)^N - 1] + r} \right) - \frac{t}{T}$$

where:

r = yield to maturity

t = number of days from the last coupon payment date to the settlement date

T = number of days in the coupon period

c = coupon rate per period

N = number of periods to maturity

Instructor's Note:

Understanding how the Macaulay duration works is more important than memorizing the formula.

Summary

LO: Calculate and interpret the sources of return from investing in a fixed-rate bond.

A bond investor has three sources of return:

- Receiving the full coupon and principal payments on the scheduled dates.
- Reinvesting the interest payments. This is also known as interest-on-interest.
- Potential capital gain or loss on sale of the bond, if the bond is sold before maturity date.

Interest rate risk affects the realized rate of return for any bond investor in two ways: coupon reinvestment risk and market price risk.

- Reinvestment income is directly proportional to interest rate movements. The value of reinvested coupons increases when the interest rate goes up.
- Bond price is inversely proportional to interest rate movements. Bond price decreases when the interest rate goes up.
- Coupon reinvestment risk matters when an investor has a long time horizon. If the investor buys a bond and sells it before the first coupon payment, then this risk is irrelevant.
- Market price risk matters when an investor has a short time horizon.

LO: Describe the relationships among a bond's holding period return, its Macaulay duration, and the investment horizon.

The Macaulay duration can be interpreted as the investment horizon for which coupon reinvestment risk and market price risk offset each other.

Duration gap = Macaulay duration – Investment horizon

- If the investment horizon is greater than the Macaulay duration of the bond, coupon reinvestment risk dominates price risk. The investor's risk is that interest rates will fall. The duration gap is negative.
- If the investment horizon is equal to the Macaulay duration of the bond, coupon reinvestment risk offsets price risk. The duration gap is zero.
- If the investment horizon is less than the Macaulay duration of the bond, price risk dominates coupon reinvestment risk. The investor's risk is that interest rates will rise. The duration gap is positive.

LO: Define, calculate, and interpret Macaulay duration.

Macaulay duration is the weighted average of the time to receipt of coupon interest and principal payments.

LM11 Yield-Based Bond Duration Measures and Properties

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Version 1.0

1. Introduction

This learning module covers:

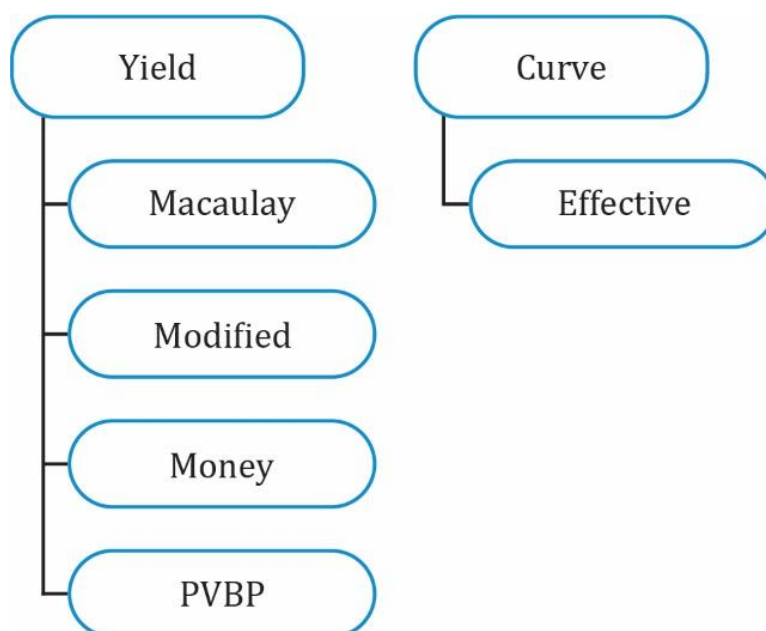
- Measures of interest rate risk: Modified duration, money duration, and the price value of a basis point (PVBP)
- How a bond's maturity, coupon, and yield levels affect its interest rate risk

2. Modified Duration

The duration of a bond measures the sensitivity of the bond's full price (including accrued interest) to changes in interest rates. In other words, duration indicates the percentage change in the price of a bond for a 1% change in interest rates. The higher the duration, the more sensitive the bond is to change in interest rates.

There are two categories of duration: yield duration and curve duration.

- Yield duration measures a bond's price sensitivity to changes in its own yield-to-maturity and assumes underlying cash flows are certain.
- Curve duration measures a bond's price sensitivity to changes in a benchmark yield curve and accounts for the possibility that a bond may default.



As indicated in the diagram above, the main yield duration measures are: Macaulay duration, modified duration, money duration, and the price value of the basis point (PVBP). Curve duration measures are covered in later learning modules.

Macaulay duration is a weighted average of the time to receipt of the bond's promised payments, where the weights are the shares of the full price that correspond to each of the bond's promised future payments.

Modified duration provides an estimate of the percentage price change for a bond given a change in its yield to maturity. It represents a simple adjustment to Macaulay duration as shown in the equation below:

$$\text{Modified duration} = \frac{\text{Macaulay duration}}{1 + r}$$

where: r is the yield per period.

Therefore, percentage price change for a bond given a change in its YTM can be calculated as:

$$\% \Delta PV^{\text{FULL}} \approx -\text{AnnModDur} \times \Delta \text{Yield}$$

The AnnModDur term is the *annual* modified duration, and the ΔYield term is the change in the *annual* yield to maturity. The $\approx$ sign indicates that this calculation is estimation. The minus sign indicates that bond prices and yields to maturity move inversely.

Example: Calculating the modified duration of a bond

A 2-year, annual payment, \$100 bond has a Macaulay duration of 1.87 years. The YTM is 5%. Calculate the modified duration of the bond.

Solution:

$$\text{Modified duration} = 1.87 / (1 + 0.05) = 1.78$$

The percentage change in the price of the bond for a 1% increase in YTM will be:
 $-1.78 \times 0.01 \times 100 = -1.78\%$.

Approximate Modified Duration

Modified duration is calculated if the Macaulay duration is known. But there is another way of calculating an approximate value of modified duration: estimate the slope of the line tangent to the price-yield curve. This can be done by using the equation below:

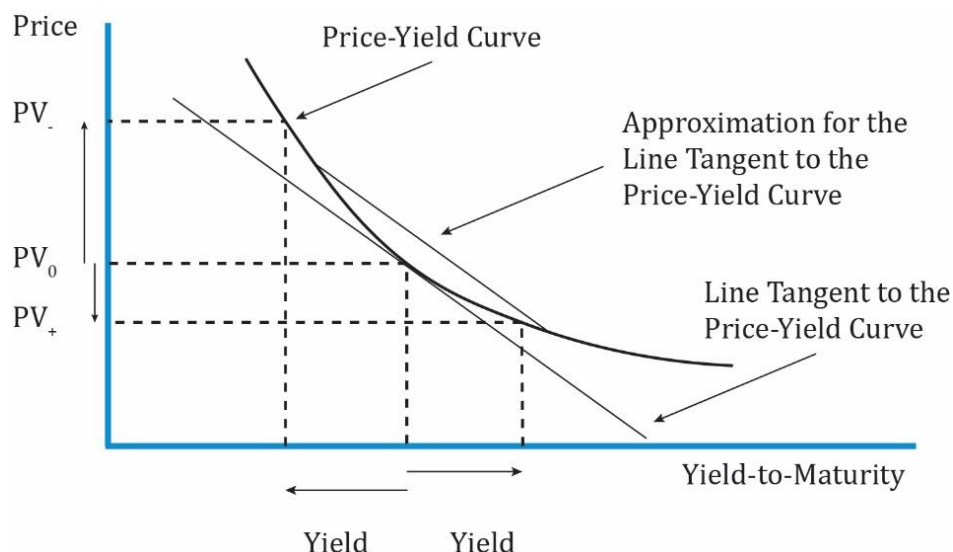
$$\text{Approximate Modified Duration} = \frac{(PV_-) - (PV_+)}{2 * \Delta \text{yield} * PV_0}$$

where:

PV_- = price of the bond when yield is decreased;

PV_0 = initial price of the bond

PV_+ = price of the bond when yield is increased



Source: CFA Program Curriculum, Understanding Fixed-Income Risk and Return

Interpretation of the diagram:

- PV_0 denotes the original price of the bond.
- PV_+ denotes the price of the bond when YTM is increased and PV_- denotes the bond price when YTM is decreased.
- Change in yield (up and down) is denoted by Δ yield.
- For slope calculation: vertical distance = $PV_- - PV_+$ and horizontal distance = $2 \times \Delta$ yield.

How to calculate approximate modified duration (or estimate the slope of the price-yield curve):

- The yield to maturity is changed (increased/decreased) by the same amount.
- Calculate the bond price for a decrease in yield (PV_-).
- Calculate the bond price for an increase in yield (PV_+).
- Use these values to calculate the approximate modified duration.

Once the approximate modified duration is known, the approximate Macaulay duration can be calculated using the formula below:

$$\text{Approximate Macaulay Duration} = \text{Approximate Modified Duration} \times (1 + r)$$

Example: Calculating the approximate modified duration and approximate Macaulay duration

Assume that the 6% U.S. Treasury bond matures on 15 August 2017 is priced to yield 10% for settlement on 15 November 2014. Coupons are paid semiannually on 15 February and 15 August. The yield to maturity is stated on a street-convention semiannual bond basis. This settlement date is 92 days into a 184-day coupon period, using the actual/actual day-count convention. Compute the approximate modified duration and the approximate Macaulay duration for this Treasury bond assuming a 50bps change in the yield to maturity.

Solution:

The yield to maturity per semiannual period is 5% ($=10/2$). The coupon payment per period is 3% ($= 6/2$). When the bond is purchased, there are 3 years (6 semiannual periods) to maturity. The fraction of the period that has passed is 0.5 ($=92/184$).

The full price (including accrued interest) at an YTM of 5% is 92.07 per 100 of par value.

$$PV_0 = \left[\frac{3}{(1.05)^1} + \frac{3}{(1.05)^2} + \frac{3}{(1.05)^3} + \frac{3}{(1.05)^4} + \frac{3}{(1.05)^5} + \frac{103}{(1.05)^6} \right] * (1.05)^{\left(\frac{92}{184}\right)}$$

$$\Rightarrow 89.85 \times (1.05)^{0.5} = 92.07$$

Increase the yield to maturity from 10% to 10.5% - therefore, from 5% to 5.25% per semiannual period, and the price becomes 90.97 per 100 of par value.

$$PV_+ = \left[\frac{3}{(1.0525)^1} + \frac{3}{(1.0525)^2} + \frac{3}{(1.0525)^3} + \frac{3}{(1.0525)^4} + \frac{3}{(1.0525)^5} + \frac{103}{(1.0525)^6} \right]$$

$$* (1.0525)^{\left(\frac{92}{184}\right)}$$

$$\Rightarrow 88.67 \times (1.0525)^{0.5} = 90.97$$

Decrease the yield to maturity from 10% to 9.5% - therefore, from 5% to 4.75% per semiannual period, and the price becomes 93.19 per 100 of par value.

$$PV_- = \left[\frac{3}{(1.0475)^1} + \frac{3}{(1.0475)^2} + \frac{3}{(1.0475)^3} + \frac{3}{(1.0475)^4} + \frac{3}{(1.0475)^5} + \frac{103}{(1.0475)^6} \right]$$

$$* (1.0475)^{\left(\frac{92}{184}\right)}$$

$$\Rightarrow 91.05 \times (1.0475)^{0.5} = 93.19$$

The approximate annualized modified duration for the Treasury bond is 2.41

$$\text{ApproxModDur} = \frac{93.19 - 90.97}{2 * 0.005 * 92.07} = 2.41$$

The approximate annualized Macaulay Duration is 2.53

$$\text{ApproxMacDur} = 2.41 * 1.05 = 2.53$$

Therefore, from these statistics, the investor knows that the weighted average time to receipt of interest and principal payments is 2.53 years (the Macaulay Duration) and that the estimated loss in the bond's market value is 2.41% (the Modified Duration) if the market discount rate were to suddenly go up by 1.0%.

3. Money Duration and Price Value of a Basis Point

Money Duration

The money duration of a bond is a measure of the price change in units of the currency in which the bond is denominated, given a change in annual yield to maturity.

$$\text{Money Duration} = \text{AnnModDur} \times PV_{\text{FULL}}$$

$$\Delta PV_{\text{FULL}} \approx -\text{MoneyDur} \times \Delta \text{yield}$$

Consider a bond with a par value of \$100 million. The current yield to maturity (YTM) is 5% and the full price is \$102 per \$100 par value. The annual modified duration of this bond is 3. The money duration can be calculated as the annual modified duration (3) multiplied by the full price (\$102 million): $3 \times \$102 \text{ million} = \306 million . If the YTM rises by 1% (100 bps) from 5% to 6%, the decrease in value will be approximately $\$306 \text{ million} \times 1\% = \3.06 million . If the YTM rises by 0.1% (10 bps), the decrease in value will be $\$306 \text{ million} \times 0.1\% = \0.306 million .

Example: Calculating money duration of a bond

A life insurance company holds a USD 1 million (par value) position in a bond that has a modified duration of 6.38. The full price of the bond is 102.32 per 100 of face value.

1. Calculate the money duration for the bond.
2. Using the money duration, estimate the loss for each 10 bps increase in the yield to maturity.

Solution:

1. First calculate the full price of the bond: $\$1,000,000 \times 102.32\% = \$1,023,200$. The money duration for the bond is: $6.38 \times \$1,023,200 = \$6,528,000$.
2. 10 bps corresponds to $0.10\% = 0.0010$. For each 10 bps increase in the yield to maturity, the loss is estimated to be: $\$6,528,000 \times 0.0010 = \$6,528.02$.

Price Value of a Basis Point (PVBP)

An important measure which is related to money duration is the price value of a basis point (PVBP). The PVBP is an estimate of the change in the full price given a 1 bp change in the yield to maturity. The formal equation is given below.

$$\text{PVBP} = \frac{PV_- - PV_+}{2}$$

where PV_- and PV_+ are full prices calculated by decreasing and increasing the YTM by 1 basis point.

A quick way of calculating the price value of a basis point is to take the money duration and multiply by 0.0001. For example, if the money duration of a portfolio is \$200,000 the price value of a basis point is $\$200,000 \times 0.0001 = \20 . (1 bp = 0.01% = 0.0001)

Example: Calculating PVBP for a bond

Consider a \$100, five-year bond that pays coupons at a rate of 10% semi-annually. The YTM is 10% and it is priced at par. The modified duration of the bond is 3.81. Calculate the PVBP for the bond.

Solution:

Money duration = $\$100 \times 3.81 = \381.00

PVBP = $\$381 \times 0.0001 = \0.0381

Exhibit 3 from the curriculum provides a summary of the various yield duration statistics.

Measure	Calculation	Use
Macaulay duration	Average time to receipt of promised cash flows, weighted by shares of the full price corresponding to each promised future cash flow	Holding period that would balance reinvestment and price risks for an investor
Modified duration	First derivative of price with respect to yield; Macaulay duration divided by $1 + \text{yield per period}$	Estimate the percentage price change for a bond given a change in its yield-to-maturity
Money duration	Modified duration multiplied by full price of bond or bond position	Estimate price change in bond investment for a given yield change
Price value of a basis point	Difference in price of a 1 bp yield decrease and a 1 bp yield increase, divided by 2	Estimate of the change in the bond price given a 1 bp change in the yield-to-maturity

Yield Duration of Zero-Coupon and Perpetual Bonds

Since zero-coupon bonds have a single payment, the face value at maturity the present weighted-average time to receipt of cash flows is the same as the time-to-maturity. This is because that single cash flow has a present value weight of 1. Therefore, the Macaulay duration of a zero-coupon bond is its time to maturity. Its modified duration is time-to-maturity divided by $(1 + r)$.

A perpetual bond is one that does not mature. There is no principal to redeem. It makes a fixed coupon payment forever. The Macaulay duration of a perpetual bond is $(1 + r)/r$.

Duration of Floating-Rate Notes and Loans

Interest rate risk for floating-rate instrument arises only between reset dates. Because at the next reset date, the coupon payments will adjust to the new MRR. Therefore, the Macaulay duration for a floating-rate bond is simply the fraction of a period remaining until the next rest date. $(T - t)/T$

For example, if there are 180 days in the coupon period and 57 days have passed since the last coupon, the Macaulay duration is: $(180 - 57)/180 = 0.6833$

4. Properties of Duration

The input variables for determining Macaulay and modified yield duration of fixed-rate bonds are:

- Coupon rate or payment per period

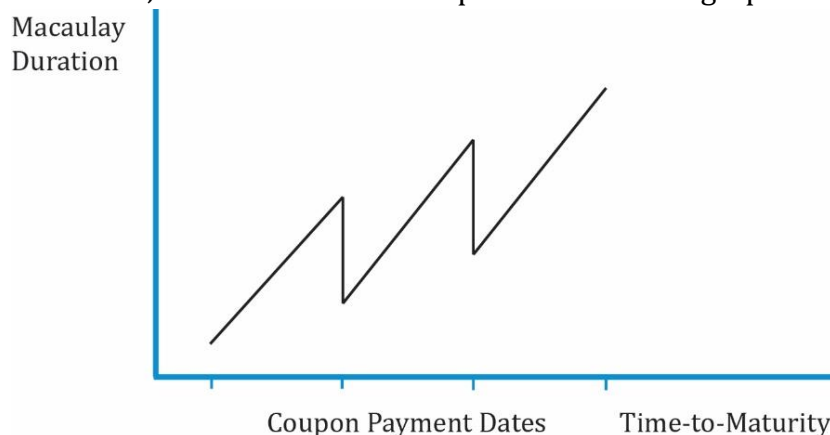
- Yield to maturity per period
- Number of periods to maturity
- Fraction of the coupon period that has gone by

By changing one of the above variables while holding others constant, we can analyze the properties of bond duration, which, in turn, helps us assess the interest rate risk. We will use the formula for Macaulay duration to understand the relationship between each variable and duration:

$$\text{MacDur} = \left\{ \frac{1+r}{r} - \frac{1+r + [N * (c-r)]}{c * [(1+r)^N - 1] + r} \right\} - (t/T)$$

The fraction of the coupon period that has gone by (t/T)

First, let us consider the relationship between fraction of time that has gone by (t/T) and duration. There is no change to the expression in braces {} as time passes by. Fraction of time (t/T) increases as time passes by. Assume T is 180. If 50 days have passed, then t/T = 0.277. If 90 days have passed, then t/T = 0.5. If 150 days have passed, then t/T = 0.83. As t/T increases from t = 0 to t = T with passing time, MacDuration decreases in value. Once the coupon is paid, t/T becomes zero and MacDuration jumps in value. When time to maturity is plotted against MacDuration, it creates a saw tooth pattern as shown graphically below:

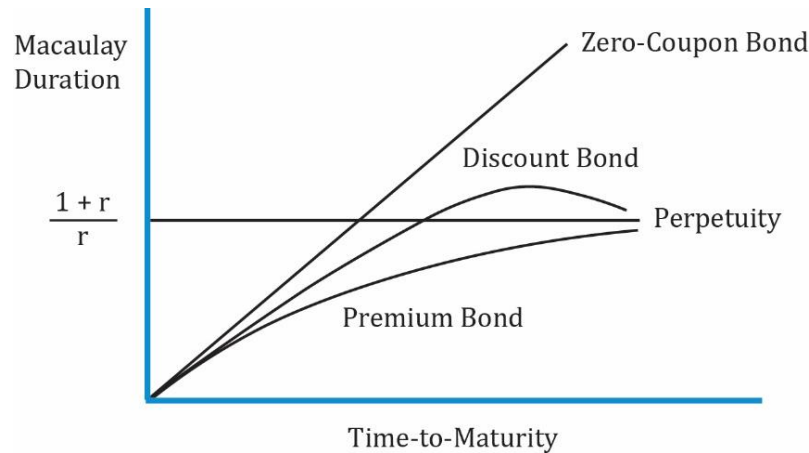


Interpretation of the Macaulay duration between coupon payments with a constant yield to maturity:

- Note: Read the diagram from right to left. As time passes between coupon periods, duration decreases in value.
- Once the coupon is paid, it jumps back up creating a saw tooth pattern.

Time to maturity

The following exhibit illustrates the relationship between Macaulay duration and the time to maturity for premium, discount, zero-coupon and perpetual bonds.



- **Zero-coupon bond:** Pays no coupon so $c = 0$. Plugging in $c = 0$ and $t/T = 0$ in MacDur formula, we get Macaulay Duration = N . For a zero-coupon bond, duration is its time to maturity.
- **Perpetuity:** A perpetual bond is one that does not mature. There is no principal to redeem. It makes a fixed coupon payment forever. If N is a large number in the above equation, then the second part of the expression in the braces becomes zero.
Denominator: N is an exponent. $(1 + r)^N$ is a very large number. So $\frac{1}{(1 + r)^N}$ must be zero and the value in the numerator will not matter here. Macaulay duration of a perpetual bond is $\frac{1+r}{r}$ as N approaches infinity.
- **Premium bond:** Bonds are trading at a premium above par or at par. The coupon rate is greater than or equal to yield to maturity (r). The numerator of the second expression in braces is always positive because $c - r$ is positive. The denominator of the second expression in braces is always positive. Second expression as a whole is always positive. MacDuration = less than $\frac{1+r}{r}$ because the second expression in braces is positive. As time passes, it approaches $\frac{1+r}{r}$.
- **Discount bond:** For a discount bond, the coupon rate is below yield to maturity. The Macaulay Duration increases for a longer time to maturity. The numerator of the second expression in braces is negative because $c - r$ is negative. Put together, the duration at some point exceeds $\frac{1+r}{r}$, reaches a maximum, and approaches $\frac{1+r}{r}$ (the threshold) from above. This happens when N is large and coupon rate (c) is below the yield to maturity (r). As a result, for a long-term discount bond, interest rate risk can be lesser than a shorter-term bond.

Coupon rate and Yield-to-maturity

All else being equal, a lower-coupon bond has a higher duration and more interest rate risk than a higher-coupon bond.

The same pattern holds for yield-to-maturity, a bond with a lower YTM has a higher duration and more interest rate risk than a bond with a higher YTM.

This is because lower coupons and lower yields increase the weight of the maturity value or final cash flow and reduce the weight of the nearer-term cash flows in the calculation for Macaulay duration.

The relationship between a bond's duration and its features are summarized in Exhibit 4 of the curriculum.

	Effect on duration (interest rate risk) from an increase in feature
Coupon rate, c	↓ (Inverse relationship)
Yield to maturity, r	↓ (Inverse relationship)
Time-to-maturity, T	↑ (Direct relationship)
Fraction of current coupon period that has elapsed, t/T	↓ (Inverse relationship)

Summary

LO: Define, calculate, and interpret modified duration, money duration, and the price value of a basis point (PVBP).

Macaulay duration is the weighted average of the time to receipt of coupon interest and principal payments.

Modified duration is a linear estimate of the percentage price change in a bond for a 100 basis points change in its yield to maturity.

$$\text{Modified duration} = \frac{\text{Macaulay Duration}}{(1 + r)}$$

$$\% \Delta PV^{\text{FULL}} \approx -\text{AnnModDur} * \Delta \text{Yield}$$

$$\text{Approximate Modified Duration} = \frac{(PV_-) - (PV_+)}{2 * \Delta \text{yield} * PV_0}$$

The money duration of a bond is a measure of the price change in units of the currency in which the bond is denominated, given a change in annual yield to maturity.

$$\text{Money Duration} = \text{AnnModDur} * PV^{\text{FULL}}$$

$$\Delta PV^{\text{FULL}} \approx -\text{MoneyDur} * \Delta \text{yield}$$

The PVBP is an estimate of the change in the full price, given a 1bp change in the yield to maturity.

$$\text{PVBP} = \frac{PV_- - PV_+}{2}$$

LO: Explain how a bond's maturity, coupon, and yield level affect its interest rate risk.

All else equal, a longer time-to-maturity, a lower coupon rate, or a lower YTM results in a higher duration and higher interest rate risk.

Conversely, a shorter time-to-maturity, a higher coupon rate, or a higher YTM results in a lower duration and lower interest rate risk.

LM12 Yield-Based Bond Convexity and Portfolio Properties

1. Introduction.....	2
2. Bond Convexity and Convexity Adjustment.....	2
3. Bond Risk and Return Using Duration and Convexity	4
4. Portfolio Duration and Convexity	6
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Version 1.0

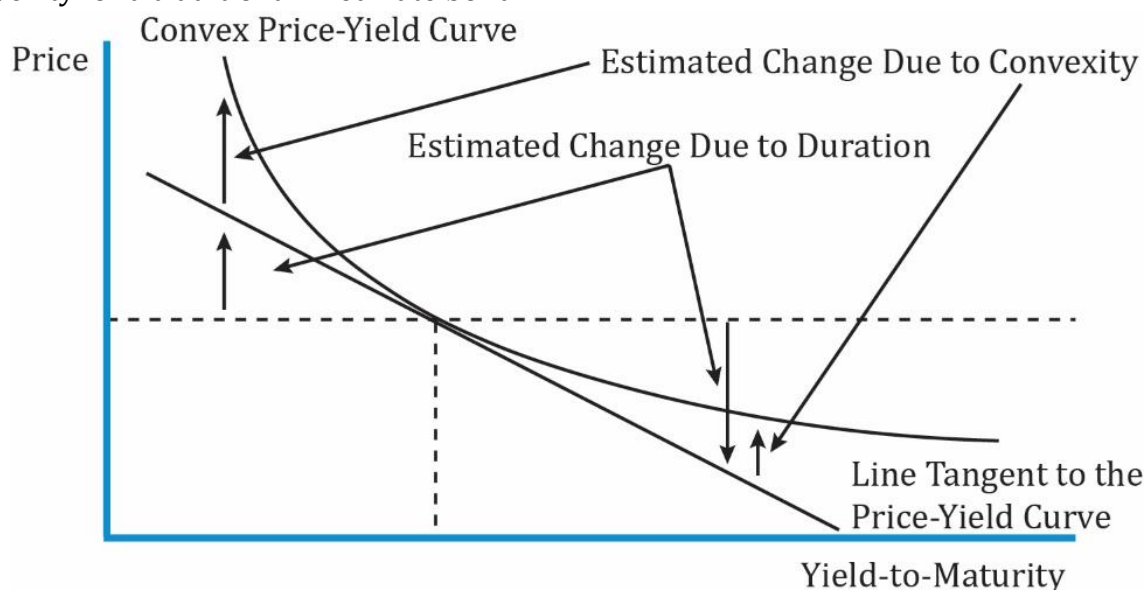
1. Introduction

This learning module covers:

- Bond convexity and convexity adjustment
- Calculating the percentage price change of a bond given its duration and convexity
- Calculating the portfolio duration and convexity for a portfolio of bonds

2. Bond Convexity and Convexity Adjustment

The graph below shows the relationship between bond price and YTM. It shows the convexity for a traditional fixed-rate bond.



Interpretation of the diagram:

- Duration assumes there is a linear relationship between the change in a bond's price and change in YTM. For instance, assume the YTM of a bond is 10% and it is priced at par (100). According to the duration measure, if the YTM increases to 11% the price moves down to a point on the straight line.
- Similarly, the price moves up to a point on the straight line if the YTM decreases.
- The curved line in the above exhibit plots the actual bond prices against YTM. So, in reality, the bond prices do not move along a straight line but exhibit a convex relationship.
- For small changes in YTM, the linear approximation is a good representation for change in bond price. That is, the difference between the straight and curved line is not significant.
- In other words, modified duration is a good measure of the price volatility.
- However, for large changes in YTM or when the rate volatility is high, a linear approximation is not accurate and a convexity adjustment is needed.

Here we need to factor in the convexity. The percentage change in the bond's full price with convexity-adjustment is given by the following equation:

Change in the price of a full bond:

$$\% \Delta PV^{\text{FULL}} = (-\text{AnnModDur} * \Delta \text{yield}) + \left[\frac{1}{2} * \text{AnnConvexity} * (\Delta \text{yield})^2 \right]$$

Expression in first braces: duration adjustment

Expression in second braces: convexity adjustment

Approximate convexity can be calculated using this formula:

$$\text{Approx. Convexity} = \frac{PV_- + PV_+ - 2 * PV_0}{(\Delta \text{yield})^2 * PV_0}$$

where:

PV_- and PV_+ = new full price when YTM is decreased and increased by the same amount

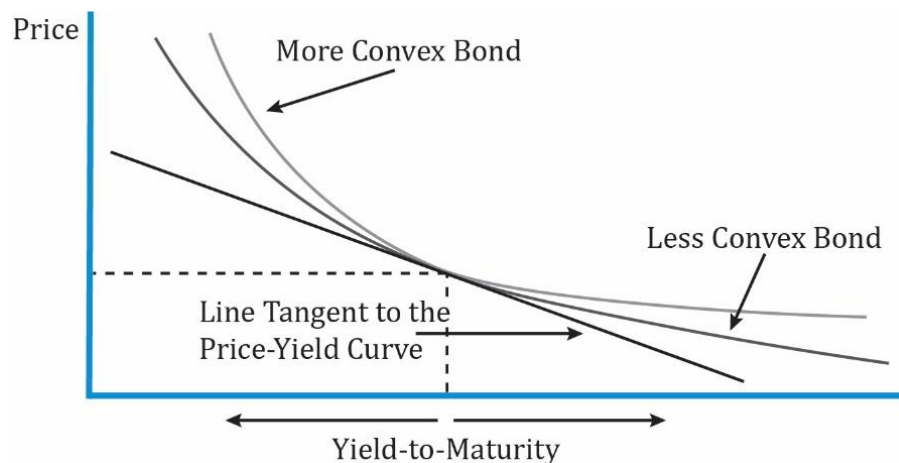
PV_0 = original full price

The change in the full price of the bond in units of currency, given a change in YTM, can be calculated using this formula:

$$\Delta PV_{\text{FULL}} \approx -(\text{MoneyDur} * \Delta \text{yield}) + \left[\frac{1}{2} * \text{MoneyCon} * (\Delta \text{yield})^2 \right]$$

Convexity is good

The following exhibit shows the price-yield curves for two bonds with the same YTM, price, and modified duration, and why greater convexity is good for an investor.



Interpretation of the diagram:

- Both the bonds have the same tangential line to their price-yield curves.
- When YTM decreases by the same amount, the more convex bond appreciates more in price.
- When YTM increases by the same amount, the more convex bond depreciates less in price than the less convex bond.
- The bond with greater convexity outperforms when interest rates go up/down.

The relationship between various bond parameters with convexity is the same as with duration.

For a fixed-rate bond,

- The lower the coupon rate, the greater the convexity.
- The lower the yield to maturity, the greater the convexity.
- The longer the time to maturity, the greater the convexity.
- The greater the dispersion of cash flow or cash payments spread over time, the greater the convexity.

3. Bond Risk and Return Using Duration and Convexity

In this section we will see how to estimate the percentage price change of a bond for a specified yield change, given the bond's duration and convexity.

Example: Calculating the full price and convexity-adjusted percentage price change of a bond

A German bank holds a large position in a 6.50% annual coupon payment corporate bond that matures on 4 April 2029. The bond's yield to maturity is 6.74% for settlement on 27 June 2014, stated as an effective annual rate. That settlement date is 83 days into the 360-day year using the 30/360 method of counting days.

1. Calculate the full price of the bond per 100 of par value.
2. Calculate the approximate modified duration and approximate convexity using a 1 bp increase and decrease in the yield to maturity.
3. Calculate the estimated convexity-adjusted percentage price change resulting from a 100 bp increase in the yield to maturity.
4. Compare the estimated percentage price change with the actual change, assuming the yield to maturity jumps to 7.74% on that settlement date.

Solution:

There are 15 years from the beginning of the current period on 4 April 2014 to maturity on 4 April 2029.

1. The full price of the bond is 99.2592 per 100 of par value.
 $FV = 100, I/Y = 6.74, PMT = 6.50, N = 15, CPT PV; PV = -97.777.$

$$\text{Full Price} = 97.777 \times 1.0674^{\frac{83}{360}} = 99.2592.$$

1. $PV_+ = 99.1689.$

$$FV = 100, PMT = 6.5, I/Y = 6.75, N = 15, CPT PV; PV = -97.687.$$

$$PV_+ = 97.687 \times 1.0675^{\frac{83}{360}} = 99.1689.$$

$$PV_- = 99.3497.$$

$$FV = 100, I/Y = 6.73, PMT = 6.5, N = 15, CPT PV; PV = -97.869.$$

$$PV_- = 97.869 \times 1.0673^{\frac{83}{360}} = 99.3497.$$

$$\text{ApproxModDur} = \left(\frac{99.3497 - 99.1689}{2 \times 99.2592 \times .0001} \right) = 9.1075.$$

The approximate modified duration is 9.1075.

$$\text{ApproxCon} = \frac{99.1689 + 99.3497 - (2 \times 99.2592)}{.0001^2 \times 99.2592} = 201.493$$

The approximate convexity is 201.493.

3. The convexity-adjusted percentage price drop resulting from a 100 bp increase in the yield to maturity is estimated to be -8.1% $(-9.1075 + 1.00746)$.

Modified duration alone estimates the percentage drop to be 9.1075%. The convexity adjustment adds 100.746 bps $(0.5 \times 201.493 \times .01^2 = 1.00746\%)$.

4. The new full price if the yield to maturity goes from 6.74% to 7.74% on that settlement date is 90.7623. The actual percentage change in the bond price is -8.5603%. The convexity-adjusted estimate is -8.1%.

Example: Calculating the approximate modified duration and approximate convexity

The investment manager for a US defined-benefit pension scheme is considering two bonds about to be issued by a large life insurance company. The first is a 25-year, 5% semiannual coupon payment bond. The second is a 75-year, 5% semiannual coupon payment bond. Both bonds are expected to trade at par value at issuance.

Calculate the approximate modified duration and approximate convexity for each bond using a 5 bp increase and decrease in the annual yield to maturity.

Solution:

In the calculations, the yield per semiannual period goes up by 2.5 bps to 2.525% and down by 2.5 bps to 2.475%.

The 25-year bond has an approximate modified duration of 14.18.

$$PV_+: FV = 100, I/Y = 2.525, PMT = 2.5, N = 50, \text{CPT PV}, PV = -99.2945.$$

$$PV_-: FV = 100, I/Y = 2.475, PMT = 2.5, N = 50, \text{CPT PV}, PV = -100.7126.$$

$$\text{ApproxModDur} = \frac{100.7126 - 99.2945}{2 \times 100 \times 0.0005} = 14.18 \text{ and an approximate convexity of 284.}$$

$$\text{ApproxCon} = \left(\frac{100.7126 + 99.2945 - (2 \times 100)}{100 \times 0.0005^2} \right) = 284. \text{ Similarly, the 75-year bond has an approximate modified duration of 19.51 and an approximate convexity of 708.}$$

4. Portfolio Duration and Convexity

In the previous section, we saw how to calculate the duration and convexity for an individual bond. What if a portfolio consists of a number of bonds, how will its duration and convexity be calculated?

There are two ways to calculate the duration of a bond portfolio:

- Weighted average of time to receipt of the aggregate cash flows.
- Weighted average of the durations and convexities of the individual bonds that comprise the portfolio.

The key points related to each method are outlined in the table below.

Weighted Average Time to Receipt of Aggregate Cash Flows	Weighted Average of Individual Bond Durations and Convexities in Portfolio
Theoretically correct, but difficult to use in practice.	Commonly used in practice.
Cash flow yield not commonly used. Cash flow yield is the IRR on a series of cash flows.	Easy to use as a measure of interest rate risk.
Amount and timing of cash flows might not be known because some of these bonds may be MBS, or with call options.	More accurate as difference in YTMs of bonds in portfolio become smaller.
Interest rate risk is usually expressed as a change in benchmark interest rates, not as a change in the cash flow yield.	Assumes parallel shifts in the yield curve, i.e., all rates change by the same amount in the same direction. That seldom happens in reality.
Change in the cash flow yield is not necessarily the same amount as the change in yields to maturity on the individual bonds.	

Example: Portfolio Duration and Convexity

(This is Example 2 from the curriculum.)

An institutional investor considers adding a new USD50 million par value position in a 10-year US Treasury bond to its existing portfolio of BRWA and government of Romania bonds. The relevant data are shown below:

Bond	Maturity (Years)	Coupon (%)	Price	Yield (%)	Duration	Convexity
BRWA	5	3.200	100.000	3.200	4.58676	24.23896
Romanian	30	4.625	98.022	4.750	15.90637	369.64203
US Treasury	10	1.500	98.168	1.700	9.23693	93.87376

Bond	Par Value (USD)	Market Value (USD)	Portfolio Weight
BRWA	50,000,000	50,000,000	0.33762
Romania	50,000,000	49,011,224	0.33094
US Treasury	50,000,000	49,083,948	0.33144

1. Calculate the weighted-average duration and convexity for the proposed portfolio.

Solution:

Given the data, the weighted-average modified duration and convexity are calculated as follows:

Weighted-average modified duration = $(4.58676 \times 0.33762) + (15.90637 \times 0.33094) + (9.23693 \times 0.33144) = 9.87415$.

Weighted-average convexity = $(24.23896 \times 0.33762) + (369.64203 \times 0.33094) + (93.87376 \times 0.33144) = 161.62749$.

2. Compare and interpret duration and convexity for the proposed portfolio versus the current portfolio.

Solution:

Adding the US Treasury position would decrease both the portfolio duration (from 10.19004 to 9.87415) and convexity (from 195.21581 to 161.62749).

The reduction in duration would reduce the price risk of the portfolio against an upward parallel shift in the yield curve, but due to the lower convexity, this reduction in risk would be lessened for large shifts.

3. Recommend whether the US Treasury bond position should be added if the investor expects a 100 bp parallel shift downward in yields.

Solution:

Given an expected 100 bp parallel shift down in yields, the investor should not add the position in US Treasury bonds. Adding the position lowers both the portfolio duration and convexity, which would also reduce the expected increase in the value of the portfolio. Given the investor's yield curve view, it should seek to increase both portfolio duration and convexity.

Summary

LO: Calculate and interpret convexity and describe the convexity adjustment.

Convexity measures the non-linear effect of yield changes on the price of an option free fixed rate bond. The convexity adjustment adds to the linear price estimate provided by modified duration.

LO: Calculate the percentage price change of a bond for a specified change in yield, given the bond's duration and convexity.

The percentage change of a bond can be calculated using the following formula if modified duration and convexity are given:

$$\% \Delta PV^{FULL} = (-\text{AnnModDur} * \Delta \text{yield}) + \left[\frac{1}{2} * \text{AnnConvexity} * (\Delta \text{yield})^2 \right]$$

LO: Calculate portfolio duration and convexity and explain the limitations of these measures.

Portfolio duration and convexity can be calculated (1) as the weighted average of time to receipt of the aggregate cash flows or (2) by using the weighted averages of the durations and convexities of the individual bonds that make up the portfolio

Method 1 is theoretically correct but difficult to use in practice. Method 2 is commonly used by portfolio managers, but it implicitly assumes parallel shifts in the yield curve, which seldom happens in reality.

LM13 Curve-Based and Empirical Fixed-Income Risk Measures

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Version 1.0

1. Introduction

This learning module covers:

- Curve based interest rate risk measures – Effective duration and Effective convexity
- Calculating the percentage price change of a bond, given the bond's effective duration and convexity
- Key rate duration
- Empirical duration versus analytical duration

2. Curve-Based Interest Rate Risk Measures

Effective Duration

Yield duration and convexity measures assume that a bond's cash flows are certain. However, for bonds with embedded options (e.g. callable or putable bonds) the future cash flows are uncertain. The option to exercise depends on the level of market interest rates relative to the coupon interest of the bond. Therefore, these securities do not have a well-defined YTM. They may be prepaid well before the maturity date.

Hence, yield duration statistics are not suitable for these instruments. For such instruments, the best measure of interest rate sensitivity is the effective duration, which measures the sensitivity of the bond's price to a change in a benchmark yield curve (instead of its own YTM).

$$\text{Effective Duration} = \frac{(PV_-) - (PV_+)}{2 * \Delta \text{ curve} * PV_0}$$

Difference between approx. modified duration and effective duration:

The denominator for approx. modified duration has the bond's own yield to maturity. It measures the bond's price change to changes in its own YTM. But, the denominator for effective duration has the change in the benchmark yield curve. It measures the interest rate risk in terms of change in the benchmark yield curve.

Example: Calculating the effective duration

A Pakistani defined-benefit pension scheme seeks to measure the sensitivity of its retirement obligations to market interest rate changes. The pension scheme manager hires an actuarial consultant to model the present value of its liabilities under three interest rate scenarios:

1. a base rate of 10%
2. a 50 bps drop in rates, down to 9.5%
3. a 50 bps increase in rates to 10.5%.

The following chart shows the results of the analysis:

Interest Rate Assumption	Present Value of Liabilities
9.5%	PKR 10.5 million
10%	PKR 10 million
10.5%	PKR 9 million

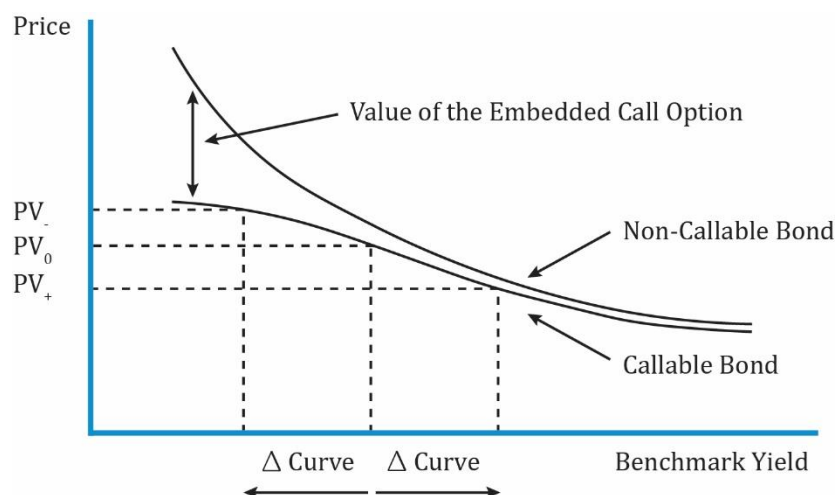
Compute the effective duration of pension liabilities. $PV_0 = 10$, $PV_+ = 9$, $PV_- = 10.5$, and $\Delta \text{ curve} = 0.005$. The effective duration of the pension liabilities is 15.

$$\frac{10.5 - 9}{2 * 0.005 * 10} = 15$$

This effective duration statistic for the pension scheme's liabilities might be used in asset allocation decisions to decide the mix of equity, fixed income, and alternative assets.

Interest Rate Risk Characteristics of a Callable Bond

A callable bond is one that might be called by the issuer before maturity. This makes the cash flows uncertain, so the YTM cannot be determined with certainty. The exhibit below plots the price-yield curve for a non-callable/straight bond and a callable bond. It also plots the change in price for a change in the benchmark yield curve. Bond price is plotted on the y-axis and the benchmark yield on the horizontal axis.



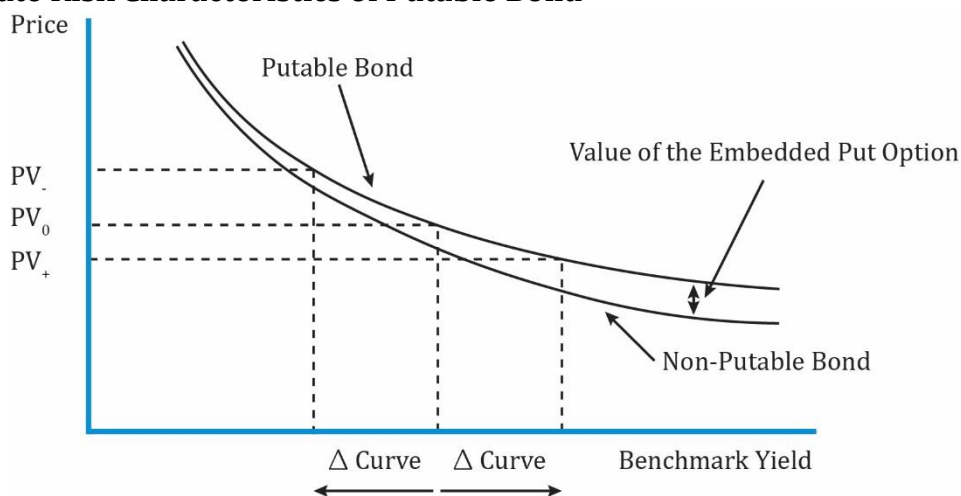
Interpretation of the diagram:

- The two bonds share the same features: credit risk, coupon rate, payment frequency, and time to maturity, but their prices are different when interest rates are low.
- When interest rates are low (left side of the graph), the price of the non-callable bond is always greater than the callable bond.
- The difference between the non-callable bond and callable bond is the value of the embedded call option.
- The call option is more valuable at low interest rates because the issuer has an option to refinance debt at lower interest rates than paying a higher interest to existing investors. For instance, if a bond is callable at 102% of par, then its price cannot

increase beyond 102 of par even if interest rates decrease. This is the reason behind the callable bond's negative convexity.

- From an investor's perspective a call option is risky. At low interest rates if the issuer calls the bond (meaning the issuer pays back the money borrowed), then investors must reinvest the proceeds at lower rates. Notice that at higher interest rates, there is not much of a difference in price because the probability of calling the bond is less.
- At relatively low interest rates, the effective duration of a callable bond is lower than that of non-callable bond, i.e., interest rate sensitivity is low.

Interest Rate Risk Characteristics of Puttable Bond



Interpretation of the diagram:

- The diagram above shows the price-yield curve for a puttable bond and a straight bond.
- A puttable bond allows the investor to sell the bond back to the issuer; usually at par.
- At low interest rates, there is not much of a difference between a straight bond and puttable bond.
- At high interest rates, the difference between the puttable bond and straight bond is the value of the put option.
- When interest rates rise, the price of the bond decreases. Investors buy puttable bonds to protect against falling prices as rates rise.
- The value of the put option increases as rates rise. This also limits the sensitivity of the bond price to changes in benchmark rates.

Effective Convexity

For bonds whose cash flows were unpredictable, we used effective duration as a measure of interest rate risk. Similarly, we use effective convexity to measure the change in price for a change in benchmark yield curve for securities with uncertain cash flows. The effective convexity of a bond is a curve convexity statistic that measures the secondary effect of a change in a benchmark yield curve. It is used for bonds with embedded options.

$$\text{Effective Convexity} = \frac{PV_- + PV_+ - 2 * PV_0}{(\Delta \text{curve})^2 * PV_0}$$

Here is a summary of some important points related to bonds with embedded options:

- Option-free bonds always have positive convexity.
- Callable bonds have positive convexity at high yields but they can have negative convexity at low yields. This is because at low yields, the call option becomes valuable and puts a limit on how much the bond price can appreciate.
- Puttable bonds always have positive convexity.

Example: Effective Duration and Convexity

(This is Example 1 from the curriculum.)

The portfolio manager asks you, the analyst on a fixed-income team, to determine the interest rate sensitivity of a callable bond she is considering. The portfolio manager is seeking to invest in a bond with a duration between 7.0 and 8.0 and positive convexity.

Suppose the full price of the callable bond is 101.060 per 100 of par value. When the government par curve is raised and lowered by 25 bps, the new full prices for this callable bond from your option valuation model are 99.050 and 102.891, respectively. Therefore,

- $PV_0 = 101.060$,
- $PV_+ = 99.050$,
- $PV_- = 102.891$, and
- $\Delta \text{Curve} = 0.0025$.

You determine effective duration and effective convexity for this callable bond as follows:

$$\text{Effective Duration} = \frac{(PV_-) - (PV_+)}{2 * \Delta \text{curve} * PV_0}$$

$$\text{Effective Duration} = \frac{(102.891) - (99.050)}{2 * (0.00025) * (101.060)} = 7.601$$

$$\text{Effective Convexity} = \frac{PV_- + PV_+ - 2 * PV_0}{(\Delta \text{curve})^2 * PV_0}$$

$$\text{Effective Convexity} = \frac{102.891 + 99.050 - 2 * 101.060}{(0.00025)^2 * 101.060} = -283$$

You recommend that the portfolio manager not buy this callable bond. While its effective duration meets her criteria, the bond has negative effective convexity.

3. Bond Risk and Return Using Curve-Based Duration and Convexity

Like yield-based interest rate risk measures, effective duration and effective convexity can also be used to estimate the percentage change in a bond's full price for a given shift in the benchmark yield curve.

$$\% \Delta PV^{\text{FULL}} = (-\text{EffDur} * \Delta \text{curve}) + \left[\frac{1}{2} * \text{EffConvexity} * (\Delta \text{curve})^2 \right]$$

Example:

(This is based on the Question Set from the curriculum.)

From a bond pricing model, the following information on effective duration and effective convexity were obtained.

Bond A: Effective duration = 9.369; Effective convexity = -353.752.

Calculate the estimated percentage change in the full price of bond A for the 200 bps curve shift.

Solution:

$$\% \Delta PV^{\text{FULL}} = (-\text{EffDur} * \Delta \text{curve}) + \left[\frac{1}{2} * \text{EffConvexity} * (\Delta \text{curve})^2 \right]$$

$$\% \Delta PV^{\text{FULL}} = (-9.369 * 0.02) + \left[\frac{1}{2} * -353.752 * (0.02)^2 \right]$$

$$\% \Delta PV^{\text{FULL}} = -25.81\%$$

Yield duration and convexity are only useful for small changes in yields. Effective duration and effective convexity are useful for both small and large changes in yield.

4. Key Rate Duration as a Measure of Yield Curve Risk

Key rate duration is a measure of a bond's sensitivity to a change in the benchmark yield curve at a specific maturity. This topic is covered in detail at Level II.

The duration measures we have seen so far assume a parallel shift in the yield curve. But what if the shift in the yield curve is not parallel? Here it is appropriate to use 'Key rate duration'.

Key rate durations are used to identify "shaping risk" of a bond, which is a bond's sensitivity to changes in the shape of the benchmark yield curve. For instance, analysts can analyze the interest rate sensitivity if the yield curve flattens or if the yield curve steepens.

The sum of weighted key rate durations of the bonds in a portfolio are equal to the effective duration of the entire portfolio.

Suppose we have an index zero-coupon portfolio invested in 2-, 5-, and 10-year bonds.

"Index" Zero-Coupon Portfolio

Tenor	Coupon	Annualized Yield	Price (per \$100)	Position (\$ MM)	ModDur	KeyRateDur
2y	0.00%	1%	98.03	98.03	1.980	0.738
5y	0.00%	2%	90.57	90.57	4.902	1.688
10y	0.00%	3%	74.40	74.40	9.709	2.747

The index portfolio is simply weighted by the price of the respective 2-, 5-, and 10-year bonds. The total portfolio value is $98.03 + 90.57 + 74.40 = \263 million.

We can calculate the 2y, 5y and 10y key rate durations (last column) for this portfolio as follows:

$$\text{KeyRateDur} = \text{ModDur} \times \text{Position size} / \text{Total portfolio size}$$

$$\text{KeyRateDur}_2 = 1.980 \times 98.03 / 263 = 0.738$$

$$\text{KeyRateDur}_5 = 4.902 \times 90.57 / 263 = 1.688$$

$$\text{KeyRateDur}_{10} = 9.709 \times 74.40 / 263 = 2.747$$

The sum of the key rate durations is equal to the total duration of the portfolio.

$$\text{Portfolio duration} = \text{KeyRateDur}_2 + \text{KeyRateDur}_5 + \text{KeyRateDur}_{10}$$

$$\text{Portfolio duration} = 0.738 + 1.688 + 2.747 = 5.173$$

We verify this by calculating the portfolio duration using the ModDur of each bond.

$$\text{Portfolio duration} = [1.98 \times (98.03/263)] + [4.902 \times (90.57/263)] + [9.709 \times (74.40/263)] = 5.173$$

5. Empirical Duration

The approaches used so far to estimate duration and convexity with mathematical formulas is called **analytical duration**. This approach implicitly assumes that benchmark yields and credit spreads are uncorrelated with one another.

However, in practice, changes in benchmark yields and credit spreads are often correlated. So, many fixed income analysts use an alternate approach – **Empirical duration**. This approach uses statistical methods and historical bond prices to derive the price-yield relationship for specific bonds or bond portfolios.

For a government bond with little or no credit risk, the analytical and empirical duration would be similar because bond prices are largely driven by changes in the benchmark yield.

However, for a high-yield bonds with significant credit risk, the analytical and empirical duration will be different. In a market stress scenario, many investors switch to high quality government bonds due to which their yields (i.e., benchmark yields) fall. But at the same time the credit spreads on high-yield bonds will widen (i.e., credit spreads and benchmark yields are negatively correlated). The wider credit spreads will fully or partially offset the decline in government benchmark yields. Thus, the empirical duration for high yield bonds will be lower than their analytical duration.

Summary

LO: Explain why effective duration and effective convexity are the most appropriate measures of interest rate risk for bonds with embedded options.

The difference between modified duration and effective duration is that modified duration measures interest rate risk in terms of a change in the bond's own yield to maturity, whereas effective duration measures interest rate risk in terms of changes in the benchmark yield curve.

Bonds with an embedded option do not have a meaningful internal rate of return (YTM) because future cash flows are contingent on interest rates.

Therefore, effective duration and effective convexity are the most appropriate measures of interest rate risk for bonds with embedded options.

$$\text{Effective Duration} = \frac{(PV_-) - (PV_+)}{2 * \Delta\text{curve} * PV_0}$$

$$\text{Effective Convexity} = \frac{PV_- + PV_+ - 2 * PV_0}{(\Delta\text{curve})^2 * PV_0}$$

LO: Calculate the percentage price change of a bond for a specified change in benchmark yield, given the bond's effective duration and convexity.

Like yield-based interest rate risk measures, effective duration and effective convexity can also be used to estimate the percentage change in a bond's full price for a given shift in the benchmark yield curve.

$$\% \Delta PV^{\text{FULL}} = (-\text{EffDur} * \Delta\text{curve}) + \left[\frac{1}{2} * \text{EffConvexity} * (\Delta\text{curve})^2 \right]$$

LO: Define key rate duration and describe its use to measure price sensitivity of fixed-income instruments to benchmark yield curve changes.

Key rate duration is a measure of a bond's sensitivity to a change in the benchmark yield curve at a specific maturity.

Key rate durations are used to identify "shaping risk" of a bond, which is a bond's sensitivity to changes in the shape of the benchmark yield curve. For instance, analysts can analyze the interest rate sensitivity if the yield curve flattens or if the yield curve steepens.

The sum of weighted key rate durations of the bonds in a portfolio are equal to the effective duration of the entire portfolio.

LO: Describe the difference between empirical duration and analytical duration.

- Analytical duration approach uses mathematical formulas to estimate duration and convexity. The approach implicitly assumes that benchmark yields and credit spreads are uncorrelated with one another.

- The empirical duration approach does not rely on this assumption. It uses statistical methods and historical bond prices to derive the price-yield relationship for specific bonds or bond portfolios.
- For government bonds, both approaches will give similar results.
- For high yield bonds, the empirical duration will be lower than analytical duration.

LM14 Credit Risk

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2. Sources of Credit Risk	2
3. Credit Rating Agencies and Credit Ratings	5
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Version 1.0

1. Introduction

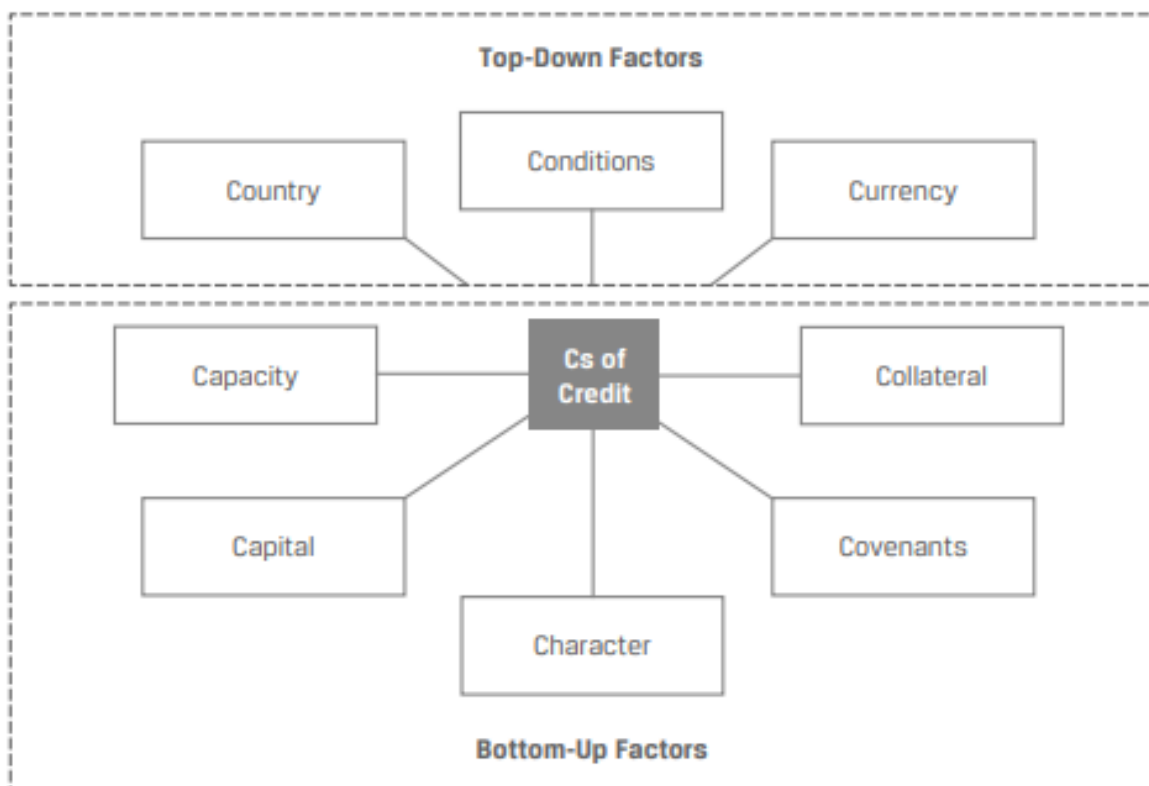
This learning module covers:

- Credit risk and its components – probability of default and loss given default
- Role of credit ratings in debt markets
- Factors that influence the level and volatility of yield spreads

2. Sources of Credit Risk

Credit risk is the risk that the borrower will fail to make principal and/or interest payments on time.

Traditionally, many analysts assessed creditworthiness using what are known as the "Cs of credit analysis," as illustrated in Exhibit 1:



The five bottom-up factors that are applicable to an individual borrower are:

- 'Capacity' refers to the ability of the borrower to make its debt payments on time.
- 'Capital' refers to other company resources available that reduce reliance on debt.
- 'Character' refers to the quality of the management and the willingness to pay debt.
- 'Covenants' refers to the terms and conditions of the debt agreement that the issuer must comply with.

- ‘Collateral’ refers to the quality and value of the assets pledged against the debt.

The remaining three factors are general top-down factors that apply to all borrowers:

- ‘Conditions’ refers to general economic, competitive, and business environment faced by all borrowers that may impact their ability to service or refinance debt.
- ‘Country’ refers to the geopolitical environment as well as the legal and political system faced by all issuers in a jurisdiction.
- ‘Currency’ refers to cash flows affected by exchange rate changes, such as import/export or borrowings in foreign currency.

Sources of Credit Risk

Exhibit 3 from the curriculum shows the main borrower types, their primary sources of repayment and credit risk.

Borrower Type	Primary Source of Cash Flow Generation	Secondary Source of Cash Flow Generation	Sources of Credit Risk
Corporate	▪ Business operations ▪ Investment and financing activities	▪ Asset sales ▪ Divestitures ▪ Additional debt issuance ▪ Equity issuance	▪ Economic contraction ▪ Strategic shifts in the business and market environment ▪ Increased competition ▪ Reduced pricing power ▪ Shrinking operating margins, increased losses ▪ Excessive debt service needs
Sovereign or public entity	▪ Corporate and personal income taxes, sales tax and VAT revenue, capital gains and wealth-based taxes ▪ Tariffs, fees, and other government revenue	▪ Newly issued debt ▪ Sale of public assets, privatization	▪ Economic contraction ▪ Political uncertainty ▪ Excessive debt service needs ▪ Expansionary economic policies ▪ Budget deficits ▪ Tax cuts ▪ Limited ability to collect taxes

As shown in the exhibit, many factors can affect the primary and secondary sources of repayment for different fixed-income issuers.

Measuring Credit Risk

Credit risk has two components:

- Probability of default (POD): The likelihood that an issuer fails to make full and timely payments of principal and interest.
- Loss given default (LGD): How severe is the loss incurred by the investor? What portion of the bond’s value (including interest) is not paid by the issuer? A default

leads to various severity of losses; for instance, the loss may be total, or the bondholders may recover some value.

We combine both the components into a single term called the 'expected loss'.

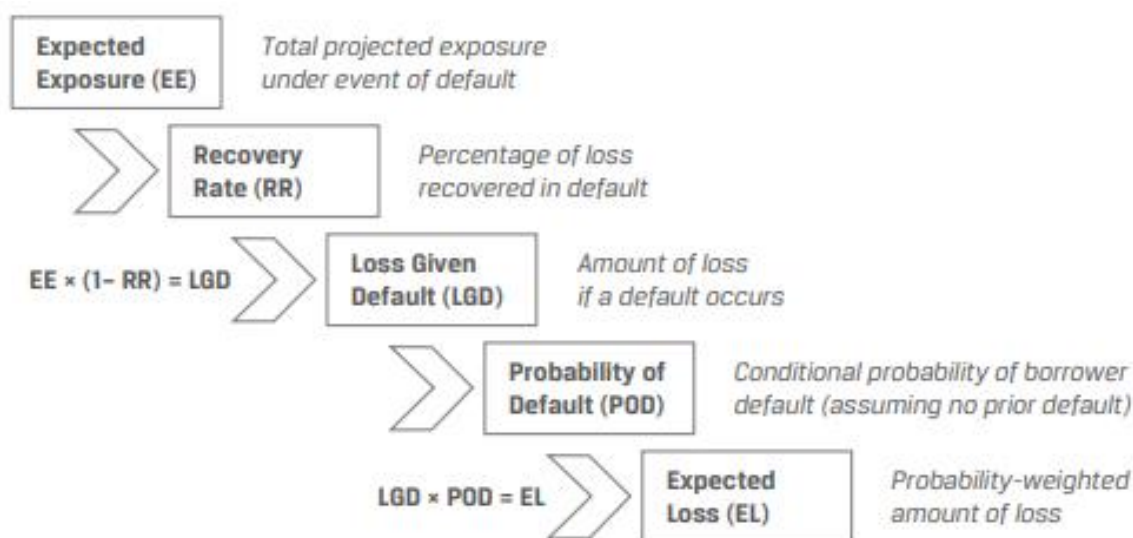
$$\text{Expected Loss} = \text{POD} \times \text{LGD}$$

The loss given default depends on:

- Exposure at default (EAD): The size of the investor's claim at the time of default.
- Recovery rate (RR): Represents the percentage of an outstanding debt claim recovered when an issuer defaults. A related term is loss severity (1-RR) which represents the unrecovered portion of the claim.

$$\text{LGD} = \text{EE} \times (1 - \text{RR})$$

Exhibit 4 from the curriculum illustrates these concepts:



One way to interpret a fixed-income security's expected loss for a given period is to compare it to the compensation an investor expects for taking on a borrower's credit risk over that period, which is the credit spread. We can say that an investor is fairly compensated if the expected loss is equal to the credit spread for a given period.

$$\text{Credit Spread} \approx \text{POD} \times \text{LGD}$$

Example:

(This is based on Question Set – Q#4 from the curriculum.)

A bond investor analyzing a company's unsecured debt estimates a POD of 2% and an LGD of 80%. The observed actual credit spread for this bond is 200 bps per year. Is the investor fairly compensated/ less than fairly compensated/more than fairly compensated for assuming the company's credit risk?

Solution:

Since $\text{POD} \times \text{LGD} = 1.60\%$ and the credit spread is 2.00% ,

Credit Spread $>$ $\text{POD} \times \text{LGD}$ and the investor would expect to be more than fairly compensated for assuming the company's credit risk.

3. Credit Rating Agencies and Credit Ratings

Moody's, S&P, and Fitch are the three main rating agencies. At least two of these agencies assign a rating to the majority of the bonds.

Credit Ratings

Be familiar with what the ratings are (Aaa represents the highest quality among investment grade securities, etc.). The rating system is almost similar across agencies starting from A for investment grade to C/D for junk.

Long-Term Ratings Matrix: Investment Grade vs. Non-Investment Grade

		Moody's	S&P	Fitch
Investment Grade	High-Quality Grade	Aaa	AAA	AAA
		Aa1	AA+	AA+
		Aa2	AA	AA
		Aa3	AA-	AA-
	Upper-Medium Grade	A1	A+	A+
		A2	A	A
		A3	A-	A-
	Low-Medium Grade	Baa1	BBB+	BBB+
		Baa2	BBB	BBB
		Baa3	BBB-	BBB-

		Moody's	S&P	Fitch
Non-Investment Grade "Junk" or "High Yield"	Low Grade or Speculative Grade	Ba1	BB+	BB+
		Ba2	BB	BB
		Ba2	BB-	BB-
		B1	B+	B+
		B2	B	B
		B3	B-	B-
		Caa1	CCC+	CCC+
		Caa2	CCC	CCC
		Caa3	CCC-	CCC-
		Ca	CC	CC
		C	C	C
	Default	C	D	D

Source: CFA Program Curriculum, Fundamentals of Credit Analysis

- Bonds rated AAA/Aaa have the highest quality, lowest credit risk, and lowest probability of default. They have extremely low default rates.
- Bonds rated Baa3/BBB- or higher are investment grade bonds. Investment grade is further divided into high-quality, upper-medium and low-medium.
- Bonds rated lower than Baa3/BBB- are non-investment grade bonds. Non-investment grade includes speculative and default bonds. These have higher default risk.
- Outlook: This is an additional indicator on a rating. The outlook can be *positive*, *stable*, or *negative*.
- The rating agencies may also indicate the potential direction of the ratings by including comments like “on review for a downgrade” or “on credit watch for an upgrade.”
- ‘Credit migration risk’ is the risk that a bond issuer’s creditworthiness deteriorates, or migrates to a lower rating, leading investors to believe that the risk of default is higher.

Note: “Investment-grade ratings focus more on timeliness, while non-investment grade ratings give more weight to recovery”.

Credit Rating Considerations

There is sufficient evidence that rating agencies, in general, have done a good job of assigning ratings that reflect the risks involved. However, these ratings are not foolproof and have their own limitations. Analysts should conduct their own credit analysis, as sole reliance on credit rating to make investment decisions has several pitfalls:

- Credit ratings can be very dynamic: Credit rating of a bond issue may change over time. For instance, a bond may have had “AAA” rating when it was issued. But five years later, it may be A. There have been instances where an “AAA” rated bond at issuance has turned into non-investment grade or junk within three years.
- Rating agencies are not infallible: The high ratings on subprime-backed mortgage securities are the best example of this. If the rating agencies had rated the issue appropriate to their risk level, it would have made investors cautious.
- Idiosyncratic or event risks are difficult to capture in ratings: Rating agencies do a poor job of foreseeing event risks because it is not possible and these cannot be reflected in a credit rating. A recent example is of the March 2011 tsunami that hit Japan, which in turn affected the credit risk of Tokyo Electric Power Company.
- Ratings tend to lag market pricing of credit: Bond prices move on a daily basis based on the perceived credit risk in them, as the market sees it. But rating agencies do not revise their ratings that often. Investors who time their buy/sell decisions solely on the ratings issued by agencies are at the risk of underperforming the market. But a sudden downgrade has a disastrous effect on bond prices.

4. Factors Impacting Yield Spreads

The yields on extremely liquid bonds with virtually no default risk (e.g., government bonds) will be equal to real interest rate plus a premium for expected inflation.

The yields on corporate bonds will include an additional risk premium that compensates investors for credit and liquidity risk and possibly the tax impact of holding a specific bond. This additional risk premium is called a spread.

Factors that affect the spread on corporate bonds:

- Credit cycle: Credit spreads narrow when the credit cycle improves; a spread is narrower at the top of a credit cycle. Credit spreads widen when the credit cycle deteriorates as seen during the recent credit crisis after 2008.
- Broader economic conditions: Credit spreads widen in a weak economy and narrow in a strong economy.
- Funding availability in the financial sector: Unlike stocks that primarily trade on exchanges, bonds usually trade over the counter. Brokers and dealers are market makers in the debt market. Yield spreads are narrow when brokers-dealers provide sufficient capital. However, at times of crisis as seen during the post-2008 period, many broker-dealer firms either closed down or were acquired. The capital available for making markets reduced substantially leading to wider spreads.
- General market supply and demand: If the supply is more and demand is less, credit spreads will widen. Conversely, if the demand is high and supply is low, then spreads tighten.
- Financial performance of the issuer: Earnings releases, news, and other developments associated with the issuer affect credit spreads. Good news will narrow spreads while bad news will widen spreads.

A number of these factors caused spreads to widen dramatically during the 2008-2009 global financial crisis.

Reasons for investing in HY bonds

Aside from the higher coupons typically offered by high-yield bonds to compensate for their higher risk, there are several reasons to invest in HY bonds:

- Portfolio diversification: HY bonds typically have a low correlation with IG bonds and default risk-free interest rates. Their inclusion in a fixed-income portfolio may therefore provide diversification benefits.
- Capital appreciation: Economic recovery or improved issuer-specific performance has a more sizeable positive impact on HY than IG bond prices.
- Equity-like return with lower volatility: While HY bond and equity prices tend to fluctuate similarly over the economic cycle, HY bond returns tend to be more stable due to their higher income component versus stocks.

The Price Impact of Spread Changes

How changes in spread affect the price and return of a bond:

Two factors that affect the return on a bond are:

- Modified duration: Price sensitivity to interest rate changes.
- Magnitude of the spread change.

Return impact is calculated using these formulae:

Return impact with and without convexity adjustment

For small changes in yield:

$$\text{Return impact (without convexity adjustment)} \approx -\text{ModDur} \times \Delta\text{spread}$$

For large changes in yield:

$$\text{Return impact (with convexity adjustment)} \approx -(\text{ModDur} \times \Delta\text{spread}) + \frac{1}{2} \times \text{Convexity} \times (\Delta\text{spread})^2$$

Note: The negative sign indicates that bond prices and yields move in opposite directions.

For option-free bonds, convexity should be scaled so it has the same order of magnitude as duration squared and the spread change is expressed as a decimal. For example, if a bond has duration of 5.0 and reported convexity of 0.235, then first re-scale convexity to 23.5, and then apply the formula. For a 1% increase in spread, the result would be

$$\% \Delta \text{PV}_{\text{Full}} = (-5.0 \times 0.01) + \frac{1}{2} \times 23.5 \times (0.01)^2 = -0.048825 \text{ or } -4.8825\%.$$

Example:

(This is Question Set- Q#4 from the curriculum.)

For a bond with a modified duration of 4 and a convexity of 0.25, which of the following changes in credit spread would result in a price decrease closest to 7.5%?

- A. 1% decrease
- B. 1% increase
- C. 2% increase

Solution:

The correct answer is C, as shown by solving for ΔSpread in Equation 4:

$$\begin{aligned} \% \Delta \text{PV}_{\text{Full}} &= -(\text{AnnModDur} \times \Delta\text{Spread}) + \frac{1}{2} \text{AnnConvexity} \times (\Delta\text{Spread})^2 \\ &= -(4 \times 0.02) + \frac{1}{2}(25) \times (0.02)^2 = -0.075 \text{ or } -7.5\%. \end{aligned}$$

The spread change is inversely related to the price effect, with a spread increase leading to a fall in bond price. Note that since duration was 4, we had to rescale the convexity from 0.25 to 25.

Summary

LO: Describe credit risk and its components, probability of default and loss given default.

Credit risk is the risk of loss if the borrower fails to make scheduled payments of interest and/or principal.

Credit risk has two components:

- Probability of Default (POD): The probability that the borrower will default.
- Loss given Default (LGD): If the borrower does default, how severe is the loss.

Expected Loss = $POD \times LGD$

LO: Describe the uses of ratings from credit rating agencies and their limitations.

Credit rating agencies, such as Moody's, Standard & Poor's, and Fitch issue credit ratings for bonds which are opinions about a bond issue's creditworthiness. These ratings help investors compare the credit risk of different debt issues.

The risks of relying on ratings from credit rating agencies are:

- Credit ratings can be very dynamic.
- Rating agencies are not foolproof.
- Idiosyncratic or event risks are difficult to capture in the ratings.
- Ratings tend to lag market pricing of credit.

LO: Describe macroeconomic, market, and issuer-specific factors that influence the level and volatility of yield spreads.

Factors that affect the spread on corporate bonds:

- Credit cycle: Credit spreads narrow when the credit cycle improves; it is narrower at the top of a credit cycle.
- Broader economic conditions: Credit spreads widen in a weak economy and narrow in a strong economy.
- Funding availability in the financial sector: The amount of capital available from brokers and dealers can influence spreads.
- Supply and demand: If the supply is more and demand is less, credit spreads will widen.
- Financial performance of the issuer: Earnings releases, news, and other developments associated with the issuer affect credit spreads.

LM15 Credit Analysis for Government Issuers

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2. Sovereign Credit Analysis.....2

3. Non-Sovereign Credit Risk.....6

Summary.....8

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Version 1.0

1. Introduction

This learning module covers special considerations for evaluating the credit worthiness of sovereign and other public issuers.

2. Sovereign Credit Analysis

Both sovereign and non-sovereign governments issue debt which is used to conduct fiscal policy and meet budgetary needs, such as providing public goods and services (e.g. infrastructure, health care, and education.) The primary source of repayment for sovereign debt is taxes and other government revenue such as fees, tariffs, and in some cases the profits of state-owned enterprises.

The ability of a sovereign government to tax all private economic activity within its jurisdiction is the primary reason that such bonds typically have the lowest credit risk of any issuer within a specific country. While sovereign bonds of advanced economies are often considered default risk-free, emerging and frontier market government bonds do involve default risk.

A combination of qualitative and quantitative factors can be used to assess a sovereign issuer's ability and willingness to pay.

Qualitative Factors

Qualitative factors included in the assessment are:

Government Institutions & Policy: Do the sovereign government institutions and policies foster political and economic stability?

Policies range from basic legal protections (e.g., maintaining the rule of law, enforcement of property rights) to promoting a culture of debt repayment, transparency, the consistency of financial data reporting, and the relative ease of doing business.

Political factors include stability of domestic political institutions and the absence of conflict with neighboring countries.

These factors are captured using a relative ranking or scoring approach, i.e., countries with better institutions and policies will get a higher rank.

In addition to a sovereign government's ability to pay, we must also evaluate its willingness to pay. Willingness to pay is important because of the principle of sovereign immunity, which limits investor's ability to force sovereign government to declare bankruptcy or liquidate its assets to settle debt claims.

Fiscal Flexibility: How well does a sovereign government establish and maintain fiscal discipline over time and under different economic conditions?

Are tax collections properly enforced? Is the allocation to public expenditures prudent? Is the outstanding sovereign debt properly managed in relation to economic activity?

Monetary Effectiveness: Monetary policy is conducted by the central bank of a country. At one extreme, central banks may implement policy almost entirely independent of government intervention and influence; at the other extreme, they may simply act as the government's agent.

Central bank independence from the public Treasury reduces the likelihood that a sovereign government will monetize its domestic debt, raising domestic inflation and lowering the domestic currency's external value.

Economic Flexibility: Key factors to gauge the economic flexibility of a country include not only the size of the economy and level of per capita income, but also the degree of economic diversification and growth potential.

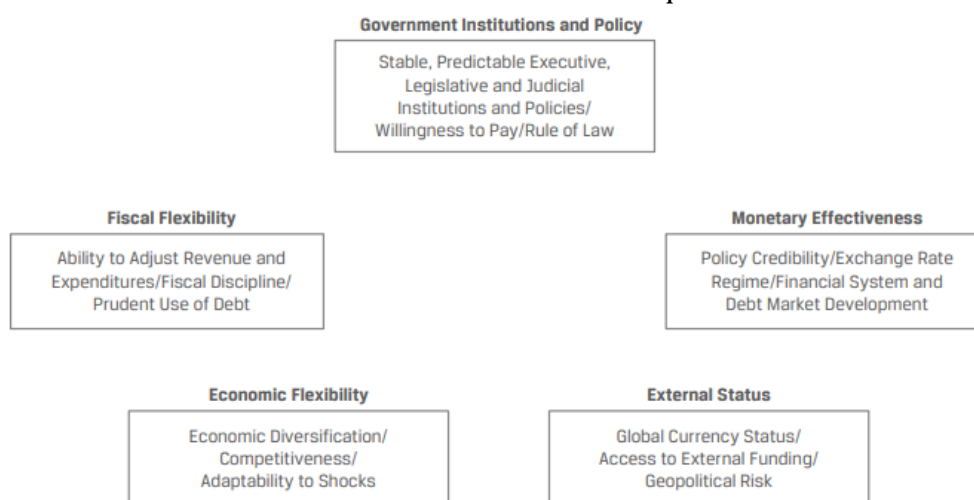
Highest rated sovereign borrowers typically have an advanced and highly diversified domestic economy with strong, sustainable growth prospects. Whereas, emerging or frontier market economies often depend on a single industry or commodity. They may also have a sizable informal economy and may face challenges in collecting tax revenue.

External Status: How do a sovereign government's international trade, capital, and foreign exchange policies influence its ability to support and service outstanding debt?

A key point to consider is whether a country's domestic currency is considered to be a 'reserve currency', i.e., one that is fully convertible and frequently held by foreign central banks and other investors as a portion of their foreign exchange reserves. A reserve currency status greatly expands a government's ability to access foreign investors using domestic currency debt. This flexibility minimizes the likelihood of sovereign default and increases a government's ability to sustain higher level of debt.

In contrast, many emerging and frontier market countries have exchange rate restrictions, capital controls, and/or a lack of full convertibility, which limits their ability to borrow from foreign entities.

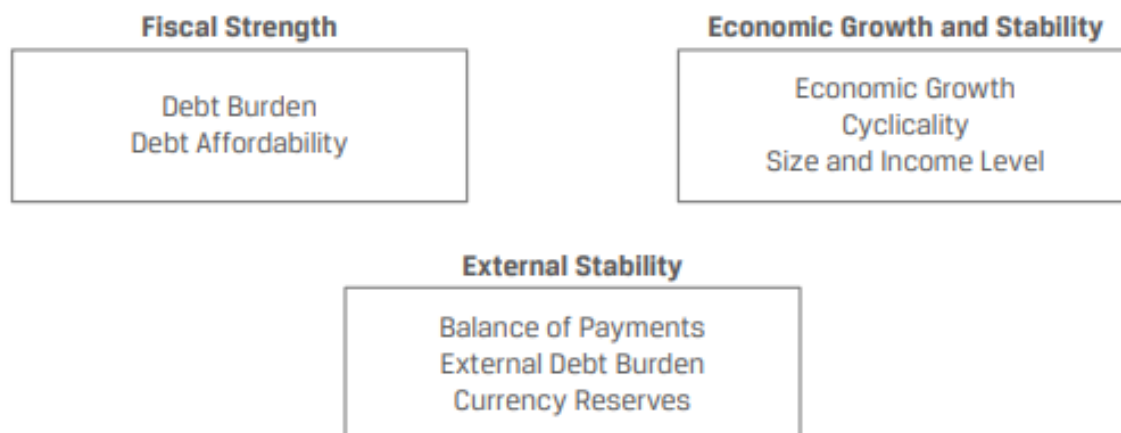
Exhibit 2 from the curriculum summarizes these qualitative factors.



The factors are often interrelated, e.g., a weak legal system often goes hand in hand with a limited ability to collect taxes or enforce debt contracts.

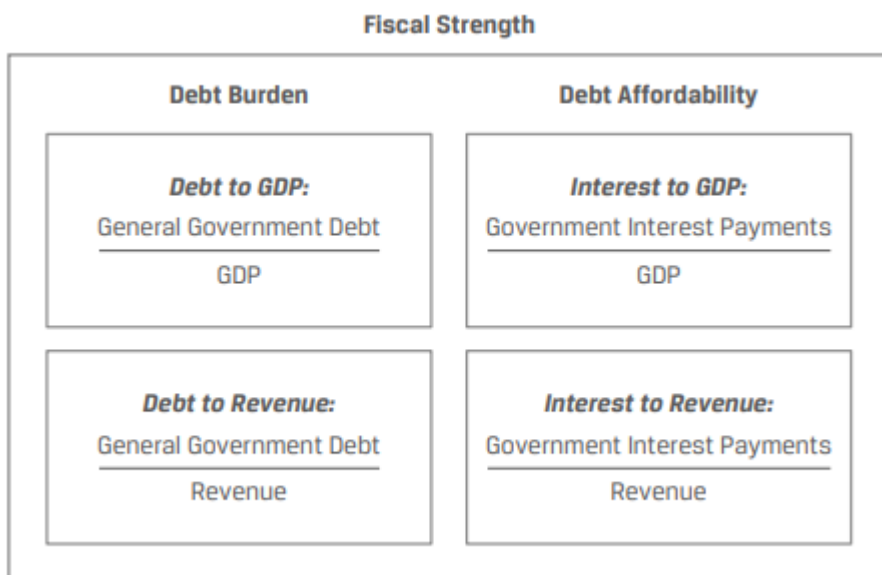
Quantitative Factors

Quantitative factors used to analyze a sovereign's creditworthiness are shown in Exhibit 3.



Fiscal Strength:

Key financial ratios that measure sovereign fiscal strength are shown in Exhibit 4.

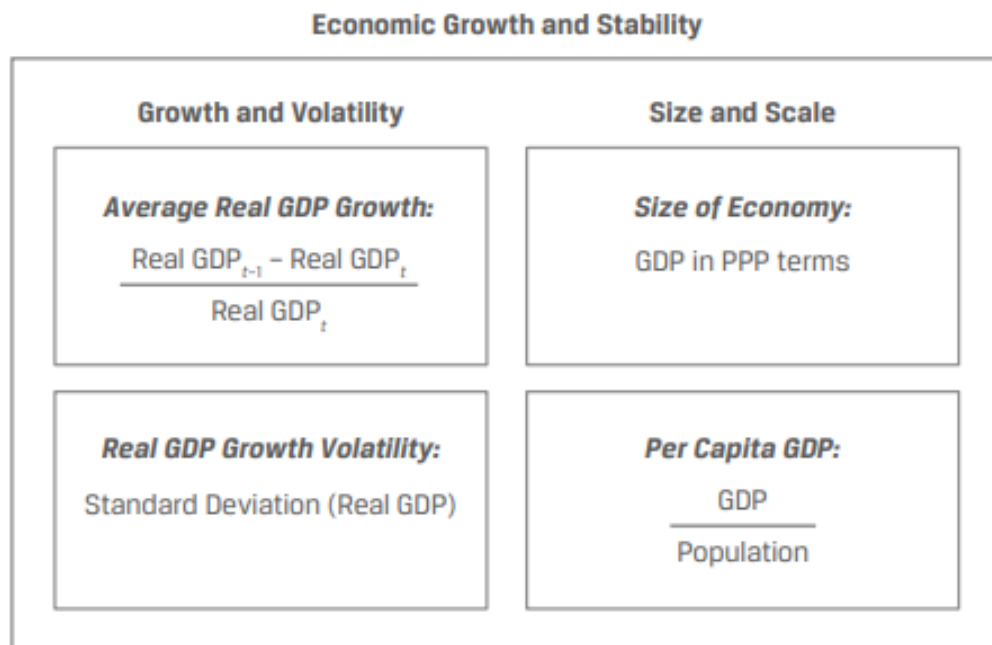


Debt burden provides an indication of a government's solvency. A higher debt burden indicates lower credit quality. Analysts should monitor whether a country's debt burden is improving or deteriorating over time.

Debt affordability provides a measure of debt coverage. A higher ratio indicates lower credit quality.

Economic Growth and Stability:

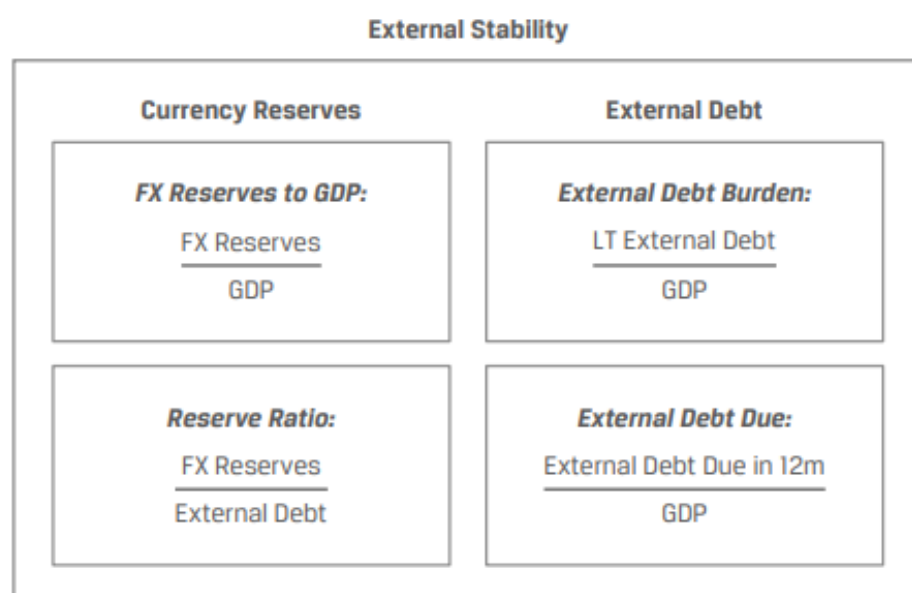
Key quantitative measures of economic growth and stability are shown in Exhibit 6.



These measures are important because larger, wealthier economies are better positioned to achieve sustainable growth and withstand economic and political shocks.

External Stability:

Key measures of external stability are shown in Exhibit 7.



These measures focus on the short- and long-term ability to generate sufficient, stable foreign currency cash inflows that can be used to meet interest and principal payments on external debt as well as other obligations to foreign investors.

3. Non-Sovereign Credit Risk

The main type of non-sovereign government issuers include agencies, public banks, supranational, and regional governments.

Agencies

Agencies are quasi-government entities whose primary function is to carry out a government-sponsored mission to provide public services, that is frequently based on a specific sovereign law or statute. For example, Airport Authority Honk Kong SAR (AAHK) is a statutory body owned by the Honk Kong SAR government that is responsible for operating and developing the Honk Kong International Airport.

Agencies can finance their activities using debt. Rating agencies typically grant agencies the same rating as the sovereign entity.

Government Sector Banks and Development Financing Institutions

Sovereign governments often sponsor the establishment of specialized financial intermediaries to operate in a specific market or to promote specific sovereign political, economic, social, or other growth and policy objectives. For example, Kreditanstalt für Wiederaufbau (KfW) is the largest national development bank established in the Federal Republic of Germany. It is 80% owned by the German federal government and 20% by the federal states.

These institutions also typically get the same rating as the sovereign entity.

Supranational Issuers

Supranational entities are organizations that are established and owned by sovereign governments that join as members to pursue a common objective. For example, the World Bank, Asian Development bank (ADB), the Development Bank of Latin America.

The rating of these entities depends on its members. For example, The Indonesia Infrastructure Finance (IIF) was established by the Government of the Republic of Indonesia (30% shareholding) along with World Bank (20% shareholding), ADB (20% shareholding), and other multilateral institutions, like KfW (15% shareholding) and Sumitomo Mitsui Banking Corporation (15% shareholding). . IIF's credit rating is BBB, equivalent to the sovereign rating of the Government of Indonesia.

Regional Government Issuers

This category includes provincial, state, and local governments. In the US, municipal issuers (state and local governments) are rated individually. They typically issue either general obligation or revenue bonds:

General obligation bonds

- GO bonds are unsecured and issued with the full faith and credit of issuing government.
- Credit analysis focuses on employment, per capita income, per capita debt, tax base, etc.
- Over-reliance on one or two types of tax revenue (e.g., capital gains or sales tax) can signal credit risk.

Revenue bonds

- Issued for specific project financing like toll road, bridge, etc.
- Credit analysis is similar to corporate bonds.
- Higher risk than GO bonds because of a single source of revenue.
- A key metric to calculate: debt-service-coverage ratio (DSCR), i.e., how much revenue to cover debt service payments. The higher the DSCR, the stronger the creditworthiness.

Summary

LO: Explain special considerations when evaluating the credit of sovereign and non-sovereign government debt issuers and issues.

Sovereign bonds:

A combination of qualitative and quantitative factors is used to assess both:

- an issuer's ability (Ability to pay will be high for debt issued in the country's own currency than for debt issued in foreign currency).
- willingness to pay its debt obligations.

Non-sovereign government debt

- Non-sovereign debt is usually either general obligation bonds or revenue bonds.
- General obligation (GO) bonds are backed by the taxing authority of the issuing non-sovereign government. Therefore, credit analysis of GO bonds is similar to sovereign analysis.
- Revenue-backed bonds support specific projects, such as toll roads, bridges, airports, and other infrastructure. Therefore, credit analysis of revenue bonds depends on the income generating capacity of the project.

LM16 Credit Analysis for Corporate Issuers IFT Notes

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2. Assessing Corporate Creditworthiness.....	2
3. Financial Ratios in Corporate Credit Analysis.....	4
4. Seniority Rankings, Recovery Rates, and Credit Ratings	5
Summary.....	8

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Version 1.0

1. Introduction

This learning module covers:

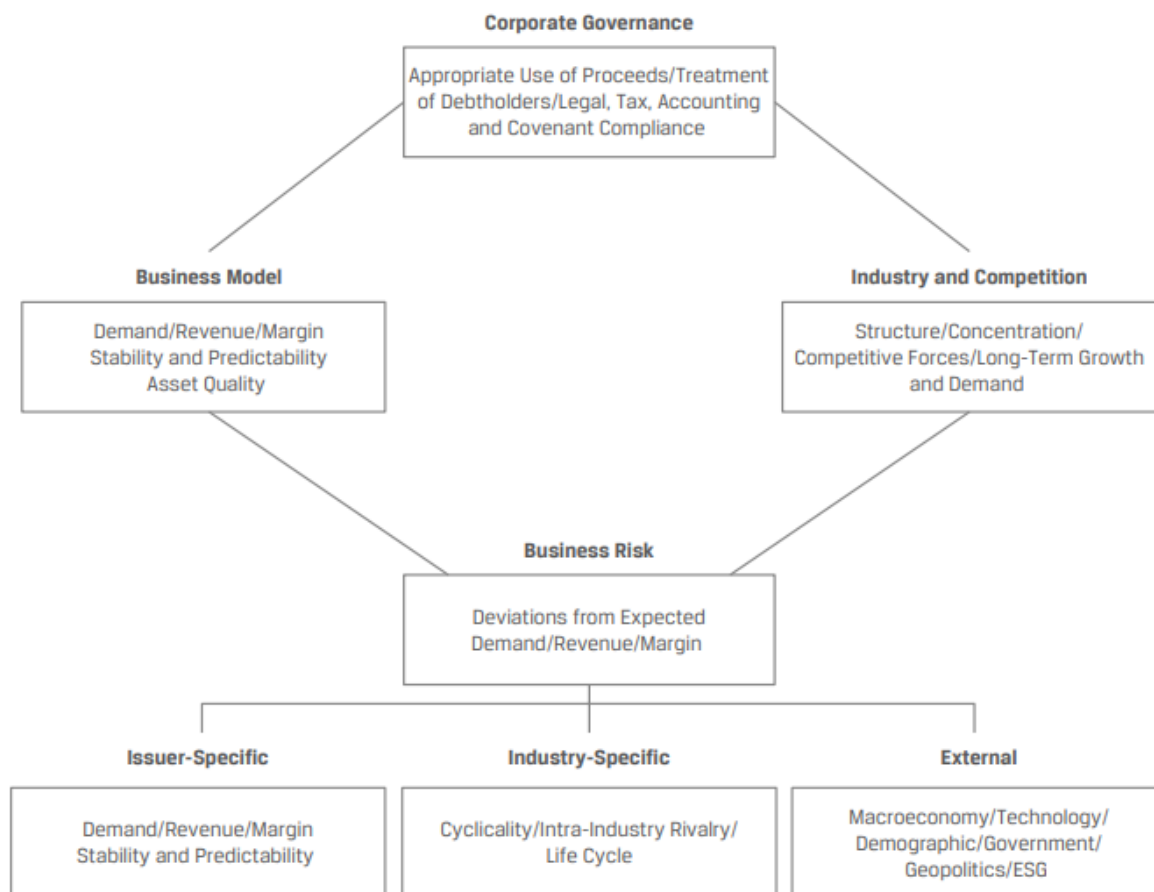
- The qualitative and quantitative factors used to evaluate the creditworthiness of a corporate borrower
- Financial ratios used in credit analysis
- Seniority rankings of debt and their impact on credit ratings

2. Assessing Corporate Creditworthiness

The creditworthiness of a company primarily depends on its ability to generate profits and cash flows sufficient to meet interest and principal payments. A combination of qualitative and quantitative factors is used to evaluate a corporate's creditworthiness.

Qualitative Factors

Exhibit 1 from the curriculum shows the key qualitative factors used to gauge a company's ability to satisfy its debt obligations.



Established firms with a business model characterized by stable and predictable cash flows, low business risk, and less competitive pressures have a greater capacity to use debt in their capital structure and a lower likelihood of default. In contrast, firms with lower and less

stable cash flows, higher business risk, and/or greater competition have a lower capacity to use debt in their capital structure and a higher likelihood of default.

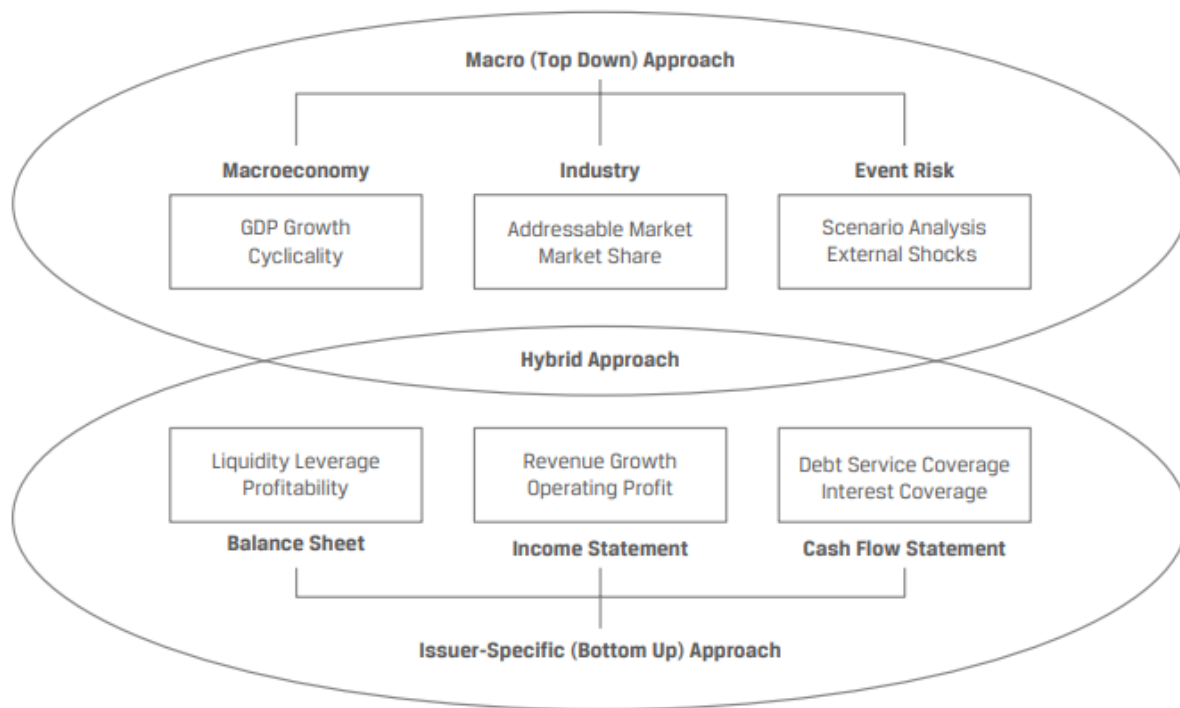
Equity vs Fixed Income analysts: While equity analysts focus on all future cash flows, fixed-income analysts focus on how a company's creditworthiness may change over a shorter time frame given the finite nature of debt claims.

Secured vs Unsecured debt: In case of unsecured debt, the company cash flows are the primary source of repayment. In case of secured debt, specific company assets are designated as collateral and serve as a secondary source of repayment. Secured lenders prefer to have tangible (or hard) collateral rather than intangible (or soft) collateral. The use of collateral and secured debt can reduce the cost of borrowing as compared to unsecured alternatives.

Corporate governance is also an important factor to consider in the analysis. Unlike shareholders who have voting rights on important company matters, debtholders seek to specify what borrower restrictions will apply and how the debt proceeds will be used at the time of issuance of debt. Analysts should also evaluate if the company uses aggressive accounting policies (e.g., use of significant off-balance sheet financing, capitalizing versus expensing items, early and premature revenue recognition). These items are considered potential warning flags.

Quantitative Factors

Exhibit 2 from the curriculum shows the key quantitative factors used to measure a company's credit risk.



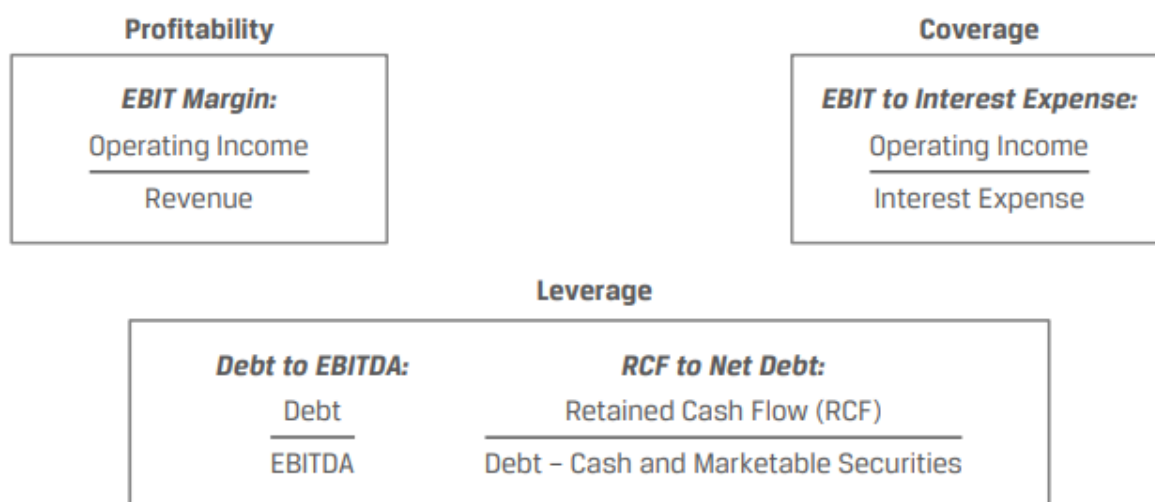
Top-down analysis typically begins with a macroeconomic forecast, which gauges a company's growth relative to GDP, estimates a firm's addressable market and market share, and assesses the likelihood and impact of potential adverse events.

Bottom-up analysis involves forecasting key revenue drivers and balance sheet positions.

A hybrid approach combines expected cyclicalities with changing bottom-up features to forecast a company's cash flows.

3. Financial Ratios in Corporate Credit Analysis

Exhibit 3 from the curriculum shows the common financial ratios used to evaluate a company's credit risk.



EBIT Margin: Earnings before interest and taxes (EBIT) helps determine a company's operating performance prior to capital costs and taxes. Higher EBIT margins increase profits available to service debt.

EBIT to Interest Expense: This measures the degree to which operating profit covers periodic interest payments. A higher coverage ratio represents less credit risk.

Debt to EBITDA: A higher debt to (Earnings before interest, taxes, depreciation and amortization) EBITDA ratio indicates increased leverage and higher credit risk.

Retained cash flow (RCF) to Net Debt: This ratio focuses on cash flow rather than accounting earnings measures. The denominator uses 'net debt' which is debt reduced by available cash. A high RCF to net debt ratio implies lower leverage and lower credit risk.

The ratios are compared with the industry average and credit rating peers. Credit analysts also evaluate how these ratios may change over time as revenue drivers change.

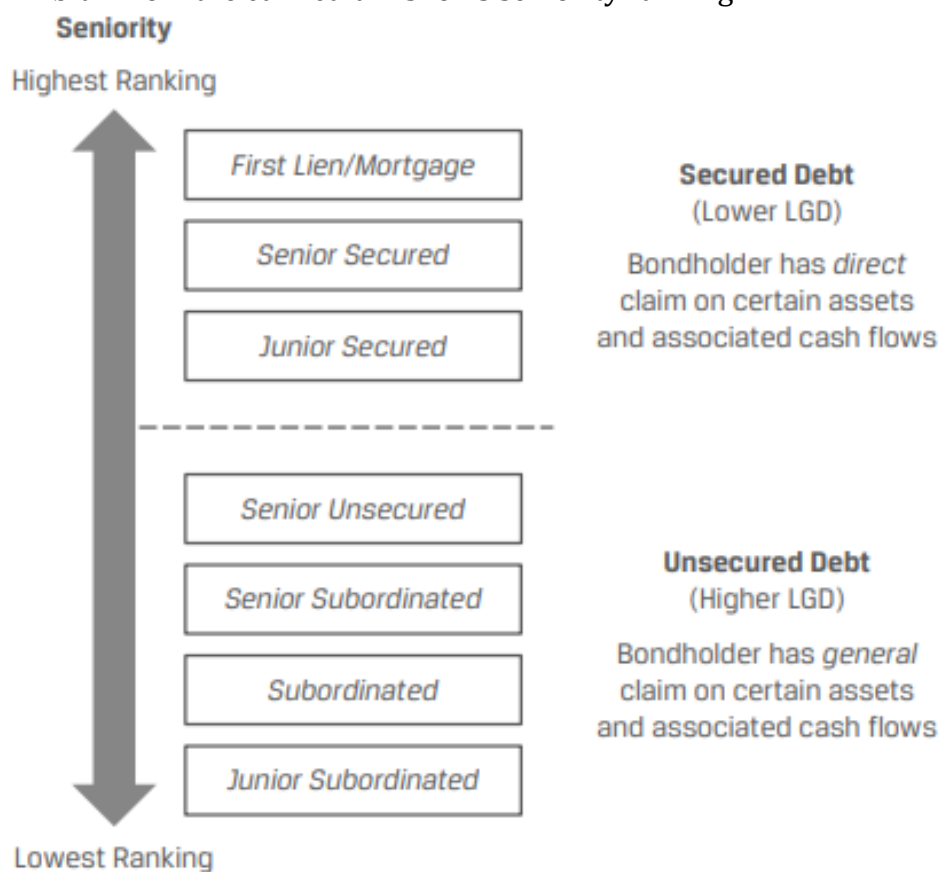
4. Seniority Rankings, Recovery Rates, and Credit Ratings

Seniority Rankings

A single borrower may issue debt with different maturity dates and coupons. These various bond issues may also have different seniority rankings. Seniority ranking determines who gets paid first, or who has the first claim on the cash flows of the issuer, in the event of default/bankruptcy/restructuring. Bondholders are broadly classified into secured and unsecured.

- **Unsecured bonds are known as debentures.** They have a general claim on an issuer's assets and cash flows.
- *Secured bondholders* have the first (direct) claim on cash flows in the event of a default.
- Priority of claims determines who gets paid first.

Exhibit 4 from the curriculum shows seniority ranking.



Secured versus Unsecured Debt

Within each category of debt, there are types and sub-rankings. Let us look at each of the items in detail now (in the order of priority for repayment):

Secured debt: Highest ranked debt in which some asset is pledged as collateral.

- *First mortgage debt:* A specific property is pledged.
- *First lien debt:* A pledge of certain assets such as buildings, patents, brands, property, and equipment, etc.
- *Second lien:* Some more assets could be pledged as a second lien to rank below the first mortgage/first lien. Most second lien loans are senior secured obligations of the borrower. They rank above the junior secured obligations.

Unsecured debt: Loss severity can be as high as 100%. Lowest priority of claims. No asset is pledged as collateral. The types are:

- Senior unsecured
- Senior Subordinated
- Subordinated
- Junior subordinated

Companies issue debt with different ranking for the following reasons:

- Cost effective: Secured debt has a lower cost due to reduced credit risk.
- Less expensive than equity: From a borrower's perspective, even though the cost of issuing subordinate debt may not be as low as secured debt, it is less restrictive and is relatively less expensive than equity.
- Investors invest in subordinated debt because the higher yield compensates the high risk assumed.

Recovery Rates

Recovery rate is the percentage of the principal amount recovered in the event of default.

Key points related to recovery rates are given below:

- *Pari passu:* All creditors at the same level of capital structure fall under the same class, and they will have the same recovery rates. For example, assume there are two creditors in the senior secured class; one with a 3-year bond and the other with a 1-year bond. The debt of the 1-year bondholder matures in 3 months while the debt of the 3-year bondholder matures in 2.5 years. If there is a default, both the investors will have the same recovery rates irrespective of the time left to maturity. This type of equal ranking for the same class of bonds is called *pari passu* (on equal footing).
- *Recovery rates vary by seniority ranking:* If the seniority ranking is high, recovery rate will also be high.
- *Recovery rates vary by industry:* If the industry is on a decline like the newspaper publishing industry, then the recovery rates will be low. There may be companies going bankrupt in industries during a recession (e.g., the steel industry in 2011-12), but in general the recovery rates here are higher than those on a decline.
- *Recovery rates depend on when they occur in a credit cycle:* For instance, if a default happened in 2008 at the peak of the recent credit crisis, then the recovery rates would have been lower relative to, let us say, in 2004-05.
- *Recovery rates vary across industries and across companies within an industry.*

- *Priority of claims is not always absolute:*
 - Secured holders have a claim to cash flows/asset in the event of a default before anyone else. However, if the value of the pledged asset is less than the claim, then the unpaid amount becomes the senior unsecured claim.
 - Unsecured creditors have a right to be paid before common/preferred shareholders.
 - Senior creditors take priority over junior/subordinated creditors.

Issuer and Issue Ratings

Rating agencies provide two ratings:

- Issuer rating: This is based on the creditworthiness of the issuer. It applies to senior unsecured debt. Also known as **corporate family rating (CFR)**.
- Issue rating: This rating is assigned to a specific debt/financial obligation. It takes into consideration the seniority ranking of the debt within the capital structure. Also known as **corporate credit rating (CCR)**.

Cross-default provision: It is believed that the probability of default on one issue is linked to other issues of the same issuer. That is, a default on one issue triggers default on other issues with the same default probability.

Notching: A rating methodology to distinguish rating between different liabilities (bond issues) of an issuer. The objective is that two securities with the same rating should have the same expected loss rate (probability of default * expected severity loss). Credit rating on issues can be moved up or down by a notch based on the risk of default/severity loss.

Factors the rating agencies consider while assigning ratings (notching up/down):

- Probability of default: it is the primary factor that drives the rating.
- Priority of payment: who gets paid first in the event of a default; secured/senior unsecured/subordinated?
- Structural subordination: this is when the issuer is a holding company rather than an operating company. Assume both the holding company and each of its operating subsidiaries has issued bonds. The debt of the subsidiaries gets serviced by its cash flows and assets first before any of it can be passed on to the holding company. All else equal, debt issued by the holding company will have a lower rating than the debt issued by the subsidiary.

Summary

LO: Describe the qualitative and quantitative factors used to evaluate a corporate borrower's creditworthiness.

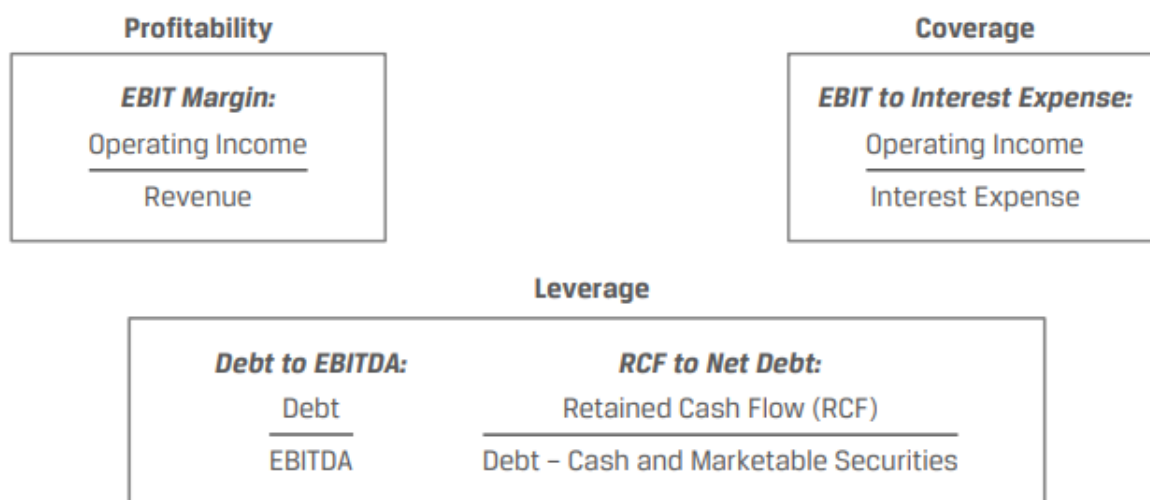
A combination of qualitative and quantitative factors is used to evaluate a company's creditworthiness.

Qualitative factors include: a company's business model, its industry, and its competitive position and business risks, as well as its corporate governance.

Quantitative factors include:

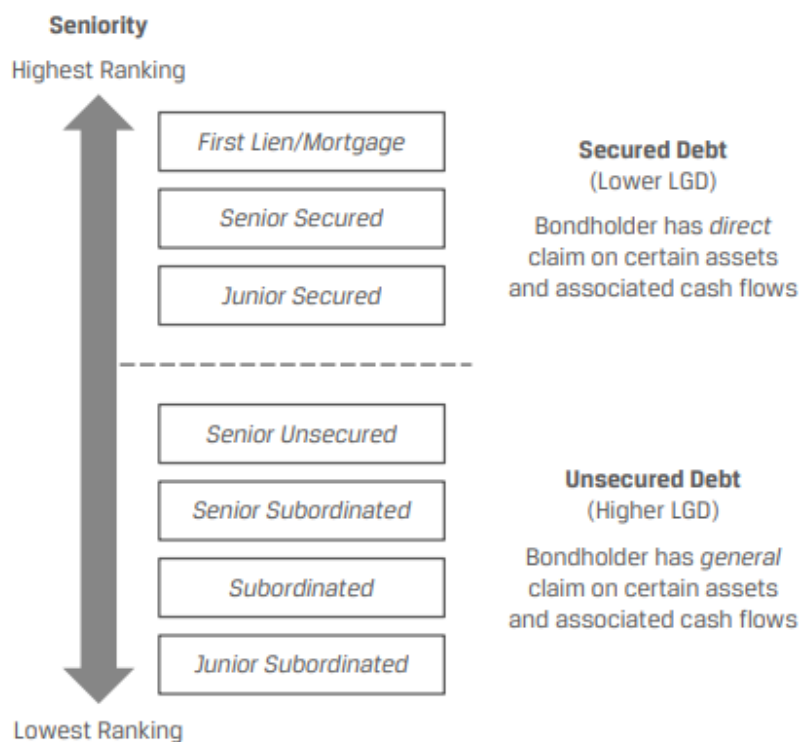
- Top-down analysis that typically begins with a macroeconomic forecast. It gauges a company's growth relative to GDP, estimates a firm's addressable market and market share, and assesses the likelihood and impact of potential adverse events.
- Bottom-up analysis which involves forecasting key revenue drivers and balance sheet positions.
- A hybrid approach that combines expected cyclicalities with changing bottom-up features to forecast a company's cash flows.

LO: Calculate and interpret financial ratios used in credit analysis.



LO: Describe the seniority rankings of debt, secured versus unsecured debt and the priority of claims in bankruptcy, and their impact on credit ratings.

Corporate debt is ranked by seniority or priority of claims; the following diagram shows this:



The priority of claims in bankruptcy is not always absolute. In bankruptcy, the court may approve a repayment plan that does not follow the priority of claims order.

Rating agencies typically provide both issuer and issue ratings.

- Issuer ratings reflect an issuer's overall creditworthiness—its risk of default.
- Ratings for issues reflect factors such as their rankings in the capital structure.

LM17 Fixed-Income Securitization IFT Notes

1. Introduction.....2

2. The Benefits of Securitization.....2

3. The Securitization Process.....3

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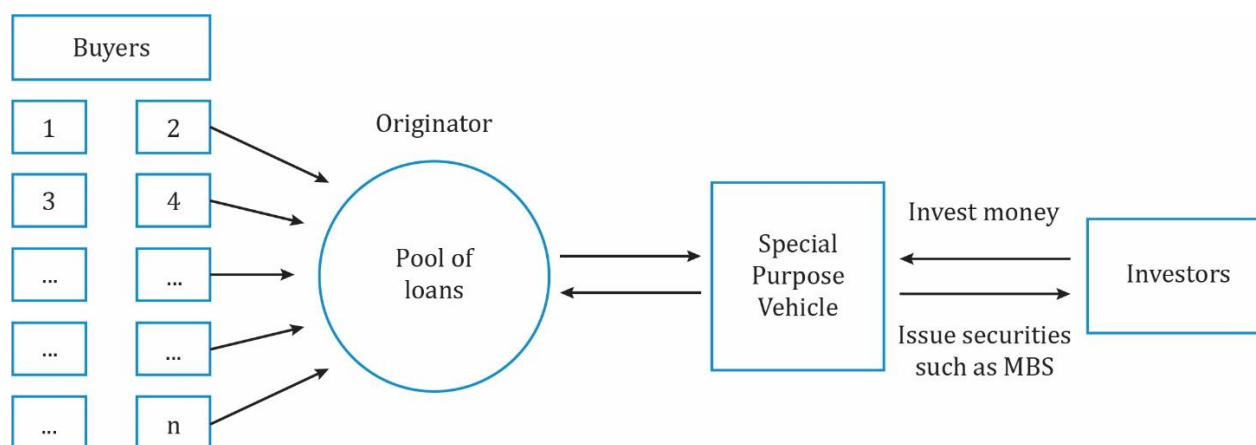
Version 1.0

1. Introduction

This learning module covers:

- The benefits of securitization
- The securitization process – the parties involved and the roles they play

Asset-backed securities (ABS) are based on a principle called **securitization**. The securitization process involves pooling relatively straightforward debt obligations, such as loans or bonds, and using the cash flows from the pool of debt obligations to pay off the bonds created in the securitization process. The instruments which become part of the pool are called securitized assets. A simple securitization process is illustrated in the figure below:



In this illustration, a mortgage bank sells mortgage loans to thousands of homeowners. The mortgage bank bundles the individual loans into a pool which is sold to a separate legal entity generally referred to as a special purpose vehicle (SPV). The special purpose vehicle issues bonds to investors. The collateral for the bonds is the pool of mortgage loans.

The term **mortgage-backed security** (MBS) is commonly used for securities which are backed by high quality real estate mortgages. The term “asset-backed securities,” or ABS, is a broader concept that refers to securities backed by other types of assets. In the example above, we can say that the SPV issues MBS.

2. The Benefits of Securitization

In this section, we look at the benefits from the perspective of the three parties involved in the securitization process: borrowers who are the homeowners, investors who want to buy mortgages, and the intermediary connecting these two parties which is a commercial bank/financial institution. Investors cannot lend directly to homeowners because they may be willing to lend/invest only a small amount of money, say \$10,000, whereas the homeowner may require \$100,000 as a mortgage loan. Second, the investor may not have all the information needed to assess the risk of the property.

Benefits to investors are as follows:

- Securitization converts an illiquid asset into a liquid security.
- It gives investors direct access to the payment streams of the underlying mortgage loans that would otherwise be unattainable.
- There are higher risk-adjusted returns to investors: pooling loans results in diversification and lower risk for investors.
- It gives investors an opportunity to buy a small part of the home buyers' mortgage in the form of a security issued by the SPV.
- It gives exposure to the market, real estate in this example, without directly investing in it.

Benefits to the bank or loan originator are as follows:

- It enables banks to increase loan origination, monitoring, and collections.
- It reduces the role of the intermediaries (known as disintermediation) like the bank. However, note that an intermediary is still required to package and distribute securities.
- Banks have the ability to lend more money if the demand for ABS and MBS is high relative to if the money was self-financed (from deposits, debt, equity, etc.).
- There is greater efficiency and profitability for the banking sector: the mortgage-backed securities, at least in the US market, trade actively in the secondary market which improves the efficiency and liquidity of the financial market.

Benefits to the borrowers of the loan are as follows:

- It lowers the risk as the pooled loans offer a diversification benefit.
- The lower risk decreases the cost of borrowing for homeowners.

3. The Securitization Process

We look at the securitization process in detail in this section.

An Example of a Securitization

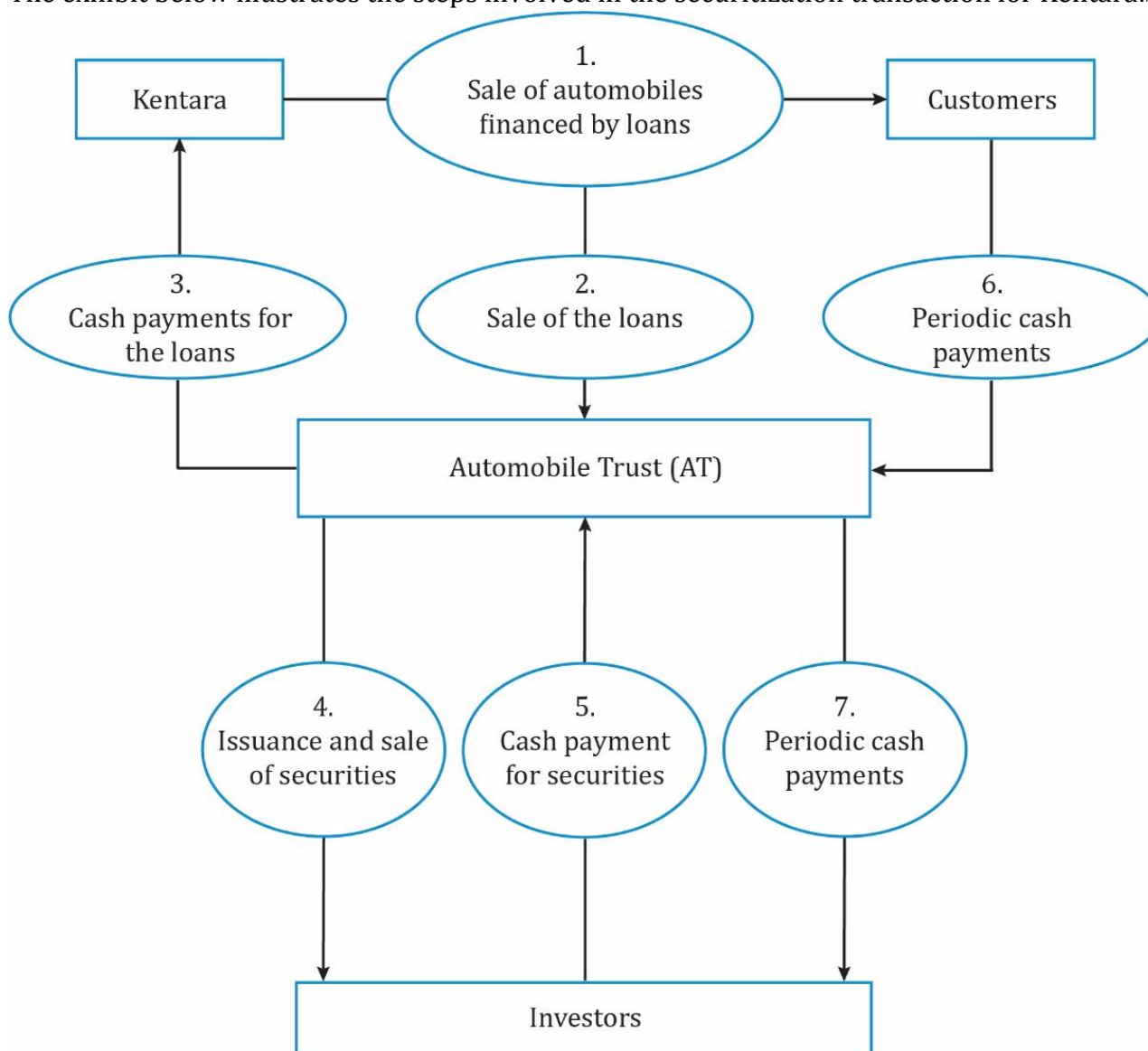
Kentara is a manufacturer of automobiles that range from \$20,000 to \$200,000. The majority of sales are made through loans granted by the company to its customers, and the automobiles serve as collateral for the loans. These loans, which represent an asset to Kentara, have maturities of five years, carry a fixed interest rate and are fully amortizing with monthly payments. Although the servicer of such loans need not be the originator of the loans, the assumption is that Kentara is the servicer.

Steps in the securitization transaction:

- Now assume that Kentara has \$100 million of loans, shown on its balance sheet as an asset, and Kentara wants to raise another \$100 million.
- Kentara can do this by securitizing the loans and selling them to a special purpose entity (SPE), Automobile Trust (AT).

- The SPE is a separate legal entity and is also called a special purpose vehicle (SPV) or a special purpose company. The legal form of the SPE varies by jurisdiction but, in almost all cases, the ultimate owner of the loans - AT - is legally independent and considered bankruptcy remote from the seller of the loans.
- AT gets \$100 million in cash from investors by selling asset-backed securities, which it pays to Kentara.
- The customers who bought the automobiles on loan make monthly payments. AT uses this cash flow to pay investors of the ABS.
- Setting up a separate legal entity ensures that if Kentara files for bankruptcy, the loans backing the ABS that are issued by AT are secure within the SPE and creditors of Kentara have no claim on the loan.

The exhibit below illustrates the steps involved in the securitization transaction for Kentara:



Parties to a Securitization

Let us look at the parties involved in a securitization transaction.

<i>Party</i>	<i>Role</i>	<i>Party in our example</i>
Seller of pool of securities	Originates the loans and sells to a special purpose entity (SPE).	Kentara
SPE or trust or issuer	Buys the loans from the seller and issues ABS.	Automobile Trust (AT)
Servicer	Services loans such as collecting payments from borrowers, notifying borrowers who may be delinquent and, if necessary, seizing automobiles from borrowers who do not make payments on time.	Kentara

Note that a third party or the issuer may act as a servicer.

Other parties involved include independent accountants, underwriters, trustees, rating agencies, and guarantors.

- Lawyers: Responsible for creating the legal documentation.
- Underwriters: Facilitate bond issuance.
- Rating agencies: Rate the securities issued.

Trustee: Safeguards the assets placed in the trust, and hold funds to be paid to bondholders.

Structure of Securitization

In the Kentara example, assume 100,000 securities of Bond Class A were issued with a par value of \$1,000 per security to raise \$100 million. All the certificate holders in this case are treated equal because there is just one class of bondholders and there is no distinction between bondholders with respect to payment time or credit risk.

However, in reality not all bond issues are created with a similar structure. The motivation for the creation of different types of structures is to redistribute prepayment risk and credit risk efficiently among different bond classes in the securitization.

Prepayment risk is the uncertainty that the actual cash flows will be different from the scheduled cash flows, as set forth in the loan agreements, because borrowers may alter payments to take advantage of interest rate movements.

Instructor's Note

Prepayment risk cannot be eliminated, but can be redistributed.

In the Kentara example, assume the buyers of automobiles are scheduled to make monthly payments towards the loan over 5 years. Instead, they prepay it in two years, perhaps because the interest rates decline, or due to any other reason. Since the loan is prepaid

quicker than planned, the investors of ABS will also be paid quickly thereby reducing their interest income.

Time tranching: Creation of bond classes to distribute the prepayment risk is called time tranching. Cash flow received from customers is distributed among the tranches based on certain parameters. For example, assume AT issued the following four bond classes, with a total par value of \$100 million, instead of one bond class:

- A1 (\$30 million)
- A2 (\$20 million)
- A3 (\$30 million)
- A4 (\$20 million)

Since the motive is to distribute prepayment risk, the A1 class may have a lower prepayment risk than A4. If customers prepay, then A4 bond-holders will get prepaid before A1.

Subordination and credit tranching: Subordination is another layering structure in securitization. The bond classes differ in their exposure to credit risk, i.e., how they share losses if the borrowers of the original loans default. An ABS is made up of a pool of loans. So, any default in payment will have a cascading effect on the investors. Here, several tranches of senior and subordinated classes are created and the credit risk is distributed to each class in a disproportionate manner based on the investor's choice.

<i>Bond class</i>	<i>Par Value (\$ millions)</i>
A (senior)	80
B (subordinated)	14
C (subordinated)	6
Total	100

In this example, all the losses are first absorbed by class C, then class B, and then class A. However, class C can accept a loss of up to \$6 million. Beyond that, it is absorbed by class B. The risk is highest for class C and lowest for class A, in this example. Based on the high risk high return rule, the expected return of class C bondholders will be higher than that of class A bondholders.

The Role of the SPE

The securitization of a company's assets may include some bond classes that have better credit ratings than the company itself or its corporate bonds. Thus, in the aggregate, the company's funding cost is often lower when raising funds through securitization than by issuing corporate bonds.

To understand why the funding cost is lower, we will go back to the Kentara example and consider two scenarios for raising \$100 million: one in which Kentara issues a corporate bond, and another in which it issues ABS by securitizing loans/receivable.

Corporate bond scenario: Kentara issues corporate bonds for \$100 million with auto loans as collateral. Assume credit-rating agencies such as Moody's assign Kentara a credit-rating of

BB (below investment-grade). The corporate bond rating will also be based on the company's credit rating as it reflects the creditworthiness of debt securities. Kentara's credit spread depends on the following two factors:

- Primarily, credit rating (BB, in this case).
- Collateral, to a lesser extent.

The cost of funding for Kentara will be higher if it issues a corporate bond, and not an ABS, for the following reasons:

- *Higher risk:* Investors perceive a higher risk given the company's creditworthiness. In case the company goes bankrupt or is reorganized, their claim to assets will follow the absolute priority rule (defined in the next section). Though, in reality the absolute priority rule has not been upheld in case of reorganizations. This means that it is not necessary for the bondholders to be paid off before the other parties (equity holders, other creditors). Hence, the credit spread for a corporate bond backed by a collateral does not decrease substantially.
- *Higher return:* To compensate for the high risk, investors expect a high return.
- *Higher credit spread:* Credit spread is the difference between the interest rate the issuer has to pay on the corporate bond and the benchmark interest rate. The riskier the bonds, the larger the spread demanded by investors as compensation for risk.

Securitization Scenario

- *Funding cost is low:* The collateral (loans/receivable) is legally an asset of AT. Any cash flow from the pool of loans will be paid to the investors of ABS. When an investor buys a bond class, he has to evaluate the credit risk of the class he is investing in. The credit rating of the bond class will depend on the quality of the collateral and capital structure of the SPV, and not the credit rating of the company as in the corporate bond.
- *The lower the risk, the lower the funding cost:* The assets belong to the SPV. If the company goes bankrupt, the absolute priority rule is followed. The principle is that senior creditors are paid in full before subordinate bondholders are paid anything. So, investors demand a lower return than a corporate bond. Lower return means lower funding cost for the issuer.
- *The SPV is a bankruptcy-remote vehicle unlike corporate bonds.* It means bankruptcy has no effect on an SPV. If the company goes bankrupt, the loans/receivable do not belong to Kentara anymore and the investors will be paid based on the securitization structure.

Summary

LO: Explain benefits of securitization for issuers, investors, economies, and financial markets.

The benefits to investors are:

- Securitization converts an illiquid asset into a liquid security.
- It gives investors direct access to the payment streams of the underlying mortgage loans that would otherwise be unattainable.
- There are higher risk-adjusted returns to investors: pooling of loans results in diversification and lower risk for investors.
- It gives investors anywhere an opportunity to buy a small part of homebuyers' mortgage in the form of a security issued by the SPV.
- Exposure to the market, real estate for example, without directly investing in it.

The benefits to the bank or loan originator are:

- It enables banks to increase loan origination, monitoring, and collections.
- It reduces the role of the intermediaries like the bank.
- Banks have the ability to lend more money if the demand for ABS and MBS is high relative to if the money was self-financed.
- There is greater efficiency and profitability for the banking sector.

The borrowers of the loan benefit from securitization in the following ways:

- It lowers the risk as the pooled loans offer a diversification benefit. The lower risk, in turn, decreases the cost of borrowing for homeowners.

LO: Describe securitization, including the parties and the roles they play.

The parties involved in the securitization process include:

- SPV (Special Purpose Vehicle): It is also called the trust or the issuer.
- Seller of pool of securities: It is also known as the originator or depositor.
- Servicer: Servicing involves collecting payments from borrowers, notifying borrowers who may be delinquent, and if necessary, seizing equipment from borrowers who default.
- Other parties: Independent accountants, underwriters, trustees, rating agencies, and guarantors.

The motivation for the creation of different types of structures is to redistribute prepayment risk and credit risk efficiently among different bond classes in the securitization.

Prepayment risk is the uncertainty that the actual cash flows will be different from the scheduled cash flows.

Time tranching is the creation of bond classes to distribute the prepayment risk.

Subordination is another layering structure in securitization. The bond classes differ in their exposure to credit risk.

LM18 Asset-Backed Security (ABS) Instrument and Market Features

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Version 1.0

1. Introduction

This learning module covers:

- Covered bonds
- Typical credit enhancement structures used in securitizations
- Non-mortgage asset-backed securities
- Collateralized debt obligations

2. Covered Bonds

Covered bonds are senior debt obligations issued by a financial institution and backed by a segregated pool of assets that typically consist of commercial or residential mortgages or public sector assets.

Covered bonds are similar to ABS, but they differ because of their:

- **Dual recourse nature:** Investors have claims against both the issuing financial institution and the underlying asset pool.
- **Balance sheet impact:** The underlying asset pool remains on the issuing financial institution's balance sheet. It is not transferred to a separate SPV. The covered pool bondholders retain a top-priority claim against the pool.
- **Dynamic cover pool:** The underlying asset pool is not static. The issuing financial institution must replace any prepaid or non-performing assets in the cover pool to ensure sufficient cash flows until maturity. In contrast, ABS pass through default and prepayment risk to investors.
- **Redemption regimes in the event of sponsor default:** In the event of sponsor default, redemption regimes align the covered bond's cash flows as closely as possible to the original maturity schedule.
 - In the case of hard-bullet covered bonds, if payments do not occur according to the original schedule, a bond default is triggered and bond payments are accelerated.
 - Soft-bullet covered bonds delay the bond default and payment acceleration of bond cash flows until a new final maturity date, which is usually up to a year after the original maturity date.
 - Conditional pass-through covered bonds convert to pass-through securities after the original maturity date if all bond payments have not yet been made.

Because of these additional safety features, covered bonds usually have lower credit risks and therefore lower yields as compared to otherwise similar ABS.

3. ABS Structures to Address Credit Risk

Credit Enhancement

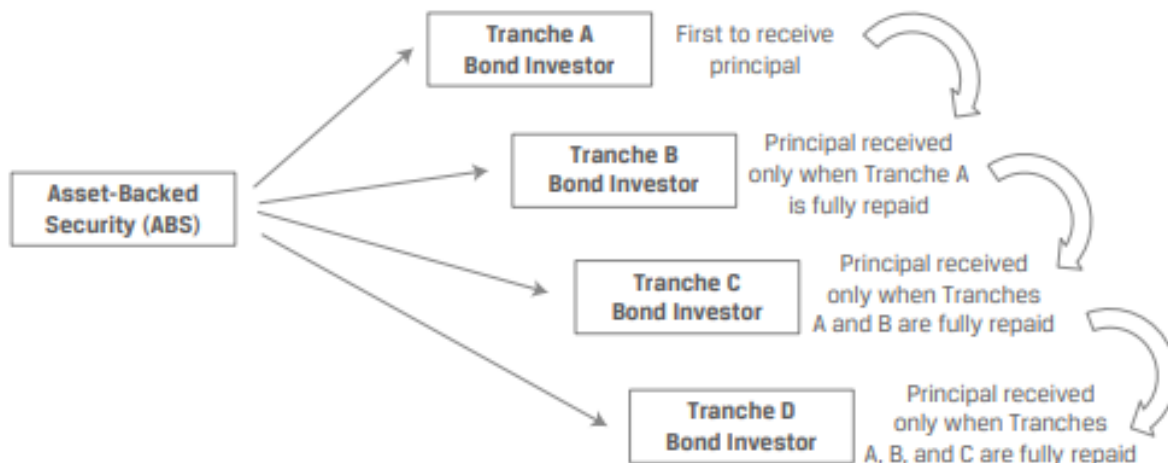
Three main types of internal credit enhancements widely used in securitization transactions are:

- Overcollateralization: The value of the collateral exceeds the face value of the issued bonds. This provides a safety cushion to investors.
- Excess spread: This is the difference between the coupon on the underlying collateral (say 8%) and the coupon paid on the securities (say 6%). The excess spread (8% - 6% = 2%) can absorb collateral shortfall or be used to build up reserves.
- Subordination or credit tranching: More than one bond class or tranches are created, and they differ in how they share any losses resulting from defaults in the collateral pool.

External credit enhancements like financial guarantees by banks or insurance companies, letters of credit, and cash collateral accounts are also commonly used. These are not covered in the curriculum.

Credit Tranching

Credit tranching was covered in the previous learning module. Exhibit 3 from the curriculum illustrates this 'waterfall structure'.



The four tranches have differing yields and degrees of risk, and they are affected differently in the event of a default.

Class A, the most senior investor tranche with the highest ranking in the capital structure, receives principal payments first, carries the least risk, and provides the lowest return. Class B mezzanine tranche assumes slightly higher risk but offers higher returns. Class D absorbs losses first and has the highest risk, but also the highest potential yield. Because of the

residual nature of its claims to cash flows from the asset pool, the lowest tranche is sometimes referred to as the "equity" tranche.

4. Non-Mortgage Asset-Backed Securities

A wide range of assets apart from mortgage loans are used as collateral for asset-backed securities (e.g. auto loans, credit card receivables, personal loans, commercial loans).

Based on the way the collateral pays, ABS can be categorized into two types: amortizing and non-amortizing. Examples of amortizing loans backing an ABS: mortgage loans and automobile loans. An example of non-amortizing loans backing an ABS: credit card receivables.

This learning module focuses on credit card receivable ABS and a residential solar ABS.

Credit Card Receivable ABS

Credit cards such as Visa and MasterCard are used to finance the purchase of goods and services, as well as for cash advances. When a cardholder makes a purchase using a credit card, he is agreeing to repay the amount borrowed (purchase amount) to the issuer of the card within a certain period, typically a month. If the outstanding amount is not repaid within this grace period, then a finance charge (interest rate) is applied to the balance not paid in full each month.

Credit card receivables are pooled together to act as a collateral for credit card receivable-backed securities. Cash flow, on a pool of credit card receivables consists of:

- Finance charges: These represent the periodic interest the credit card borrower is charged on the unpaid balance after the grace period.
- Fees and principal repayments. Fees include any late payment fees and any annual membership fees.

Characteristics of Credit Card Receivable-backed Securities

Payment Structure

- The security holders are paid an interest periodically (e.g., monthly, quarterly, or semiannually). The interest rate may be fixed or floating.
- The principal payments made by borrowers do not flow through to investors during a period known as the lockout period. Instead, the repayments are reinvested to issue new loans. As a result, credit card receivables increase during the lockout period. During this period, the cash flow to security holders comes from finance charges and fees.

Amortization Provision

- Credit card receivable-backed securities are non-amortizing loans. The principal is not amortized during the lockout period.
- Certain provisions in credit card receivable-backed securities require early amortization of the principal if certain events occur. Such provisions are referred to as

“early amortization” or “rapid amortization” provisions and are included to safeguard the credit quality of the issue. The only way the principal cash flows can be altered is by triggering the early amortization provision. For example, if issuers believe there may be a default in credit card repayments, then the principal repayments will be used to pay security holders (investors) instead of reinvesting to issue new loans.

Solar ABS

An increasing number of homeowners are installing solar energy systems, so many specialty finance companies have begun to offer specialized home improvement financing options: solar loans or solar leases. Solar loans involve borrowing money to purchase the system from an installer. Solar leases involve renting equipment directly from a solar company.

A solar ABS is created by securitizing solar loans. They have captured the interest of institutional investors because they provide the opportunity to contribute to sustainability while generating attractive risk-adjusted yields. Institutional investors looking for ESG or climate finance investment alternatives can invest in solar ABS.

ABS are typically collateralized by the underlying debt, such as mortgages, loans, or receivables. The loans can be further secured by pledging a lien on the installed systems, the property itself, or both. This effectively makes them a junior mortgage on the property and increases their safety for investors.

Solar loan borrowers are typically prime borrowers who own their homes and have a track record of timely payments. Solar ABS investors are also typically protected by overcollateralization, subordination, and excess spreads. Collectively, these features reduce the default risk in these securitizations even further.

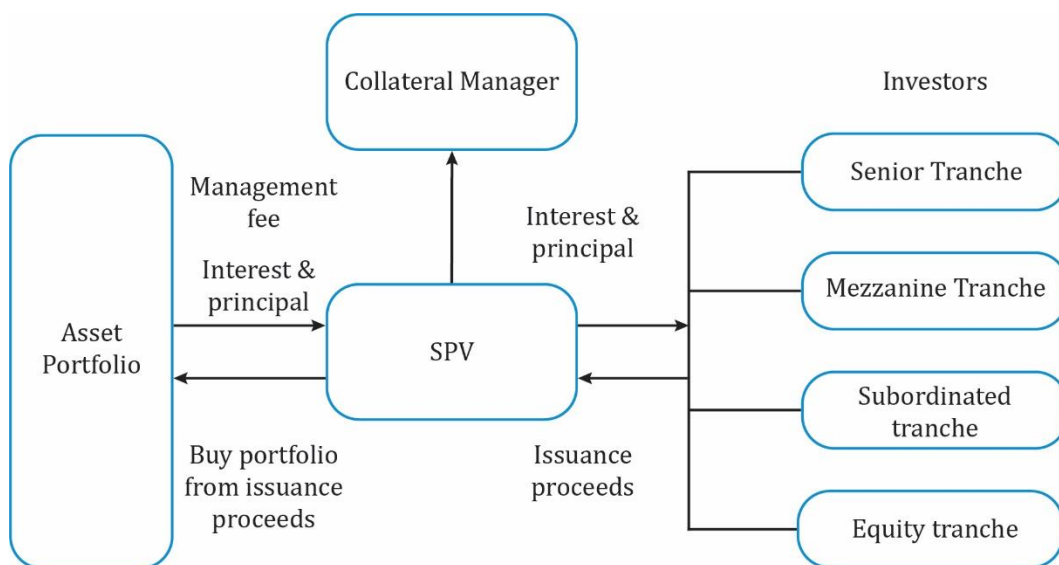
Many solar ABS contain a pre-funding period, which allows the trust to acquire additional qualifying transactions, during a certain period of time after the close of the initial transaction.

5. Collateralized Debt Obligations

A collateralized debt obligation is a generic term used to describe a security backed by a diversified pool of one or more debt obligations (e.g., corporate and emerging market bonds, leveraged bank loans, ABS, RMBS, CMBS, or CDO).

Like an ABS, a CDO involves the creation of a SPV. But, in contrast to an ABS, where the funds necessary to pay the bond classes come from a pool of loans that must be serviced, a CDO requires a collateral manager to buy and sell debt obligations, for and from the CDO's portfolio of assets, to generate sufficient cash flows to meet the obligations of the CDO bondholders, and to generate a fair return for the equity holders.

The structure of a CDO includes senior, mezzanine, and subordinated/equity bond classes. The whole process is illustrated below:



Generic CDO Structure

The key components of a CDO structure are as follows:

- The CDO raises funds by issuing debt. The debt obligations are structured as bond classes or tranches, such as senior, mezzanine, or subordinated classes with varying risk and return expectations. Investors invest in a particular bond class based on their risk appetite.
- The funds raised from the issuance of bonds are used by the collateral manager to invest in assets. He seeks to earn a higher return from these assets than what is paid to the bondholders.
- The excess spread is used to pay the equity holders and the CDO manager.

The sources of cash flow to bondholders include interest, principal repayments, and sale of collateral assets.

The most common types of CDOs are:

- Cash flow CDOs, where the cash flows from interest payments and principal repayments are redistributed across the tranches.
- Market value CDOs, where the value accruing to the tranches depends on the market value of the portfolio.
- Synthetic CDOs, where the collateral pool is created synthetically through credit derivatives.

CDOs: Risks and Motivations

In the case of defaults in the collateral, there is a risk that the manager will fail to earn a return sufficient to pay off the investors in the senior and mezzanine tranches. Investors in the subordinated/equity tranche risk the loss of their entire investment.

Summary

LO: Describe characteristics and risks of covered bonds and how they differ from other asset-backed securities.

Covered bonds are similar to ABS, but they differ because of their:

- Dual recourse nature: Investors have claims against both the issuing financial institution and the underlying asset pool.
- Balance sheet impact: The underlying asset pool remains on the issuing financial institution's balance sheet. It is not transferred to a separate SPV. The covered pool bondholders retain a top-priority claim against the pool.
- Dynamic cover pool: The underlying asset pool is not static. The issuing financial institution must replace any prepaid or non-performing assets in the cover pool to ensure sufficient cash flows until maturity. In contrast, ABS pass through default and prepayment risk to investors.
- Redemption regimes in the event of sponsor default: In the event of sponsor default, redemption regimes align the covered bond's cash flows as closely as possible to the original maturity schedule.

LO: Describe typical credit enhancement structures used in securitizations.

The main types of internal credit enhancements used in securitizations are: overcollateralization, excess spread and subordination.

External credit enhancements like financial guarantees by banks or insurance companies, letters of credit, and cash collateral accounts are also commonly used.

LO: Describe types and characteristics of non-mortgage asset-backed securities, including the cash flows and risks of each type.

Credit Card Receivable-Backed Securities

The credit card receivables are pooled together to act as a collateral for credit card receivable-backed securities. Cash flow, on a pool of credit card receivables consists of finance charges and fees, and principal repayments.

Solar ABS

A solar ABS is created by securitizing solar loans. Institutional investors looking for ESG or climate finance investment alternatives can invest in solar ABS.

LO: Describe collateralized debt obligations, including their cash flows and risks.

A collateralized debt obligation is a generic term used to describe a security backed by a diversified pool of one or more debt obligations (collateral). The structure of a CDO includes senior, mezzanine, and subordinated/equity bond classes. In the case of defaults in the

collateral, there is a risk that the manager will fail to earn a return sufficient to pay off the investors in the senior and mezzanine tranches. Investors in the subordinated/equity tranche risk the loss of their entire investment.

LM19 Mortgage-Backed Security (MBS) Instrument and Market Features

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Version 1.0

1. Introduction

This learning module focuses on the largest ABS market in the world – the mortgage-backed securities (MBS) market. In this module we will cover:

- Prepayment risk and time tranching structures
- Fundamental features of residential mortgage loans that are securitized
- Types and characteristics of residential mortgage-backed securities
- Commercial mortgage-backed securities

2. Time Tranching

Prepayment Risk

The cash flow of a mortgage consists of the following three components:

- Interest
- Scheduled principal payments
- Prepayments (any principal repaid in excess of the scheduled principal).

The risk associated with uncertainty in future cash flows of MBS securities because of unscheduled principal repayments of the underlying mortgages is called prepayment risk. It has two components: contraction risk and extension risk.

- Contraction risk is the risk that when interest rates decline, the security will have a shorter maturity than was anticipated at the time of purchase because homeowners refinance at now-available lower interest rates. For instance, assume the interest rate is 8% when the Smiths take the loan. If two years later it falls to 6%, then they will prepay the loan and refinance at the lower rate.
- Extension risk is the risk that when interest rates rise, fewer prepayments will occur because homeowners are reluctant to give up the benefits of a contractual interest rate that now looks low. From an investor's perspective, the security becomes longer in maturity than it was at the time of purchase.

Instructor's Note

Contraction risk occurs when interest rates decline.

Extension risk occurs when interest rates rise.

Time Tranching

Time tranching is a securitization structure that allows the redistribution of “prepayment risk” among bond classes. For example, in a sequential tranching structure, the principal repayments flow first to one tranche until its principal is fully repaid, then principal repayments flow to the next tranche and so on. This concept is covered later in this learning module.

3. Mortgage Loans and their Characteristic Features

A mortgage loan is a loan secured by the collateral of some specified real estate property which obliges the borrower to make a predetermined series of payments to the lender. In simple words, it is a loan a buyer takes for buying a real estate property (land, apartment, house, etc.); the collateral is the property being bought. If the buyer defaults on mortgage payments, then it gives the lender the right to foreclose on the loan, take possession of the property, and sell it to recover funds given as debt.

The amount lent as loan towards the purchase of the property is always less than the purchase price. It is equal to the purchase price minus the down payment made by the buyer. The buyer's initial equity is equal to the down payment made.

The ratio of the mortgage loan amount to the property's purchase price is called the loan-to-value (LTV) ratio. From a lender's perspective, lower the LTV, the less likely the borrower is to default. Also, if the borrower does default, the lender will have better chances of recovering the amount loaned by repossessing and selling the property.

Another consideration is the capacity to sustain debt payments measured by the debt-to-income ratio (DTI). This ratio compares an individual's monthly debt payments to their monthly pre-tax, gross income. From the lender's perspective, lower the DTI, the less likely the borrower is to default.

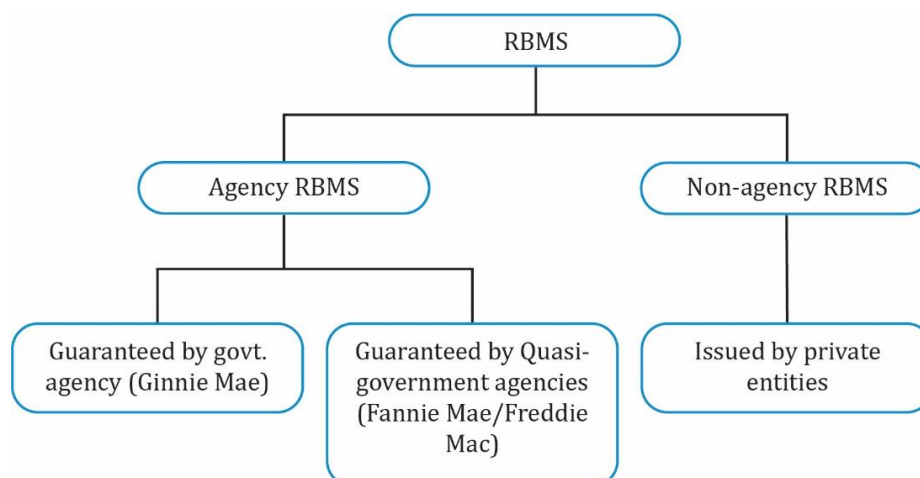
In the United States, based on the credit quality of the borrower, mortgages can be classified as:

- Prime loans – Borrower has high credit quality, strong credit history, sufficient income to service the loan, and substantial equity in the underlying property.
- Subprime loans – Borrower has low credit quality, high DTI, and/or high LTV.

Agency and Non-Agency RMBS

Residential mortgage-backed securities are bonds created from the securitization of residential mortgage loans. In the U.S., residential mortgage-backed securities are divided into the following three sectors:

1. Those guaranteed by a federal agency (Ginnie Mae) whose securities are backed by the full faith and credit of the U.S. government.
2. Those guaranteed by either of the two government-sponsored enterprises or GSEs (Fannie Mae and Freddie Mac) but not by the U.S. government. They do not carry the full faith and credit of the U.S. government.
3. Those issued by private entities that are not guaranteed by a federal agency or a GSE.



The first two sectors (guaranteed by the government or a quasi-government entity) are called the agency RMBS. The third sector is called non-agency RMBS.

Examples of agency RMBS include:

- Mortgage pass-through securities
- Collateralized mortgage obligations

The two differences between agency RMBS issued by GSEs and non-agency RMBS are as follows:

- Non-agency RMBS use credit enhancements to reduce credit risk, while agency RMBS issued by the GSEs are guaranteed by the GSEs themselves.
- For a loan to be included in a pool of loans backed by an agency RMBS, it must satisfy the underwriting standards of the government agencies.

Mortgage Contingency Features

Mortgages may contain certain features that give the borrower and the lender certain rights throughout the contract.

Prepayment option: A 'prepayment option' or an 'early repayment option' entitles the borrower to prepay all or part of the outstanding mortgage principal prior to maturity. In some countries, there may be a penalty for prepayment as it hurts the lender (recall 'prepayment risk'). The objective of imposing a penalty is to compensate the lender for the difference in the contract rate and the prevailing mortgage rate when the borrower prepays as rates decline.

Recourse and non-recourse mortgage loans: If the borrower of a loan defaults on payments, then the lender can seize the property and sell it. The proceeds from the sale may be less than the outstanding mortgage balance, and not enough to recoup the losses. There are two types of mortgage loans in such cases:

- *Recourse loans*: The lender can claim the shortfall (outstanding mortgage balance – after the property is sold) from the borrower. For instance, if the borrower has other

properties or possessions such as an expensive car, or valuable art, then these could be sold to fulfill the shortfall.

- *Non-recourse loans*: Most mortgage loans are non-recourse. The lender may sell the property in case of a default and keep the proceeds. But, unlike a recourse loan, the bank/lender cannot claim other assets of the borrower to fulfill the shortfall in the outstanding mortgage balance.

4. Residential Mortgage-Backed Securities (RMBS)

Mortgage Pass-Through Securities

A mortgage pass-through security is created when one or more holders of mortgages form a pool of mortgages and sell shares or participation certificates in the pool. The investors receive a share of cash flows from the underlying pool of mortgage loans.

Cash Flow Characteristics

- Monthly mortgage payments consist of interest, scheduled principal repayment, prepayments.
- Payments are made to security holders each month.
- The servicer collects monthly payments, sends payment notices to borrowers, sends reminders if payments are overdue, maintains records of principal balances, etc.
- The servicing fee is part of the mortgage rate.
- The amount of cash flow from mortgage loans is not equal to that received by the investors. Similarly, there is a delay in passing the cash flow from mortgage loans to the security holders.
- Monthly cash flow of a mortgage pass-through security = monthly cash flow of the underlying pool of mortgages - servicing and other fees. In other words, investors of the mortgage pass-through security receive less than the cash flow coming in from the mortgage loans because a servicing fee is collected by the servicer.

How is the rate and maturity of a mortgage loan calculated?

- *Pass-through rate*: A mortgage pass-through security's coupon rate is called the pass-through rate. For example, if the mortgage rate for a pool of mortgages is 8%, the annualized servicing fee is 0.6%, then the investors receive an average return of around 7.4%.
- *Weighted average coupon (WAC)*: Each of the mortgage loans in the securitized pool may not have the same mortgage rate. The WAC is found by weighting the rate of each mortgage loan in the pool by the percentage of the mortgage outstanding relative to the outstanding amount of all mortgages in the pool.
- *Weighted average maturity (WAM)*: Similarly, not all the loans in the pool will have the same maturity. WAM is found by weighting the remaining number of months to maturity for each mortgage loan in the pool by the amount of the outstanding mortgage balance.

Example

Assume that a pool includes four mortgages with the following characteristics:

Mortgage	Outstanding Balance	Coupon Rate	Time to Maturity
1	\$1,000	4.50%	28 months
2	\$2,000	4.75%	42 months
3	\$4,000	5.15%	37 months
4	\$3,000	3.80%	60 months

Calculate the weighted average coupon rate and weighted average maturity.

Solution:

The total outstanding amount is \$10,000 and hence, the weights are as follows: Mortgage 1 = 10%, 2 = 20%, 3 = 40% and 4 = 30%.

The weighted average coupon is:

$$0.10 \times 4.50\% + 0.20 \times 4.75\% + 0.40 \times 5.15\% + 0.30 \times 3.80\% = 4.60\%$$

The weighted average maturity (WAM) is:

$$0.10 \times 28 + 0.20 \times 42 + 0.40 \times 37 + 0.30 \times 60 = 44 \text{ months}$$

Collateralized Mortgage Obligations (CMOs)

The prepayment risk seen in mortgage pass-through securities can be reduced by distributing the cash flows of these mortgage products to different classes or tranches through a process called structuring.

Collateralized mortgage obligation (CMO) is one such security created based on this principle of structuring where the cash flows (interest and principal) are redistributed to different tranches based on a set of rules. The different classes of bondholders in a CMO have different exposures to prepayment risk. The collateral for a CMO is a pool of mortgage pass-through securities and not a pool of mortgage loans.

Advantages of a CMO

Following are the key benefits of a CMO:

- CMOs can be created to closely satisfy the asset/liability needs of institutional investors, thereby broadening the appeal of mortgage-backed products.
- Some investors may want to increase their exposure to prepayment risk, while some may want to reduce. Based on their individual needs and risk appetite, investors can choose the CMO.

The most common types of CMO tranches are sequential-pay tranches.

Sequential-Pay CMO

Each class/tranche of bond in this CMO structure is retired sequentially. Let us consider a CMO with four tranches. Note that this example is for simplicity. The coupon rate usually varies by tranche.

Tranche	Par amount(US \$)	Coupon rate (%)
A	389,000,000	5.5
B	72,000,000	5.5
C	193,000,000	5.5
D	146,000,000	5.5
Total	800,000,000	

The prepayment risk is mitigated in this CMO by following these interest and principal repayment rules:

For payment of monthly coupon interest: Disburse monthly coupon interest to each tranche on the basis of the amount of principal outstanding for each tranche at the beginning of the month.

For disbursement of principal payments:

Disburse principal payments to tranche A until it is completely paid off.

After tranche A is completely paid off, disburse principal payments to tranche B until it is completely paid off.

After tranche B is completely paid off, disburse principal payments to tranche C until it is completely paid off.

After tranche C is completely paid off, disburse principal payments to tranche D until it is completely paid off.

In this structure:

- Tranche A has the highest contraction risk while tranche D has the highest extension risk.
- Tranches A and B provide protection against contraction risk for tranches C and D.
- Similarly, tranches C and D provide protection against extension risk for tranches A and B respectively.

Other CMO Structures

- Z-tranches: They do not pay interest payments until a pre-set date, when both principal and accrued interest payments start.
- Principal-only (PO) securities: They pay only the principal repayments from the pool.
- Interest-only (IO) securities: They pay only the interest payments from the pool.
- Floating-rate tranche: Although the collateral pays a fixed rate, we can create tranches that pay floating rates. To do this a floater and an inverse floater combination is constructed from any of the fixed-rate tranches in the CMO structure. These tranches are sold to separate sets of investors with opposing views on interest rate movements. If interest rates go up, the floating rate tranche will pay a higher rate but the inverse

floaters will pay a lower rate. Thus, the two tranches offset each other and the effective rate paid will be equal to rate on original fixed rate tranche.

- **Planned Amortization Class (PAC) tranche:** PAC tranches are often accompanied by support tranches. PAC tranches make scheduled and fixed principal payments over a predetermined time period to their investors if the prepayment levels in the pool are within a certain maximum and minimum range. If the prepayment rate is within the specified range, all prepayment risk is absorbed by the support tranche. Thus, PAC tranches offer the greatest predictability and stability of the cash flows.

5. Commercial Mortgage-Backed Securities (CMBS)

Commercial mortgage-backed securities (CMBS) are backed by a pool of commercial mortgage loans on income-producing property. Important features of a CMBS are as follows:

- The underlying are loans to purchase or refinance a commercial property such as a warehouse, apartment building, office building, hotels, health care facilities, etc.
- Commercial mortgage loans are non-recourse loans. Lenders can only stake a claim to the income-producing property backing the loan in case of a default and not on any other asset of the borrower. Therefore, analysis of CMBS securities focus on the credit risk of the property and not on the credit risk of the borrower.
- It is important to study the cash flows from the underlying properties for credit analysis.
- There are two key indicators to assess the potential credit performance of a commercial mortgage loan: 1) debt-to-service coverage ratio and 2) the loan-to-value ratio

$$\text{Debt-to-service coverage ratio} = \frac{\text{Annual net operating income}}{\text{Debt service}}$$

where:

Debt service = annual interest payment and principal repayment

Net operating income = rental income – cash operating expenses – a non-cash replacement reserve

If DSC > 1.0, then cash flows from property are sufficient to service debt.

How to interpret DSC and LTV ratios:

- The higher the DSC ratio, the lower the credit risk and the better is the borrower's ability to service debt.
- A low loan-to-value ratio implies lower credit risk.
- Note: to memorize this formula, draw a parallel with interest coverage ratio from FSA.

CMBS Structure

Interest and principal repayments in a CMBS are structured as follows:

- Interest on principal outstanding is paid to all tranches.

- The highest-rated bonds are paid off first in the CMBS structure.
- Losses arising from loan defaults are charged against the principal balance of the lowest priority CMBS tranche outstanding. These tranches may be unrated by credit-rating agencies and are called the “first-loss piece”, “residual tranche”, or “equity tranche”.

Characteristics of a CMBS structure

Call Protection

RMBS investors are exposed to prepayment risk since the borrowers have a right to prepay and are not penalized for prepayment; they have an incentive to prepay. CMBS has considerable call protection, which is protection against early prepayment of mortgage principal. The call protection comes in two forms: at the structure level and at the loan level.

Call protection at the structural level:

Call protection at the structural level comes by structuring CMBS into sequential-pay tranches, by credit rating. A lower-rated tranche cannot be paid off until the higher-rated tranches are retired. But, in the case of a default, the losses must be charged to the lowest-rated tranche first and last to the highest-rated tranche.

Call protection at the loan level:

There are four mechanisms that offer investors call protection at the loan level:

1. *Prepayment lockouts*: The borrower is prohibited from any prepayments during a specific period of time.
2. *Prepayment penalty points*: The borrower must pay a fixed percentage of the outstanding loan balance as prepayment penalty if he wishes to refinance.
3. *Yield maintenance charges*: Also known as “make-whole charge”. The borrower must pay a penalty to the lender that makes refinancing uneconomical if the sole objective was to get a lower mortgage rate.
4. *Defeasance*: Defeasance is a protection at the loan level that requires the borrower to provide sufficient funds that can be invested in a portfolio of government securities to replicate the cash flows in the absence of prepayments.

CMBS Risks

Commercial mortgage-backed securities (CMBS) can consist of only a few underlying commercial mortgages, so a single default in a CMBS pool can have a significant impact on CMBS investors. Investors must evaluate this unique concentration risk by analyzing the individual loans and properties backing the CMBS.

Balloon Risk

Residential mortgages are fully-amortizing loans that are fully amortized over a long period of time. Usually, there is no principal outstanding after the last mortgage payment. But many commercial loans backing CMBS transactions are balloon loans which require a substantial

principal payment on the final maturity date. If the borrower is not able to make the lump sum payment, he may ask for an extension of the loan over a period of time called the “workout period”.

Balloon risk is a type of extension risk. The risk that a borrower will not be able to make the balloon payment either because the borrower cannot arrange for refinancing or cannot sell the property to generate sufficient funds to pay off the balloon balance is called “balloon risk”.

Summary

LO: Define prepayment risk and describe time tranching structures in securitizations and their purpose.

Prepayment risk refers to the uncertainty about the timing of the principal component of the cash flows.

Contraction risk is the risk that the loan principal will be repaid earlier than expected.

Extension risk is the risk that the loan principal will be repaid later than expected.

LO: Describe fundamental features of residential mortgage loans that are securitized.

Mortgage loans provide borrowers the funds to purchase property and require borrowers to repay lenders on a mutually agreed schedule, otherwise the lender has the right to seize the property.

Loan-to-value (LTV) ratio and Debt-to-income (DTI) ratio are two important metrics used to evaluate a mortgage.

In the United States, based on the credit quality of the borrower, mortgages can be classified as:

- Prime loans – Borrower has high credit quality, strong credit history, sufficient income to service the loan, and substantial equity in the underlying property.
- Subprime loans – Borrower has low credit quality, high DTI, and/or high LTV.

Mortgages may contain certain features that give the borrower and the lender certain rights throughout the contract.

- A prepayment option entitles the borrower to prepay all or part of the outstanding mortgage principal prior to maturity.
- Recourse loans: The lender can claim the shortfall (outstanding mortgage balance – after selling the property) from the borrower.
- Non-recourse loans: The lender may sell the property in case of a default and keep the proceeds but they cannot claim other assets of the borrower to fulfill the shortfall in outstanding mortgage balance.

LO: Describe types and characteristics of residential mortgage-backed securities, including mortgage pass-through securities and collateralized mortgage obligations, and explain the cash flows and risks for each type.

Residential mortgage-backed securities (RMBS) are bonds created from the securitization of residential mortgage loans. Examples of agency RMBS include:

- Mortgage pass-through securities
- Collateralized mortgage obligations

A mortgage pass-through security is created when one or more holders of mortgages form a pool of mortgages and sell shares or participation certificates in the pool. The cash flow of a mortgage pass-through security depends on the cash flow of the underlying pool of mortgages and consists of monthly mortgage payments representing interest, the scheduled repayment of principal, and any prepayments, net of servicing and other fees.

Collateralized mortgage obligations (CMOs) securitize mortgage pass-through securities or multiple pools of loans and are structured to redistribute cash flows to different bond classes or tranches, resulting in securities with varying prepayment risk exposures.

LO: Describe characteristics and risks of commercial mortgage-backed securities.

Two important characteristics of a CMBS structure: call protection and balloon risk.

Call protection

CMBS have considerable call protection, which is protection against early prepayment of mortgage principal. The call protection comes in two forms: at the structure level and at the loan level.

- Call protection at the structural level comes by structuring CMBS into sequential-pay tranches, by credit rating.
- The four mechanisms that offer investors call protection at the loan level are payment knockouts, prepayment penalty points, yield maintenance charges, and defeasance.

Balloon risk

Balloon risk is a type of extension risk. The risk that a borrower will not be able to make the balloon payment, either because the borrower cannot arrange for refinancing or cannot sell the property to generate sufficient funds to pay off the balloon balance, is called “balloon risk”.